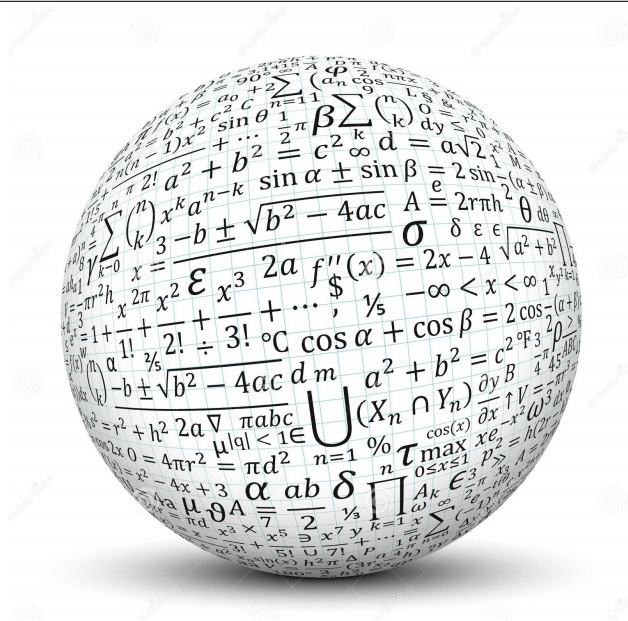




1 Instructions:

- There are some questions which have functions with discrete valued domains (such as day, month, year etc).
- For NAT type questions, enter only one right answer even if you get multiple answers for that particular question.
- **Notations:**
 - $\mathbb{R}$ = Set of real numbers
 - $\mathbb{Q}$ = Set of rational numbers
 - $\mathbb{Z}$ = Set of integers
 - $\mathbb{N}$ = Set of natural numbers
- The set of natural numbers includes 0.

AQ 1.1

Questions

1. Below is a list of numbers: 22, -17, 47, -2000, 0, 1, 43, 1729, 6174, -63, 100, 32, -9. How many natural numbers are there in the given list?

- (a) 6
- (b) 7
- (c) 8
- (d) 9

[Ans: (d)]

2. Suppose a class has M students. If we distribute 3 pens to each student, then 4 pens will remain. If there are 70 pens in total, what is the value of M ?

- (a) 22
- (b) 20
- (c) 19
- (d) 23

[Ans: (a)]

3. Consider '2' to be the first prime number. Which of the following will be the sixteenth prime number?

- (a) 47
- (b) 61
- (c) 59
- (d) 53

[Ans: (d)]

4. If $x \bmod 3 = 1$ and $x \bmod 7 = 4$, which of the following is(are) possible values for x ?

- (a) 25
- (b) 34
- (c) 58
- (d) 67

[Ans: (a), (d)]

Questions

5. Suppose Sheetal's birthday is on the P th of October and Karthik's birthday is on the Q th of November, where P and Q are perfect squares. If Sheetal was born 10 days before Karthik, then the value of $P - 2Q$ will be

[Ans: 17]

AQ 1.2

Questions

1. Which of the following option(s) is(are) true?

- (a) $\frac{4}{7} > \frac{5}{8}$
- (b) $\frac{12}{17} > \frac{7}{13}$
- (c) $\frac{11}{7} > \frac{13}{8}$
- (d) $\frac{7}{15} < \frac{2}{5}$

[Ans: (b)]

2. Which of the following option(s) is(are) in reduced form?

- (a) $\frac{5}{60}$
- (b) $\frac{12}{27}$
- (c) $\frac{11}{18}$
- (d) $\frac{13}{91}$

[Ans: (c)]

3. Let $\gcd(a, 4) = 2$ and $\gcd(a, b) = 1$. If $4 > a > b$ and a, b are natural numbers, then the values of a and b are respectively

- (a) 2, 1
- (b) 1, 2
- (c) 2, 4
- (d) 4, 2

[Ans: (a)]

AQ 1.3

Questions

1. Which of the following statement(s) is(are) false?

- (a) The sum of two natural numbers is always a natural number.
- (b) The difference between two integers is always an integer.
- (c) The product of two rational numbers is always a real number.
- (d) The product of two irrational numbers is always an irrational number.

[Ans: (d)]

2. Below is a list of real numbers:

$$\sqrt{3}, 2.5, e, \sqrt{49}, \frac{7}{12}, 22, \pi, -35, \sqrt{6}, 1729, -20000$$

How many irrational numbers are there in the given list?

- (a) 4
- (b) 5
- (c) 6
- (d) 7

[Ans: (a)]

3. How many integers are there in the given list?

- (a) 3
- (b) 4
- (c) 5
- (d) 6

[Ans: (c)]

4. Which of the following statement(s) is(are) true?

- (a) $\sqrt{-2}$ is a complex number.
- (b) Real numbers extend rational numbers.
- (c) None of these.

[Ans: (a), (b)]

Questions

5. Which of the following rational numbers are greater than $\sqrt{2}$ and less than $\sqrt{3}$?

(a) $\frac{9}{5}$

(b) $\frac{3}{2}$

(c) $\frac{5}{3}$

(d) $\frac{17}{10}$

[Ans: (b), (c), (d)]

AQ 1.4

Questions

1. Which of the following sets are same?

- (i) {Ankitha, Keerthana, Raju, Suresh}
- (ii) {Raju, Ankitha, Keerthana, Raju, Ankitha, Suresh}
- (iii) {Keerthana, Suresh, Dheeraj, Raju, Ankitha}
- (iv) {Suresh, Raju, Ankitha, Keerthana}
- (v) {Dheeraj, Raju, Soumya, Keerthana}

- (a) (i) and (ii)
- (b) (iii), (iv) and (v)
- (c) (i) and (iv)
- (d) (i), (ii) and (iv)

[Ans: (d)]

2. Suppose $X = \{3, \pi, \text{Tiger}, \text{Ball}, -40, \text{Dhoni}\}$. Which of the following statement(s) is(are) true about X ?

- (a) X is not a set because a collection of names of persons and numbers together will not be considered a set.
- (b) X is not a set because " π " cannot be an element of a set.
- (c) X is a set because X contains items that are from more than two categories.
- (d) X is a set because it is a collection of different well-defined items.

[Ans: (d)]

3. Which of the following sets is a subset of the set of prime numbers?

- (a) {2, 3, 5, 8, 11}
- (b) {3, 5, 7, 11}
- (c) {2, 5, 7, 27}
- (d) {5, 9}

[Ans: (b)]

Questions

4. Which of the following statement(s) is(are) false?
- (a) Venn diagrams are pictorial representations of sets.
 - (b) Every set is a subset of itself.
 - (c) $X \subseteq Y$ implies that X is a proper subset of Y .
 - (d) Empty set contains only one element i.e. ϕ .

[Ans: (c), (d)]

5. Which of the following are empty sets?
- (a) Set of all perfect squares between 45 and 50
 - (b) Set of all integers which are both even and odd
 - (c) Set of all years having 366 days
 - (d) Set of all prime numbers divisible by 2

[Ans: (b)]

6. If $X = \{a, b, c\}$, then the power set $P(X) =$
- (a) $\{\phi, \{a\}, b, c, ab, ac, bc, abc\}$
 - (b) $\{\phi, b, c, ab, ac, bc\}$
 - (c) $\{\phi, \{a\}, b, \{c\}, ab, \{ac\}, bc, \{abc\}\}$
 - (d) $\{\phi, \{a\}, b, \{c\}, ab, \{ac\}, bc, \{b, c\}\}$

[Ans: (a)]

7. Let M be the set of all days in a week whose starting letter is either 'S' or 'T', and N be the set of all months in a year whose ending letter is 'Y'. What are the cardinalities of M and N , respectively?
- (a) 4, 3
 - (b) 3, 4
 - (c) 4, 4
 - (d) 3, 3

[Ans: (c)]

AQ 1.5

Questions

1. Which of the following is a correct representation of set comprehension?

- (a) $\{x ; x \in \mathbb{N}, x \text{ is even}\}$
- (b) $\{x \mid x \in \mathbb{N}, x \text{ is even}\}$
- (c) $\{x \text{ is even} \mid x \in \mathbb{N} \mid x\}$
- (d) $\{x \text{ is even}; x \in \mathbb{N}, x\}$

[Ans: (b)]

2. Which of the following is the set of natural numbers that are multiples of 3 or 5?

- (a) $\{x \mid x \in \mathbb{N}, x \bmod 3 = 0 \text{ and } x \bmod 5 = 0\}$
- (b) $\{x \mid x \in \mathbb{Z}, x \bmod 3 = 0 \text{ or } x \bmod 5 = 0\}$
- (c) $\{x \mid x \in \mathbb{Z}, x \bmod 3 = 0 \text{ and } x \bmod 5 = 0\}$
- (d) $\{x \mid x \in \mathbb{N}, x \bmod 3 = 0 \text{ or } x \bmod 5 = 0\}$

[Ans: (d)]

Questions

Given below is the set of students from Madras University who are going to participate in SCHROETER, a sports fest organized by IIT Madras:

Students: {**Abbas, Akhil, Aditya, Balachandra, Bhuvan, Chidambaram, Charan, Kiran, Dayal, Deepak, Jaivardhan, Nithin, Sunil, Karthik, Rakesh**}

These students are interested in playing volleyball, basketball, and cricket. The sets for each sport are:

Volleyball: {**Akhil, Charan, Deepak, Sunil, Karthik, Rakesh**}

Basketball: {**Charan, Rakesh, Kiran, Chidambaram, Balachandra**}

Cricket: {**Abbas, Akhil, Aditya, Bhuvan, Charan, Dayal, Deepak, Kiran, Jaivardhan, Nithin, Rakesh**}

Based on the above data:

3. Which of the following sets represents students from Madras University who play both cricket and basketball?
- (a) {**Abbas, Akhil, Rakesh**}
 - (b) {**Bhuvan, Charan, Dayal, Kiran**}
 - (c) {**Kiran, Charan, Rakesh**}
 - (d) {**Aditya, Jaivardhan, Nithin**}

[Ans: (c)]

4. What is the number of students who play all three sports?
- (a) 1
 - (b) 2
 - (c) 3
 - (d) 4

[Ans: (b)]

Questions

5. Which of the following set represents the students from Madras University who play volleyball but not cricket?

- (a) {**Akhil, Rakesh, Sunil**}
- (b) {**Sunil, Kartik**}
- (c) {**Deepak**}
- (d) {} (Empty Set)

[Ans: (b)]

6. Which of the following sets represent the students from Madras University who are not playing cricket?

- (a) {**Dayal, Deepak, Kiran, Jaivardhan**}
- (b) {**Rakesh, Kiran, Chidambaram, Balachandra**}
- (c) {**Bhuvan, Jaivardhan, Sunil, Chidambaram**}
- (d) {**Sunil, Chidambaram, Karthik, Balachandra**}

[Ans: (d)]

Questions

Consider two sets that represent the items that are required to be carried by a student to an exam center to write exam A and exam B respectively:

$$A = \{\text{Hall ticket, Pen, Calculator, Ruler, Pencil}\},$$

$$B = \{\text{Compass, Pencil, Drafter, Ruler, Eraser, Sharpener, Hall ticket}\}$$

Based on the above data, answer the following questions:

7. If exams A and B are going to be conducted simultaneously, then what are the items that are required to be carried by the student to the exam center?
- (a) {Hall ticket, Ruler, Pencil, Eraser, Sharpener, Pen, Drafter, Compass, Calculator}
 - (b) {Calculator, Ruler, Pencil, Hall ticket, Pen, Paper, Compass, Drafter}
 - (c) {Pen, Calculator, Pencil, Hall ticket, Ruler, Compass, Drafter, Eraser}
 - (d) {Ruler, Pencil, Compass, Drafter, Eraser, Sharpener, Hall ticket, Calculator}
- [Ans: (a)]
8. If a student forgets to bring her calculator, which exam will she not be allowed to write?
- (a) Exam B
 - (b) She can write both the exams
 - (c) Exam A
 - (d) None of the above

[Ans: (c)]

AQ 1.6

Questions

1. Which of the following sets is(are) infinite?

- (a) Set of all Indian Nobel laureates
- (b) Set of squares of all odd natural numbers
- (c) Set of all countries in the world
- (d) Set of all leap years

[Ans: (b), (d)]

2. Which of the following set comprehension defines real numbers in interval $[-2, 0) \cup (4, 8]$?

- (a) $\{r|r \in \mathbb{R}, -2 \leq r < 8\}$
- (b) $\{r|r \in \mathbb{R}, -2 < r < 0 \text{ and } 4 < r \leq 8\}$
- (c) $\{r|r \in \mathbb{R}, -2 < r < 0 \text{ or } 4 < r \leq 8\}$
- (d) $\{r|r \in \mathbb{R}, -2 \leq r < 0 \text{ and } 4 < r \leq 8\}$

[Ans: (c)]

3. Which of the following set comprehensions define squares of the first 100 natural numbers?

- (a) $\{n|n \in \mathbb{N}, \sqrt{n} \in \mathbb{N}, n < 100\}$
- (b) $\{n^2|n \in \mathbb{N}, n < 100\}$
- (c) $\{n|n \in \mathbb{N}, \sqrt{n} \in \mathbb{N}, n < 10,000\}$
- (d) $\{n^2|n \in \mathbb{N} < 10,000\}$

[Ans: (b), (c)]

4. Which of the following statement(s) is(are) true?

- (a) Empty set contains only one element i.e. ϕ .
- (b) Set with n elements has 2^{n-1} subsets.
- (c) Empty set is an element in the power set of a set.
- (d) Power set is a set of all subsets of a set.

[Ans: (c), (d)]

Questions

5. Which of the following intervals are subsets of $[2, 3]$?

- (a) $((0, 2.3] \cup (2.3, 3]) \cap (0, 2.3)$
- (b) $((2, 2.5] \cup (2.5, 4]) \setminus (0, 2.3)$
- (c) $((2, 2.5] \cup (2.5, 4]) \cap (0, 3)$
- (d) $((0, 2.3] \cup (2.2, 3]) \setminus (0, 2.3)$

[Ans: (c), (d)]

6. Let M be the set of all real numbers that are strictly greater than 6 or strictly less than -6. How can we represent B in set comprehension form, where B is a subset of M and has only integers as elements?

- (a) $B = \{z | z \in \mathbb{Z}, z \in [-6, 6]\}$
- (b) $B = \{z | z \in \mathbb{Z}, z \in (-6, 6)\}$
- (c) $B = \{z | z \in \mathbb{Z}, z \in (-\infty, -6) \cup (6, \infty)\}$
- (d) $B = \{z | z \in \mathbb{Z}, z \in (-\infty, -6] \cup [6, \infty)\}$

[Ans: (c)]

7. Two finite sets A and B are such that the total number of subsets of A is 56 more than the total number of subsets of B . What are the cardinalities of A and B respectively?

- (a) 6, 3
- (b) 8, 4
- (c) 7, 1
- (d) 8, 3

[Ans: (a)]

AQ 1.7

Questions

1. In a class of 40 students, 20 took Mathematics, 18 took Statistics, 16 took English, 7 took Mathematics and Statistics, 7 took Mathematics and English, 9 took Statistics and English, and 3 took all the three courses. How many students are not taking any of these courses?
 - (a) 3
 - (b) 4
 - (c) 5
 - (d) 6

[Ans: (d)]

2. In a survey, it is found that 30 people like apples, 45 people like oranges, and 20 people like both apples and oranges. How many people like only one fruit?

[Ans: 35]

3. In a class, 50 students play cricket, 35 play football, 14 play both, and 9 play neither. How many students are there in the class?
 - (a) 80
 - (b) 108
 - (c) 90
 - (d) 62

[Ans: (a)]

Questions

4. A group of 25 people are planning to get together to celebrate their 10 years of friendship. Out of them, 9 suggested to go to Goa, 8 suggested to go to Bangkok, X suggested to go to Pondicherry and 3 of them did not suggest any place. Considering each person suggested only one place, what is the value of X ?

- (a) 5
- (b) 3
- (c) 4
- (d) Data is insufficient

[Ans: (a)]

5. An advertising agency finds that, of its 200 clients who use Television or Radio or both, 150 use Television. How many use only Radio?

- (a) 150
- (b) 100
- (c) 50
- (d) Data is insufficient

[Ans: (c)]

AQ 1.8

Questions

1. Let $A = \{1, 4\}$ and $B = \{2, 4, 6, 8\}$. Which of the following is the Cartesian product of A and B ?

- (a) $\{(1, 2), (1, 4), (1, 6), (1, 8), (4, 2), (4, 4), (4, 6), (4, 8)\}$
- (b) $\{(1, 2), (1, 4), (1, 6), (1, 8), (4, 2), (4, 4), (4, 6), (4, 8)\}$
- (c) $\{(1, 2), (1, 4), (1, 6), (1, 8), (2, 4), (4, 2), (4, 6), (4, 8)\}$
- (d) $\{(1, 6), (1, 8), (2, 4), (2, 6), (4, 2), (4, 4), (4, 6), (4, 8)\}$

[Ans: (b)]

2. If set M contains 4 elements and set N contains 3 elements, then $M \times N$ contains x elements. What is the value of x ?

[Ans: 12]

3. Suppose the Cartesian product $A \times B$ has 18 elements. Which of the following options may represent the number of elements in A and B ?

- (a) 3, 9
- (b) 2, 9
- (c) 9, 9
- (d) 6, 3

[Ans: (b), (d)]

4. Which of the following statement(s) is(are) true?

- (a) In a set, order of elements is not important
- (b) A relation is a subset of Cartesian product
- (c) In each pair of Cartesian product, order is not important
- (d) All of the above

[Ans: (a), (b)]

Questions

5. Let $B = \{\text{Anil, Ramu, Suraj}\}$ and $G = \{\text{Neha, Keerthi}\}$ be the sets of boys and girls respectively. Ramu is brother of Neha; Anil is brother of Keerthi; Suraj is brother of Neha and Keerthi. Let us define a relation $R = \{(a, b) | (a, b) \in B \times G, a \text{ is brother of } b\}$. Which of the following will be R ?

- (a) $\{(\text{Anil, Keerthi}), (\text{Suraj, Neha}), (\text{Ramu, Neha}), (\text{Suraj, Keerthi})\}$
- (b) $\{(\text{Anil, Keerthi}), (\text{Ramu, Keerthi}), (\text{Ramu, Neha}), (\text{Suraj, Neha})\}$
- (c) $\{(\text{Anil, Neha}), (\text{Ramu, Keerthi}), (\text{Ramu, Neha}), (\text{Suraj, Keerthi})\}$
- (d) $\{(\text{Anil, Keerthi}), (\text{Ramu, Neha}), (\text{Suraj, Neha}), (\text{Ramu, Keerthi}), (\text{Anil, Neha}), (\text{Suraj, Keerthi})\}$

[Ans: (d)]

6. Let S be a set of students who are studying B.Tech first year at IIT Madras. Which of the following relations is a symmetric relation?

- (a) $\{(a, b) | a, b \in S, a \text{ "is younger than" } b\}$
- (b) $\{(a, b) | a, b \in S, a \text{ "got more marks in mathematics course than" } b\}$
- (c) $\{(a, b) | a, b \in S, a \text{ "is classmate of" } b\}$
- (d) $\{(a, b) | a, b \in S, a \text{ "is taller than" } b\}$

[Ans: (c)]

7. Which of the following statement(s) is(are) incorrect?

- (a) An equivalence relation partitions a set
- (b) An equivalence class is same as equivalence relation
- (c) Antisymmetric relation is an equivalence relation
- (d) Elements in Cartesian product will only be in pairs

[Ans: (b), (c), (d)]

AQ 1.9

Questions

1. Suppose $f : \mathbb{Z} \rightarrow \mathbb{Z}$ is a function defined by $f(k) = k^3 + 4k - 10$. The value of $f(k)$ at $k = 4$ is _____. [Ans: 70]

2. Let $f(x) = |x| + 5$ and $\text{Dom}(f) = \{x \in \mathbb{R} | f(x) \in \mathbb{R}\}$. Which of the following is(are) true?

- (a) Domain of $f(x)$ is $\mathbb{R}$
- (b) Range of $f(x)$ is $\mathbb{R}$
- (c) Domain of $f(x)$ is $[5, \infty)$
- (d) Range of $f(x)$ is $[5, \infty)$

[Ans: (a), (d)]

3. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function and $f(x) = |(x + 4)(4x - 10)|$. Which of the following is(are) true?

- (a) f is an injective function
- (b) f is a surjective function
- (c) f is a bijective function
- (d) None of these

[Ans: (d)]

4. Which of the following statement(s) is(are) false?

- (a) Range of a function is a subset of co-domain
- (b) A function is bijective if and only if it is both injective and surjective
- (c) Every relation is a function
- (d) Co-domain of a function is a subset of domain

[Ans: (c), (d)]

Questions

5. Let $x \in \mathbb{R}$ and $\text{Dom}(f) = \{x \in \mathbb{R} \mid f(x) \in \mathbb{R}\}$. Which of the following functions is(are) injective?

(a) $f(x) = \sqrt{10 - x}$

(b) $f(x) = \frac{7x+6}{3x}$

(c) $f(x) = 2x + 9$

(d) $f(x) = (5x + 4)(2x - 3)/2$

[Ans: (a), (b), (c)]

6. Suppose $f : Z \rightarrow Z$ is a function defined by $f(x) = ax + b$. For which of the following integer values of a and b , is the given function bijective?

(a) $a = 0, b = \{z \mid z \in Z\}$

(b) $a = \{-1, 1\}, b = \{z \mid z \in Z\}$

(c) $b = \{-1, 1\}, a = 0$

(d) $b = 0, a = \{z \mid z \in Z\}$

[Ans: (b)]

AQ 1.10

Questions

Answer the questions 1–3 based on the following information:

Let $A = \{x | x \in \mathbb{N}, x < 10 \text{ and } x \text{ is odd}\}$

$B = \{y | y \in \mathbb{N}, y \text{ is a perfect square and } 15 < y < 40\}$

1. Which of the following is a subset of $B \times A$?

- (a) $\{(36, 3), (25, 5), (36, 6)\}$
- (b) $\{(1, 25), (6, 36), (7, 25), (3, 36)\}$
- (c) $\{(16, 5), (25, 9), (36, 3), (16, 1)\}$
- (d) $\{(1, 16), (5, 16), (3, 36), (9, 36), (7, 25)\}$

[Ans: (c)]

2. Which of the following is a subset of $A \times A$?

- (a) $\{(1, 3), (3, 5), (5, 9), (7, 3), (7, 7)\}$
- (b) $\{(36, 25), (25, 25), (36, 36), (16, 25)\}$
- (c) $\{(3, 3), (1, 5), (3, 6), (9, 9)\}$
- (d) $\{(3, 5), (5, 1), (9, 4), (1, 1), (7, 1)\}$

[Ans: (a)]

3. Which of the following is(are) element(s) of $A \times B \times A$?

- (a) $(2, 25, 3)$
- (b) $(5, 25, 9)$
- (c) $(3, 36, 9)$
- (d) $(5, 16, 1)$

[Ans: (b), (c), (d)]

Questions

Answer the questions 4-5 based on the following information: Suppose the Cartesian product $M \times M$ has 9 elements among which two elements are $(2, 4)$ and $(3, 2)$.

4. Which of the following sets is M ?

- (a) $\{9, 2, 4\}$
- (b) $\{3, 2, 9\}$
- (c) $\{2, 3, 4\}$
- (d) $\{3, 4, 9\}$

[Ans: (c)]

5. Remaining elements of $M \times M$ are:

- (a) $\{(2, 3), (2, 2), (3, 4), (3, 2), (4, 4), (4, 2), (3, 3)\}$
- (b) $\{(2, 3), (2, 2), (3, 4), (4, 3), (4, 4), (4, 2), (3, 3)\}$
- (c) $\{(2, 3), (2, 2), (4, 3), (2, 4), (4, 4), (4, 2), (3, 3)\}$
- (d) $\{(2, 2), (2, 3), (4, 3), (3, 2), (3, 3), (4, 4), (4, 3)\}$

[Ans: (b)]

Questions

Answer the questions 6 - 8, based on following information:

Suppose the relation $R = \{(\text{Snail}, \text{Frog}), (\text{Bird}, \text{Bird}), (\text{Fox}, \text{Frog}), (\text{Snail}, \text{Fox})\}$ is defined on the set $S = \{\text{Snail}, \text{Fox}, \text{Bird}, \text{Frog}\}$.

6. The relation R on the set S is(are):

- (a) Reflexive
- (b) Symmetric
- (c) Transitive
- (d) None of these

[Ans: (c)]

7. Let P be a subset of $[(S \times S) - R]$. What is the set of possible cardinalities of P ?

- (a) $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$
- (b) $\{\}$
- (c) $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$
- (d) $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

[Ans: (d)]

8. What is the minimum number of elements of the Cartesian product $S \times S$ that need to be added to R such that the relation becomes symmetric?

- (a) 4
- (b) 8
- (c) 2
- (d) 3

[Ans: (d)]

AQ 1.11

Questions

1. If $\text{Dom}(f) = \{x \in \mathbb{R} | f(x) \in \mathbb{R}\}$ defined by $f(x) = \frac{x+12}{4x-8}$, then the domain of the function f is -----.

- (a) $\mathbb{R}$
- (b) $\mathbb{R} \setminus \{\frac{1}{4}\}$
- (c) $\mathbb{R} \setminus \{-12\}$
- (d) $\mathbb{R} \setminus \{2\}$

[Ans: (d)]

2. The product of the minimum value of the function $f(x) = 9|x| - 8$ and the maximum value of the function $g(x) = 11 - |x + 8|$ is -----.

[Ans: -88]

3. Which of the following functions is always non-negative if $\text{dom}(f) = \{x \in \mathbb{R} | f(x) \in \mathbb{R}\}$?

- (a) $f(x) = \sqrt{3x + 15}$
- (b) $f(x) = |x|$
- (c) $f(x) = \frac{30-11x}{5x+60}$
- (d) $f(x) = 4x + 19$

[Ans: (a), (b)]

Questions

4. Let us define a function $f : \mathbb{N} \rightarrow \mathbb{N}$ as follows:

$$f(n) = \begin{cases} \frac{n+1}{2} & \text{if } n \text{ is odd,} \\ n/2 & \text{if } n \text{ is even.} \end{cases}$$

Which of the following is true?

- (a) f is one to one and onto
- (b) f is one to one but not onto
- (c) f is onto but not one to one
- (d) f is neither one to one nor onto

[Ans: (c)]

5. If $f(x) = |x + 2|$ and $g(x) = x^2 - 4$, then which of the following is true?

- (a) The value $f(x)$ is greater than $g(x)$ at $x = 1$
- (b) The value $g(x)$ is greater than $f(x)$ at $x = -2$
- (c) The value $g(x)$ is greater than $f(x)$ at $x = 3$
- (d) The value $f(x)$ is greater than $g(x)$ at $x = -3$

[Ans: (a)]

6. If $f(x) = \sqrt{14 - (3x + \frac{35-5x}{4})}$, then the domain of the function f is:

- (a) $(-\infty, -3]$
- (b) $[3, \infty)$
- (c) $[-\infty, 3)$
- (d) $[-3, \infty)$

[Ans: (c)]

Questions

The domain and range of the function that is shown in Figure 1 are $[m, x]$ and $[n, y]$ respectively, where $m, n, x, y \in \mathbb{R}$

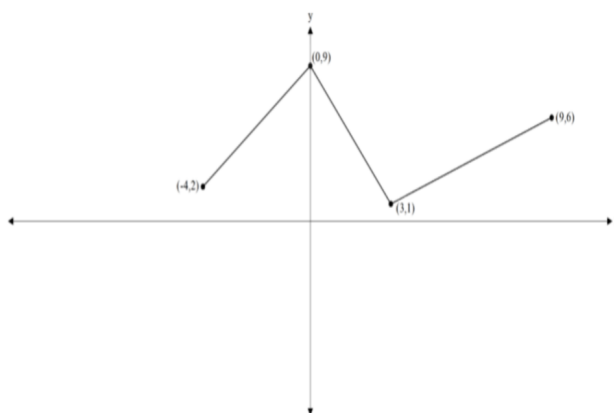


Figure 1

7. What are the values of m and n respectively?

- (a) 0, 3
- (b) -4, 1
- (c) -4, 9
- (d) 2, 6

[Ans: (b)]

8. What are the values of x and y respectively?

- (a) 9, 9
- (b) -4, 9
- (c) 1, 9
- (d) 2, 9

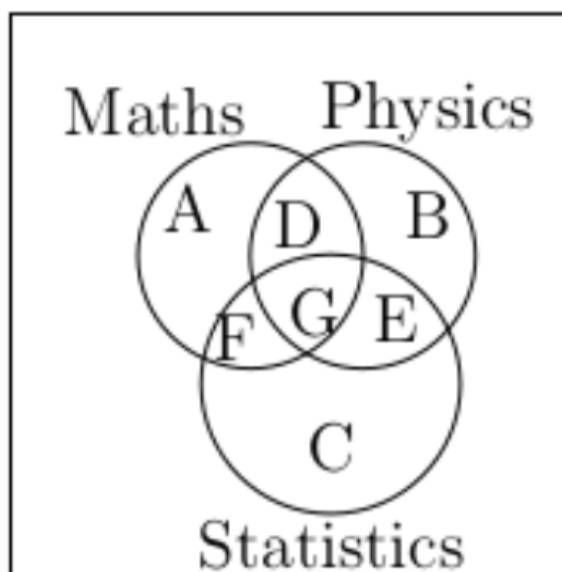
[Ans: (a)]

PA 1

Questions

1. Given below is a Venn diagram for sets of students who take Maths, Physics, and Statistics. Which of the option(s) is (are) correct?

Notation: For sets P and Q , $P \setminus Q$ indicates the set of elements in P which are not in Q .



- (a) D is the set of students who take both Maths and Statistics.
(b) $D \cup E \cup F \cup G$ is the set of all students who take at least two subjects.
(c) E is a subset of the set of the students who have not taken Maths.
(d) $Maths \setminus D$ is the set of all students who have taken only Maths.
(e) $Physics \setminus (D \cup G \cup E)$ is the set of all students who have taken only Physics.
- [Ans: (b), (c), (e)]
2. Let A be the set of natural numbers less than 6 and whose greatest common divisor with 6 is 1. Let B be the set of divisors of 6. What are the cardinalities of A , B , $A \cup B$, and $A \cap B$?

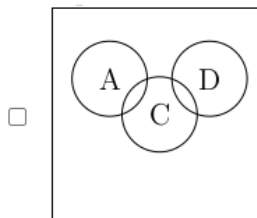
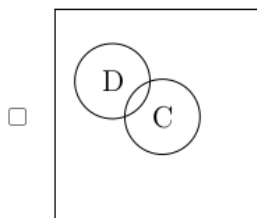
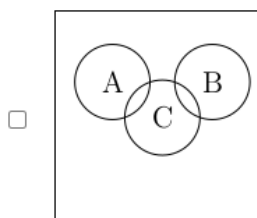
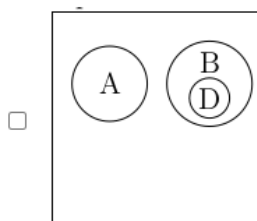
- (a) (1, 5, 6, 0)
(b) (1, 4, 5, 0)
(c) (2, 4, 5, 1)
(d) (2, 4, 6, 1)

[Ans: (c)]

Questions

3. Let A be the set of all even natural numbers (including zero), B be the set of all odd natural numbers, C be the set of all natural numbers which divide 100, and D be the set of all prime numbers less than 100. Which of the following is (are) correct representation of these sets?

Note: A region represented in a Venn diagram could be empty. Take the set of real numbers to be the universal set.



[Ans: (b),(c)]

Questions

4. Let A be the set of natural numbers which are multiples of 5 strictly less than 100, and B be the set of natural numbers which divide 100. What are the cardinalities of the following sets?

$B \setminus A$ (the set of elements in B but not in A), $A \cap B$, and B

- (a) (2, 5, 7)
- (b) (4, 5, 9)
- (c) (3, 4, 7)
- (d) (3, 5, 8)

[Ans: (b)]

5. Suppose the cardinality of set A is 2 and the cardinality of set B is 3, what are the cardinalities of the cartesian product $A \times B$ and the power set of $A \times B$?

- (a) 6 and 36
- (b) 5 and 32
- (c) 6 and 64
- (d) 5 and 25

[Ans: (c)]

6. In a survey it is found that in a particular locality 64 houses buy English newspapers, 94 houses buy Tamil newspapers, and 26 houses buy both English and Tamil newspapers. How many houses buy newspapers of only one language?

[Ans: 106]

7. Which of the following numbers is (are) irrational?

- (a) $\sqrt{2} + \sqrt{3}$
- (b) $(2 + \sqrt{3})(2 - \sqrt{3})$
- (c) $(2 + \sqrt{3}) + (2 - \sqrt{3})$
- (d) $2\sqrt{3} + 3\sqrt{2}$

[Ans: (a),(d)]

Questions

8. Which of the following is (are) true for R given below?

$R = \{(a, b) \mid \text{both } a \text{ and } b \text{ are even non-zero integers and } \frac{a}{b} \text{ is an integer} \}$

- (a) R is a reflexive relation.
- (b) R is a symmetric relation.
- (c) R is a transitive relation.
- (d) R is an equivalence relation.

[Ans: (a),(c)]

9. Find the domain and range of the following real valued function.

$f(x) = \sqrt{3 - x}$ (Note: $\sqrt{}$ denotes the positive square root)

- (a) domain = $\{x \in \mathbb{R} \mid x \neq 3\}$
range = $\{x \in \mathbb{R} \mid x \geq 3\}$
- (b) domain = $\{x \in \mathbb{R} \mid x \geq 3\}$
range = $\{x \in \mathbb{R} \mid x \geq 0\}$
- (c) domain = $\{x \in \mathbb{R} \mid x \leq 3\}$
range = $\{x \in \mathbb{R} \mid x \geq 0\}$
- (d) domain = $\{x \in \mathbb{R} \mid x \leq 3\}$
range = $\{x \in \mathbb{R} \mid x \leq 0\}$

[Ans: (c)]

10. Which of the following is (are) true for the given function?

$f : \mathbb{R} \rightarrow \mathbb{R}$

$f(x) = x^2 + 2$

- (a) f is not injective.
- (b) f is surjective.
- (c) f is not surjective.
- (d) f is bijective.

[Ans: (a),(c)]

Questions

11. Find the domain of the following real valued function.

$$f(x) = \sqrt{x+2}/(x^2-9)$$

(a) $\{x \in \mathbb{R} \mid x \geq -2, x \neq 3\}$

(b) $\{x \in \mathbb{R} \mid x \leq -2, x \geq 3\}$

(c) $\{x \in \mathbb{R} \mid x \neq -2, x \neq 3\}$

(d) $\{x \in \mathbb{R} \mid x \geq -2, x \neq 3\}$

[Ans: (a)]

END OF WEEK 1—————

AQ 2.1

Questions

1. Choose the correct option with respect to the points $P(5, -3)$, $Q(-3, 3)$, $R(0, -100)$, and $S(-2.5, 0)$ on the rectangular coordinate system.

- (a) Point R does not lie in any quadrant
- (b) Points P and R lie in Quadrant III
- (c) Points S and Q lie in Quadrant II
- (d) Points R and S cannot be represented on the rectangular coordinate system

[Ans: (a)]

2. Which of the following is/are correct with respect to the rectangular coordinate system?

- (a) The horizontal line is called Y -axis
- (b) The point of intersection of the X and Y axes is called the origin
- (c) The vertical line is called X -axis
- (d) Any point on the coordinate plane can be represented as an ordered pair (x, y)

[Ans: (b), (d)]

3. Identify the incorrect options for the representation of a point on the coordinate plane.

- (a) Quadrant I : $(+, +)$
- (b) Quadrant IV : $(-, +)$
- (c) Quadrant II : $(-, -)$
- (d) Quadrant III : $(-, +)$
- (e) On X -axis: $(0, \pm)$
- (f) On Y -axis: $(\pm, 0)$
- (g) Origin $(0, 0)$

[Ans: (b), (c), (d), (e), (f)]

AQ 2.2

Questions

1. The distance of a point $P(1, |\tan \theta|)$ from the origin is

(Hint: $|\tan^2 \theta| + 1 = |\sec^2 \theta|$)

- (a) 1
- (b) $\sec \theta$
- (c) $\tan \theta$
- (d) $-\operatorname{cosec} \theta$
- (e) 0

[Ans: (b)]

2. The distance of a point lying on the X -coordinate from the origin is

- (a) The value of the vertical distance of that point from the origin
- (b) Cannot be determined
- (c) The value of the horizontal distance of that point from the origin
- (d) 0

[Ans: (c)]

3. The distance between $P(-5, -2)$ and $Q(-2, 2)$ is

- (a) 7
- (b) 9
- (c) -6
- (d) 5

[Ans: (d)]

4. Which of the following is/are true, if D is the distance between two points (x_1, y_1) and (x_2, y_2) ?

- (a) $D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- (b) $D = \sqrt{(x_1 - x_2)^2 + (y_2 - y_1)^2}$
- (c) $D = \sqrt{x^2 + y^2}$ (if one of the points is at origin, where x, y represents any point)
- (d) $D = (x_2 - x_1) + (y_2 - y_1)$

[Ans: (a), (b), (c)]

AQ 2.3

Questions

1. The coordinates of the midpoint of points $P(4, -2)$ and $Q(-1, -1)$ are

- (a) $(0.5, -0.5)$
- (b) $(1.5, -1.5)$
- (c) $(-0.5, -0.5)$
- (d) $(-0.5, 0.5)$

[Ans: (b)]

2. The ratio in which the origin cuts the line segment $P(2, 8)$ and $Q(-3, -2)$ in $(m : n)$ form is

- (a) 0
- (b) $1 : 2$
- (c) $3 : 8$
- (d) Origin does not lie on the line segment
- (e) $2 : 1$

[Ans: (d)]

3. If the origin cuts the line segment $P(-2, -4)$, $Q(6, k)$ in the ratio $1 : 3$, then the value of k is

- (a) Cannot be determined
- (b) 0
- (c) 12
- (d) 18

[Ans: (c)]

AQ 2.4

Questions

1. Choose the correct statement based on the three points $P(0, 10)$, $Q(-20, -30)$, and $R(10, 30)$.
 - (a) The given points form a triangle of area 5 square units
 - (b) The given points form a triangle of area 15 square units
 - (c) The given points do not form a triangle
 - (d) None of the above

[Ans: (c)]

2. The area of the triangle formed by the midpoints of line segments PQ , QR , and RP where the coordinates of P , Q , and R are $(0, 0)$, $(3, 0)$, and $(3, 4)$ respectively, is -----.
 - (a) 1.5 square units
 - (b) 8.5 square units
 - (c) 2.5 square units
 - (d) 7.5 square units

[Ans: (a)]

3. Which of the following gives the area (A) of a triangle whose vertices are $P(x_1, y_1)$, $Q(x_2, y_2)$, and $R(x_3, y_3)$?
 - (a) $A = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$
 - (b) $A = \frac{1}{2} |x_2(y_3 - y_1) + x_1(y_2 - y_3) + x_3(y_1 - y_2)|$
 - (c) $A = \frac{1}{2} |x_1(y_3 - y_2) + x_3(y_1 - y_2) + x_2(y_3 - y_1)|$
 - (d) $A = \frac{1}{2} |x_3(y_1 - y_2) + x_1(y_2 - y_3) + x_2(y_3 - y_1)|$

[Ans: (a), (d)]

AQ 2.5

Questions

1. Find the slope of a line passing through the origin and the point $(-3, -2)$.

- (a) $\frac{4}{6}$
- (b) $\frac{-2}{3}$
- (c) $\frac{-1}{3}$
- (d) $\frac{3}{2}$

[Ans: (a)]

2. If the slope of a line passing through $P(1, 0)$ and $Q(-2, k)$ is 1, then the value of k is

- (a) -2
- (b) -3
- (c) 0
- (d) 2

[Ans: (b)]

3. Which of the following is/are correct with respect to the slope of a line?

- (a) The line parallel to the X -axis has slope of value 1
- (b) The slope of a line which is parallel to the Y -axis is not defined
- (c) The line parallel to the Y -axis has a slope of 90
- (d) The line parallel to the X -axis has a slope of 0
- (e) The slope of a line is always positive

[Ans: (b), (d)]

4. Through which of the following points do lines of slope 0 pass?

- (a) $(0, 0)$
- (b) $(-100, 0.5)$
- (c) $(-0.05, 0)$
- (d) $(0, 4)$

[Ans: (a), (b), (c), (d)]

AQ 2.6

Questions

1. If a line is perpendicular to the X -axis, then the slope of such a line is

- (a) 0
- (b) Not defined
- (c) 1
- (d) -1

[Ans: (b)]

2. If a line is parallel to a line which is perpendicular to the Y -axis, then the slope of the first line is

- (a) 0
- (b) Not defined
- (c) 1
- (d) -1

[Ans: (a)]

3. The angle formed by the intersection of two lines with slopes 1 and -1 is

- (a) 0 degrees
- (b) 45 degrees
- (c) 90 degrees
- (d) 180 degrees

[Ans: (c)]

Questions

4. If l_1 and l_2 are lines parallel to each other with slopes m_1 and m_2 and with inclinations α and β , respectively, then which of the following is/are correct?

- (a) $\alpha = \beta$, and $\tan \alpha = \tan \beta$
- (b) $\alpha \neq \beta$, and $\tan \alpha \neq \tan \beta$
- (c) $m_1 = m_2$, always
- (d) $m_1 \times m_2 = -1$

[Ans: (a), (c)]

5. Lines l_1 and l_2 are perpendicular to each other with slopes m_1 and m_2 and with inclinations α and β , respectively. Then, which of the following is/are correct?

- (a) $\alpha = 90^\circ + \beta$, and $\tan \alpha = -\tan \beta$
- (b) $\beta = 90^\circ + \alpha$, and $\tan \alpha = -\cot \beta$
- (c) $\alpha = 90^\circ + m_1$, and $\tan \beta = -\cot \alpha$
- (d) $m_2 = -\frac{1}{m_1}$

[Ans: (b), (c), (d)]

6. If two lines l_1 and l_2 with slopes m_1 and m_2 and inclinations α_1 and α_2 ($\alpha_2 > \alpha_1$) respectively intersect each other and make an acute angle θ and an obtuse angle ϕ , then which of the following is/are correct?

Note: $\theta = (\alpha_2 - \alpha_1)$

- (a) $\tan \theta = \frac{\tan \alpha_2 - \tan \alpha_1}{1 + \tan \alpha_1 \cdot \tan \alpha_2} \neq \frac{m_1 - m_2}{1 + m_1 \cdot m_2}$
- (b) $m_1 \times m_2 = -1$
- (c) $\tan \phi = \frac{m_1 - m_2}{1 + m_1 \cdot m_2}$
- (d) $\tan \theta = \tan(\alpha_2 - \alpha_1)$

[Ans: (a), (c), (d)]

AQ 2.7

Questions

1. Which of the following represents an equation of the horizontal line?

- (a) $y = 0$
- (b) $x = 5$
- (c) $x = -2$
- (d) $x = 0$

[Ans: (a)]

2. The equation of a line parallel to the X -axis and passing through the point $(-2, 0)$ is -----.

- (a) $y = 0$
- (b) $x = 0$
- (c) $x = -2$
- (d) $y = -2$

[Ans: (a)]

3. The equation of a line passing through the origin and with slope -1 is -----.

- (a) $y = x$
- (b) $y = -x + 1$
- (c) $y = -x$
- (d) $y = -x - 1$

[Ans: (c)]

4. The equation of a line passing through the origin and $(-1, -3)$ is -----.

- (a) $y = 3x$
- (b) $y = -3x + 3$
- (c) $y = -3x$
- (d) $y = -\frac{x}{3}$

[Ans: (a)]

Questions

5. Which of the following is/are correct with respect to a horizontal line?

- (a) It is parallel to the Y -axis
- (b) Point $(0, 0)$ can lie on a horizontal line
- (c) The equation of a horizontal line can be $x = a$
- (d) Points lying on a horizontal line are of the form (x, a)

[Ans: (b), (d)]

6. Which of the following is/are correct with respect to a vertical line?

- (a) It is parallel to the X -axis
- (b) Points lying on a vertical line are of the form (a, y)
- (c) Point $(0, 0)$ can lie on a vertical line
- (d) The equation of a line can be $y = b$

[Ans: (b), (c)]

7. Choose the correct option.

- (a) The equation of a line in point-slope form and two-point form can both be converted to $y = mx + c$ form.
- (b) $(y - y_0) = m(x - x_0)$ is a two-point form of an equation of a line.
- (c) The equation of a horizontal line can be $x = a$.
- (d) $(y - y_1) = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$ is a two-point form of an equation of a line.

[Ans: (a), (d)]

AQ 2.8

Questions

1. The equation of a line passing through $(-1, -1)$ with value of slope 1 is

- (a) $y = -x$
- (b) $y = x$
- (c) $y = -x - 1$
- (d) $y = -x + 1$

[Ans: (b)]

2. The equation of a line which cuts the X -axis at $(5, 0)$ and Y -axis at $(0, 5)$ is

- (a) $x + y = -1$
- (b) $y = -x + 5$
- (c) $-y = -x + 1$
- (d) $y = 5x + 5$

[Ans: (b)]

3. The X -intercept of a line $y = 3x + 1$ is

- (a) -1
- (b) $\frac{1}{2}$
- (c) $-\frac{1}{2}$
- (d) $-\frac{1}{3}$

[Ans: (d)]

4. Choose the correct answers:

- (a) $y = 2x - 8$ is the equation of a line in slope-intercept form where -8 is the Y -intercept.
- (b) $y = 0$ is the equation of a line passing through the origin.
- (c) $y = 2x - 8$ does not meet either the origin and the X -axis.
- (d) $y = -2(x + 1)$ is the equation of a line in intercept form.

[Ans: (a), (b)]

AQ 2.9

Questions

1. If the x -intercept and the y -intercept of a straight line are -6 and 7 respectively, then choose the correct equation of the line.

- (a) $7x - 6y + 42 = 0$
- (b) $-6x + 7y - 1 = 0$
- (c) $7x - 6y - 1 = 0$
- (d) $-6x + 7y - 2 = 0$

[Ans: (a)]

2. The slope of the line $6x - 2y + 8 = 0$ is

- (a) 2
- (b) 3
- (c) 5
- (d) 1

[Ans: (b)]

3. If the general equation of a line is $3x + 5y - 30 = 0$, then choose the set of correct options.

- (a) The intercept form of the line is $\frac{x}{10} + \frac{y}{6} = 1$
- (b) The slope intercept form of the line is $y = -3x - 30$
- (c) The intercept form of the line is $5x + 3y - 30 = 1$
- (d) The slope intercept form of the line is $y = -\frac{3}{5}x + 6$

[Ans: (a), (d)]

Questions

4. What is the slope of the line given by $4x - 2y - 12 = 0$? [Ans: 2]
5. What is the x -intercept of the line given by $4x - 2y - 12 = 0$? [Ans: 3]
6. What is the y -intercept of the line given by $4x - 2y - 12 = 0$? [Ans: -6]

AQ 2.10

Questions

1. Which of the following statements are true?

- (a) Lines $2x + 3y - 8 = 0$ and $3x - y - 2 = 0$ are parallel lines
- (b) Lines $3x + 5y - 10 = 0$ and $6x + 10y - 26 = 0$ are parallel lines
- (c) Lines $6x + 8y - 20 = 0$ and $4x - 3y = 0$ are perpendicular to each other
- (d) Lines $2x - 3y + 8 = 0$ and $3x + 2y - 18 = 0$ are not perpendicular to each other
- (e) Lines $4x + 5y + 10 = 0$ and $10x - 8y - 16 = 0$ are perpendicular to each other

[Ans: (b), (c), (e)]

2. Given lines $4x + 2ky + 5 = 0$ and $12x + 6y + 15 = 0$ are parallel, find the value of k .

[Ans: 1]

AQ 2.11

Questions

1. The equation of a line passing through the point $(3, 4)$ and perpendicular to the line $3x + 4y - 8 = 0$ is:

- (a) $8x - 6y = 0$
- (b) $2x + 8y = 38$
- (c) $8x + 4y = 5$
- (d) $x + y = 1$

[Ans: (a)]

2. Given lines $18x - 8y + 30 = 0$ and $4x + ky + 5 = 0$ are perpendicular to each other, find the value of k .

[Ans: 9]

AQ 2.12

Questions

1. If the general form of a line is $3x + 2y - 5 = 0$, then choose the correct set of options.

- (a) The slope of the given line is $-\frac{3}{2}$
- (b) The x -intercept is 3
- (c) The point where the given line cuts the X -axis is $(\frac{5}{3}, 0)$
- (d) The y -intercept is 2
- (e) The point where the given line cuts the Y -axis is $(0, \frac{5}{2})$

[Ans: (a), (c), (e)]

2. Given the point $(-2, 1)$ and the line $-3x + 4y - 7 = 0$, choose the correct set of options.

- (a) The distance between $(-2, 1)$ and the line $-3x + 4y - 7 = 0$ is 2
- (b) The slope of the given line is $\frac{3}{4}$
- (c) The distance between $(-2, 1)$ and the line $-3x + 4y - 7 = 0$ is $\frac{3}{5}$
- (d) The y -intercept of the given line is $\frac{7}{4}$

[Ans: (c), (d)]

3. Find the distance between the two parallel lines $4x + 3y + 5 = 0$ and $8x + 6y - 30 = 0$.

[Ans: 4]

4. If the distance between the point $(-1, 3)$ and the line $4x + c = 0$ is $\frac{3}{2}$, then find the value of c where c is a positive number.

[Ans: 10]

5. The coordinates of the vertices of a triangle are $(0, 3)$, $(0, 0)$, $(2, 0)$. Find the area of the triangle.

[Ans: 3]

AQ 2.13

Questions

1. If a line fit $y = x + 1$ is given for the data as shown in Table AQ-3.1, then compute the Sum of Squares Error (SSE).

x	1	2	3	4	6
y	1	1	2	5	7

Table AQ-3.1

[Ans: 9]

2. If the relation between x and y is as shown in Table AQ-3.2, then which among the following lines is the best fit?

x	0	1	2	3	4	5	6
y	1	3	4	2	5	0	8

Table AQ-3.2

- (a) $y = 2x$ with SSE = 143
- (b) $y = x$ with SSE = 40
- (c) $y = 3x$ with SSE = 428
- (d) $y = 4x$ with SSE = 895

[Ans: (b)]

PA 2

Questions

1. If R is the set of all points which are 5 units away from the origin and are on the axes then R is:

- (a) $R = \{(5, 5), (-5, 5), (-5, -5), (5, -5)\}$
- (b) $R = \{(5, 0), (5, 5), (5, -5), (-5, 0)\}$
- (c) $R = \{(5, 0), (0, 5), (5, 5), (0, -5)\}$
- (d) $R = \{(5, 0), (0, 5), (-5, 0), (0, -5)\}$
- (e) $R = \{(5, 0), (0, 5), (-5, 0), (-5, 5)\}$
- (f) There is no such set.

[Ans: (d)]

2. A point P divides the line segment MN such that $MP : PN = 2 : 1$. The coordinates of M and N are $(-2, 2)$ and $(1, -1)$ respectively. What will be the slope of the line passing through P and the point $(1, 1)$?

- (a) $\frac{4}{3}$
- (b) 1
- (c) Inadequate information.
- (d) $-\frac{4}{3}$
- (e) $\tan\left(\frac{4}{3}\right)$
- (f) None of the above.

[Ans: (b)]

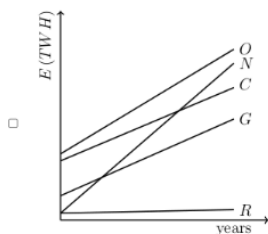
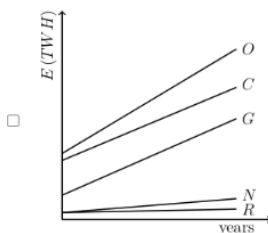
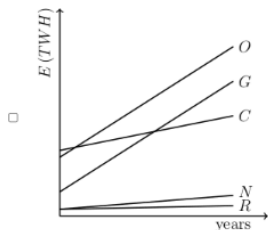
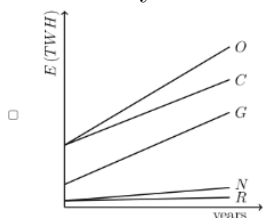
Questions

3. Table P-2.0 shows the different types of energies consumed (approximate values) in years 1965 and 2015 across the world.

Energy type	Approximate energy used (TWH)	
	1965	2015
Oil (O)	19000	49000
Coal (C)	17000	38000
Gas (G)	7000	29000
Nuclear (N)	2000	6000
Renewable (R)	2000	3000

Table P-2.0

3. A student assumes a linear relationship between energy consumed (E) and the number of years after 1965. Choose the option which best represents the linear relationships assumed by the student (from 1965 to 2015).



[Ans: (c)]

Questions

4. The student estimated the energy consumption in 2025 and created Table P-2.0. Choose the correct option.

Energy type	Approximate energy used (TWH)		
	1965	2015	2025
Oil (O)	19000	49000	o
Coal (C)	17000	38000	c
Gas (G)	7000	29000	g
Nuclear (N)	2000	6000	n
Renewable (R)	2000	3000	r

Table P-2.0

- (a) $o = 64000$
- (b) $c = 48500$
- (c) $g = 38500$
- (d) $n = 8000$
- (e) $r = 3500$
- (f) None of the above.

[Ans: None of the above.]

Questions

5. The elements of a relation R are shown as points in the graph in Figure P.2.1. Choose the set of correct options:

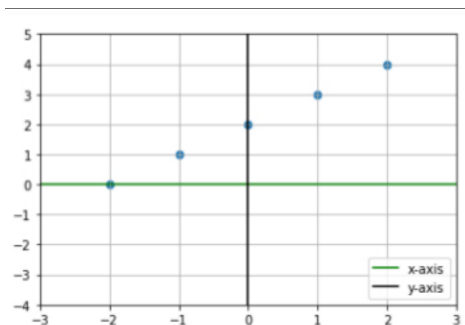


Figure P-2.1

- (a) R can be represented as $R = \{(-2, 0), (-1, 1), (0, 2), (1, 3), (2, 4)\}$
- (b) We can write R as $R = \{(a, b) | (a, b) \in X \times Y, b = a + 2\}$, where X is the set of all values on the x -axis, and Y is the set of all values on the y -axis
- (c) R cannot be a function because it is a finite set
- (d) R is a symmetric relation
- (e) R is a function because it has only one output for one input
- (f) If R is a function then it is a bijective function on $X \times Y$, where X is the set of all values on the x -axis, and Y is the set of all values on the y -axis.
- (g) We can write R as $R = \{(a, b) | (a, b) \in X \times Y, b = a + 2\}$, where $X = \{-2, -1, 0, 1, 2\}$ and $Y = \{0, 1, 2, 3, 4\}$

[Ans: (a), (e), (g)]

6. Find the values of a for which the triangle $\triangle ABC$ is an isosceles triangle, where A, B , and C have the coordinates $(-1, 1)$, $(1, 3)$, and $(3, a)$ respectively.

- (a) If $AB = BC$, then $a = 1$
- (b) If $AB = BC$, then $a = -1$ or -5
- (c) If $BC = CA$, then $a = -1$
- (d) If $BC = CA$, then $a = 1$

[Ans: (a), (c)]

Questions

7. A plane begins to land when it is at a height of 1500 metre above the ground. It follows a straight line path and lands at a point which is at a horizontal distance of 2700 metre away. There are two towers which are at horizontal distances of 900 metre and 1800 metre away in the same direction as the landing point. Choose the correct option(s) regarding the plane's trajectory and safe landing.
- The trajectory of the path could be $\frac{y}{27} + \frac{x}{15} = 100$ if $X - axis$ and $Y - axis$ are horizontal and vertical respectively
 - The maximum safe height of the towers are 1000 metre and 1500 metre respectively.
 - The trajectory of the path could be $\frac{y}{15} + \frac{x}{27} = 100$ if $X - axis$ and $Y - axis$ are horizontal and vertical respectively.
 - The maximum safe height of the towers are 1500 metre and 500 metre respectively
 - The maximum safe height of the towers are 1000 metre and 500 metre respectively
 - None of the above.

[Ans: (c), (e)]

8. Find the area of triangle ADE.

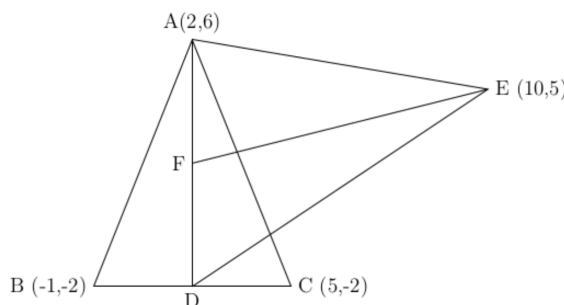


Figure P-2.2

[Ans: 32]

9. Let the slope of a line FG be 2 and the coordinate of the point G be $(a, 9)$. Then, what is the value of a ? [Ans: 5.5]
10. Leo rents a motorcycle for 2 days. The rental company provides the motorcycle at Rs. 500 per day with 100 km free per day. The additional charges after 100 km are Rs. 2 per km. Leo drives the motorcycle for a total of 500 km. How much (Rs.) will he have to pay to the rental company? [Ans: 1600]

Questions

11. A vehicle is travelling on a straight line path and passes through the points $A(-4, 2)$, $B(-1, 3)$, and $C(2, \mu)$. The value of μ is:

(a) 2
 (b) 4
 (c) -2
 (d) 10

[Ans: (b)]

12. Suppose two boats are starting their journey from the ferry ghat A (considered as the origin), one towards ferry ghat B along the straight line $y = -2x$ and the other towards the ferry ghat C along a straight line perpendicular to the path followed by B. The river is 1 km wide uniformly and parallel to the X-axis. Suppose Rahul wants to go to the exact opposite point of A along the river. (For questions 12,13)

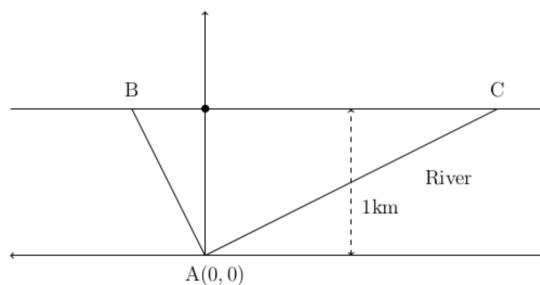


Figure P-3.1

12. How much total distance does Rahul have to travel to reach his destination if he takes the boat towards ferry ghat B?

(a) $\sqrt{5}$
 (b) $\sqrt{5} + 2$
 (c) $\frac{\sqrt{5}}{2}$
 (d) $\frac{\sqrt{5}+1}{2}$

[Ans: (d)]

Questions

13. How much total distance does Rahul have to travel to reach his destination if he takes the boat towards ferry ghat C?

- (a) $\sqrt{5}$
- (b) $\sqrt{5} + 2$
- (c) $\frac{\sqrt{5}}{2}$
- (d) $\frac{\sqrt{5}+1}{2}$

[Ans: (b)]

14. Suppose a bird is flying along the straight line $4x - 5y = 20$ on the plane formed by the path of the flying bird and the line of eye point view of a person who shoots an arrow which passes through the origin and the point $(10, 8)$. What is the point on the co-ordinate plane where the arrow hits the bird?

- (a) $(20, 12)$
- (b) $(25, 16)$
- (c) The arrow will miss the bird.
- (d) Inadequate information.

[Ans: The arrow will miss the bird.]

Questions

15. We plot the displacement (S) versus time (t) for different velocities as it follows the equation $S = vt$, where v is the velocity. Identify the best possible straight lines in the Figure P-3.2 for the given set of velocities.

Table P-3.1

v_1	v_2	v_3	v_4
1	-2	0.5	-1

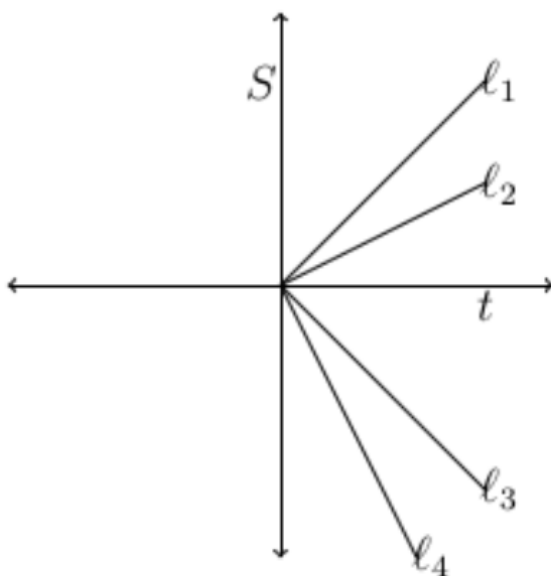


Figure P-3.2

- (a) $v_1 \rightarrow l_1, v_2 \rightarrow l_2, v_3 \rightarrow l_3, \text{ and } v_4 \rightarrow l_4$.
 (b) $v_1 \rightarrow l_1, v_2 \rightarrow l_4, v_3 \rightarrow l_3, \text{ and } v_4 \rightarrow l_2$.
 (c) $v_1 \rightarrow l_1, v_2 \rightarrow l_4, v_3 \rightarrow l_2, \text{ and } v_4 \rightarrow l_3$.
 (d) $v_1 \rightarrow l_2, v_2 \rightarrow l_4, v_3 \rightarrow l_1, \text{ and } v_4 \rightarrow l_3$.

[Ans: (c)]

Questions

16. A constructor is asked to construct a road which is at a distance of $\sqrt{2}$ km from the municipality office and perpendicular to a road which can be defined by the equation of the straight line $x - y = 8$ (considering the municipality office to be the origin). Find out the possible equations of the straight lines to represent the new road to be constructed.

- (a) $x - y - 2 = 0$
- (b) $x + y + 2 = 0$
- (c) $x - y + 2 = 0$
- (d) $x + y - 2 = 0$

[Ans: (b),(d)]

17. Suppose there are two roads perpendicular to each other which are both at the same distance from Priya's house (considered as the origin). The meeting point of the two roads is on the x -axis and at a distance of 5 units from Priya's house. Choose the correct possible equations representing the roads.

- (a) Inadequate information.
- (b) $y = \frac{1}{2}x + 5, y = -2x - 5$
- (c) $y = -x - 5, y = x + 5$
- (d) $y = 2x - 10, y = -2x - 10$
- (e) $y = 2x - 5, y = -\frac{1}{2}x - 5$
- (f) $y = -x + 5, y = x - 5$
- (g) $x = 5, x = -5$

[Ans: (c), (f)]

Questions

18. Which of the following option(s) is(are) true about the triangles $\triangle ABC$ and $\triangle DEF$ given in Figure P-3.3?

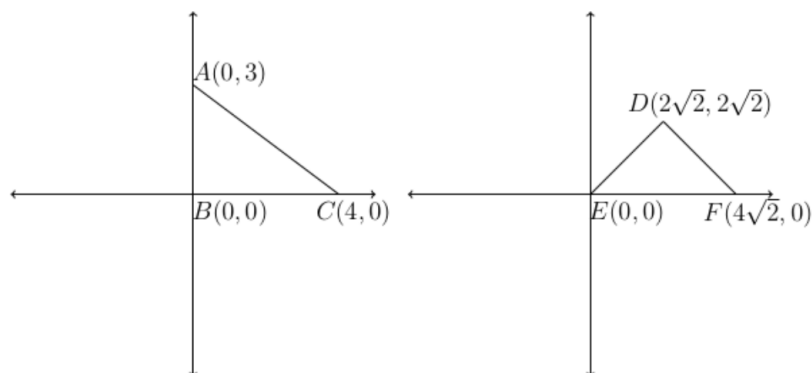


Figure P-3.3

- (a) Only $\triangle ABC$ is a right angled triangle while $\triangle DEF$ is not.
 - (b) Both $\triangle ABC$ and $\triangle DEF$ are right angled triangles.
 - (c) The area of $\triangle ABC$ is greater than the area of $\triangle DEF$.
 - (d) Both the triangles have the same area
 - (e) The area of $\triangle DEF$ is 8 sq.unit
- [Ans: (b), (e)]
19. Let the diagonals of a quadrilateral with one vertex at $(0, 0)$ bisect each other perpendicularly at the point $(1, 2)$. Further, let one of the diagonals be on the straight line $y = 2x$. Then, which of the following is (are) correct statements?
- (a) The diagonally opposite vertex of $(0, 0)$ is $(2, 4)$.
 - (b) The other diagonal is on the straight line $y = -\frac{1}{3}x$
 - (c) The other diagonal is on the straight line $y = -\frac{1}{3}x + \frac{5}{2}$
 - (d) The diagonally opposite vertex of $(0, 0)$ is $(\frac{3}{2}, 3)$
- [Ans: (a), (c)]

Questions

20. A woman is reported missing in a locality. The police department finds a human femur bone during their investigation. They estimate the height H of a female adult (in cm) using the relationship $H = 1.8f + 70$, where f is the length (in cm) of the femur bone. The length of the femur found is 35 cm, and the missing woman is known to be 130 cm tall. In the particular locality, maximum height of a female is 195 cm and the minimum length of a female femur bone is 15 cm. Based on the given data answer the following questions. (For question 20, 21)

20. Choose the set of correct options.

- (a) If an error of 1 cm is allowed, bone could belong to missing female.
- (b) If an error of 3 cm is allowed, bone could belong to missing female
- (c) If the height as a function of femur length is known to be accurate, the range of the function is $[70, 195]$.
- (d) If the height as a function of femur length is known to be accurate, the range of the function is $[97, 195]$
- (e) If the height as a function of femur length is known to be accurate, the domain of the function is $\left[15, \frac{625}{9}\right]$.

[Ans: (b), (d), (e)]

21. A new detective agency came up with a relationship $H = mf + 70$, where H is the height of a male adult (in cm) and f is the length (in cm) of the femur bone. They have used the following sample set given below in the Table P-3.2, such that the sum squared error is minimum.

height(H) (in cm)	150	160	170	180
length of femur bone(f) (in cm)	40	42	48	56

Table P-3.2

- (a) $m = 1$
- (b) $m = 1.5$
- (c) $m = 2$
- (d) $m = 2.5$

[Ans: (c)]

22. What will be the slopes of the straight lines perpendicular to the following lines?
 (a) $2x + 5y - 9 = 0$; (b) $-5x + 25y + 28 = 0$ [Ans: (a)-2.5, (b)-5]

END OF WEEK 2—————

AQ 3.1

Questions

1. Which of the graphs in Figure 1 represents the following function: $y = x^2 - x + 1$?

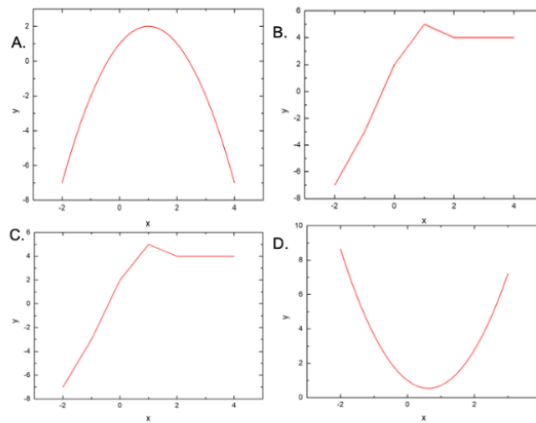


Figure 1: Options for Question 1

[Ans: D]

2. Which of the graphs in Figure 1 represents the following function: $y = -x^2 + 2x + 1$?

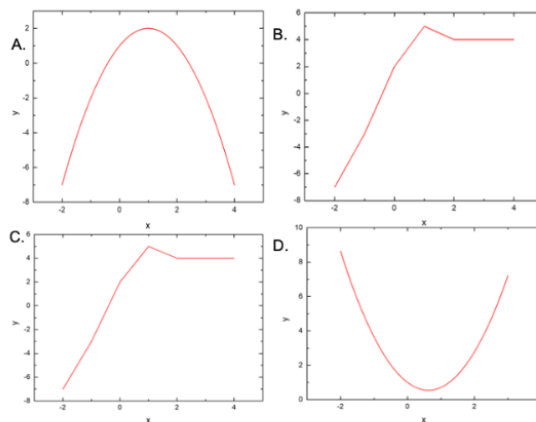


Figure 1: Options for Question 1

[Ans: A]

Questions

3. For the function $y = x^2 + 10x + 1$, determine whether y has a minimum or maximum value and identify it from the options given below:
- (a) Maximum value -24
 - (b) Minimum value -24
 - (c) Maximum value 76
 - (d) Neither maximum nor minimum value exists

[Ans: (b)]

AQ 3.2

Questions

1. The curve on the surface of the banana as shown in Figure 2 can be described using the equation $y = x^2 + 2x + 4$. An ant (shown in blue color in Figure 2) is walking from one end of the banana to the other end. What will be the x -coordinate of the ant's location once it reaches the vertex of its path?

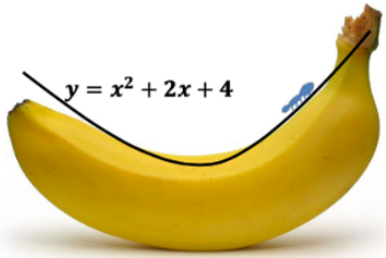


Image source: <https://www.independent.co.uk/food-and-drink/features/>

Figure 2: Path of an ant over the top surface of a banana for Question 1

- (a) 1
- (b) 0
- (c) -1
- (d) 0.5

[Ans: (c)]

2. Deb and Ananya bought 20 toys together. From these 20 toys, Deb lost 3 toys and Ananya lost 4 toys. The product of the current number of their toys is 42. Can you form an equation for Deb to know how many toys he had initially? [Let us assume Deb initially had x number of toys.]

- (a) $-x^3 + 13x - 4 = 42$
- (b) $19x - 4 = 42$
- (c) $-x^2 + 19x - 48 = 42$
- (d) $x^2 = 42$

[Ans: (c)]

Questions

3. Represent the following problem in the form of an equation: A medicine manufacturer produces y number of vaccines every day. The manufacturing cost of each vaccine is ₹100 plus the number of vaccines manufactured on that day. On a particular day, the total manufacturing cost was ₹10,000. How many vaccines were manufactured on that day?

- (a) $-y^2 + 13y - 4 = 42$
- (b) $19y - 4 = 10000$
- (c) $y^2 + 100y = 10000$
- (d) $y^2 = 100$

[Ans: (c)]

4. Find the vertex and axis of symmetry of the graph of the quadratic function: $f(x) = x^2 + 4x + 5$.

- (a) Vertex: $(1, -2)$, Axis of symmetry: $x = 1$
- (b) Vertex: $(-1, 1)$, Axis of symmetry: $x = 2$
- (c) Vertex: $(-2, 1)$, Axis of symmetry: $x = -2$
- (d) Vertex: $(0, 0)$, Axis of symmetry: $x = 3$

[Ans: (c)]

5. Ananya runs a shop selling books. The profit she makes from her shop is given by the function:

$$P(u) = 100 + 40u - 2u^2,$$

where u is the amount that she spends on bookbinding. Find the value of u in order to maximize the profit $P(u)$.

- (a) 10
- (b) 11
- (c) 20
- (d) 15

[Ans: (a)]

Questions

6. Find the maximum profit obtained by Ananya's shop.

- (a) ₹300
- (b) ₹400
- (c) ₹310
- (d) ₹315

[Ans: (a)]

AQ 3.3

Questions

1. Calculate the slope of the parabola at a point (x, y) obtained by plotting the following function: $y = x^2 + 2x + 4$.

- (a) $2x + 2$
- (b) $+2x$
- (c) $-1.5x$
- (d) 0.5

[Ans: (a)]

2. Calculate the slope of the parabola at a point (x, y) obtained by plotting the following function: $y = -5x^2 + 10x + 10$.

- (a) $-10x + 10$
- (b) $-10x$
- (c) $10x$
- (d) 9.8

[Ans: (a)]

3. Calculate the slope of the parabola given by the following function: $y = 6x^2 + 10x + 15$ at the point $(2, y)$.

- (a) 34
- (b) $-10x + 10$
- (c) $10x$
- (d) 21

[Ans: (a)]

AQ 3.4

Questions

1. A stone is thrown with an initial speed u (m/s) as shown in **Figure 3**. The height of the stone's trajectory above the ground is given by:

$$H(t) = -5t^2 + \frac{1}{2}ut,$$

where t is the time of flight. If the highest point in air that the stone can reach is 5 m above the ground, then calculate the initial speed u .

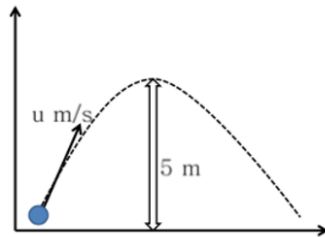


Figure 3: Diagram for Lecture-4.4 Qn. 1: Stone thrown in air at maximum height 5 metre

- (a) 20 m/s
- (b) 10 m/s
- (c) 12 m/s
- (d) 2 m/s

[Ans: (a)]

2. Find the roots of the equation: $x^2 - 8x + 15 = 0$.

- (a) Two real roots 3, 5
- (b) Two real roots 4, 15
- (c) One real root -1
- (d) One real root 3

[Ans: (a)]

3. Find the roots of the equation: $x^2 - 2x + 1 = 0$.

- (a) One repeated, real root 1
- (b) Two real roots 6, 7
- (c) One real root -5
- (d) One real root 3

[Ans: (a)]

AQ 3.5

Questions

1. The slope of a line which passes through the vertex and the y -intercept of the quadratic equation $x^2 + 10x - 5$ is:

- (a) $\sqrt{1230}$
- (b) -5
- (c) 5
- (d) $\sqrt{650}$

[Ans: (c)]

2. Identify the point at which the slope of the equation $x^2 + 2x - 5$ is 10.

- (a) $(-4, -3)$
- (b) $(-3, -4)$
- (c) $(4, 19)$
- (d) $(4, 3)$

[Ans: (c)]

3. Choose the correct statement with respect to the sensitivity of the following equations:

- (a) The equation $10x^2$ is the most sensitive.
- (b) The equation $0.01x^2$ is the least sensitive.
- (c) The equation x^2 is the most sensitive.
- (d) The equation $5x^2$ is the least sensitive.

[Ans: (a), (b)]

Questions

4. A football player kicks a football from the ground. The path of the football can be modeled by the function

$$h(x) = -\frac{1}{25}x^2 + 20,$$

where x is the horizontal distance; $h(x)$ is the height of the football above the ground, when both are measured in feet. Another player jumps and takes a header when the slope of the function $h(x)$ is -0.64 . Choose the correct set of options.

- (a) The maximum height reached by the player's head to take the header is 7.2 feet.
- (b) The maximum height achieved by the ball is 20 feet.
- (c) The maximum height reached by the player's head to take the header is 9.2 feet.
- (d) None of the above

[Ans: (a), (b)]

AQ 3.7

Questions

1. Choose the correct standard form of a quadratic equation with roots $\frac{7}{2}$ and $-\frac{10}{3}$.

- (a) $6x^2 - x - 70 = 0$
- (b) $\frac{1}{3}(6x^2 - x - 70) = 0$
- (c) $\frac{1}{6}(6x^2 - x - 7) = 0$
- (d) $(x - \frac{7}{2})(x + \frac{10}{3}) = 0$

[Ans: (a)]

2. Choose the correct option about a with the help of Figure AQ-5.1:

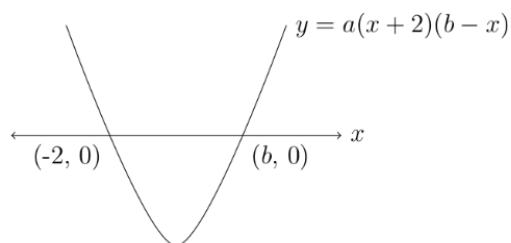


Figure AQ-5.1

- (a) $a > 0$
- (b) $a = 5$
- (c) $a = 0$
- (d) $a < 0$

[Ans: (d)]

Questions

3. Choose the set of correct options regarding quadratic functions $p(x) = x^2 - 7x + 10$ and $q(x) = x^2 + 7x + 10$.

- (a) $p(x)$ has two roots -5 and -2
- (b) $p(x)$ has two roots 5 and 2
- (c) $q(x)$ has two roots 5 and 2
- (d) $q(x)$ has two roots -5 and -2

[Ans: (b), (d)]

4. Identify the incorrect options for the quadratic equation $x^2 - 15x + 54 = 0$.

- (a) 9 is a root
- (b) Both the roots are equal
- (c) There are no roots
- (d) 6 is a root

[Ans: (b), (c)]

AQ 3.8

Questions

1. What should be added in $p(x)$ to make it a perfect square, if $p(x) = x^2 - 12x + 34$?

- (a) 1
- (b) 2
- (c) 3
- (d) 4

[Ans: (b)]

2. Choose the correct option regarding the equation $x^2 - 12x + 37 = 0$.

- (a) Can be solved using the prime factorization method
- (b) 1 is a root
- (c) Cannot be solved by completing the square method
- (d) 0 is a root

[Ans: (c)]

3. Which of the following is/are true, if $x^2 - 8x + 13 = 0$ is solved by completing the square method?

- (a) 3 should be added on both sides of the equation
- (b) $4 + \sqrt{3}$ is one of the roots
- (c) $4 - \sqrt{3}$ is one of the roots
- (d) Equal roots

[Ans: (a), (b), (c)]

AQ 3.9

Questions

1. The quadratic equation $9x^2 + 6x + 1 = 0$ has:

- (a) Two distinct roots
- (b) Equal roots
- (c) No roots
- (d) Inadequate information

[Ans: (b)]

2. If two different quadratic equations have the same discriminant, then choose the correct option.

- (a) The roots of both equations will always be the same
- (b) The roots of both equations will always be real
- (c) The roots of both equations will always be distinct
- (d) Discriminant cannot determine the relation between the roots of two different equations

[Ans: (d)]

3. What will the roots of the equation $2x^2 - x - 10 = 0$ be?

- (a) Cannot be determined
- (b) 2 and $\frac{5}{2}$
- (c) -2 and $\frac{5}{2}$
- (d) 2 and $-\frac{5}{2}$

[Ans: (c)]

4. Which of the following gives distinct real roots?

- (a) $69x^2 - 42x - 56 = 0$
- (b) $69x^2 - 10x + 560 = 0$
- (c) $9x^2 - 10x + 1 = 0$
- (d) $9x^2 - 10x - 1 = 0$

[Ans: (a), (c), (d)]

PA 3

Questions

1. What will be the equation of the tangent to the curve $f(x) = 2x^2 + 9x + 20$ at point $(-3, 11)$?

- (a) $y = 3x$
- (b) $y = -3x + 2$
- (c) $y = -3x + 20$
- (d) $y = -\frac{x}{3} + 2$
- (e) $y = \frac{x}{3} + 20$
- (f) $y = -\frac{x}{3}$

[Ans: (b)]

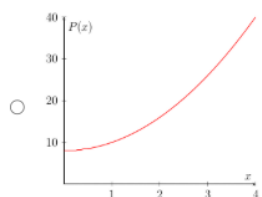
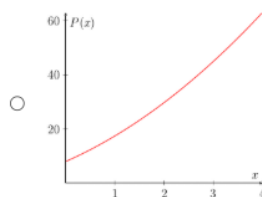
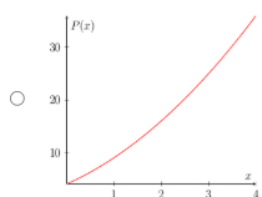
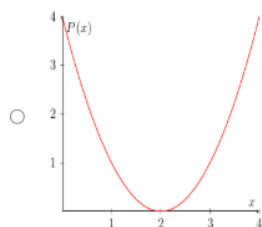
2. Find the length of the line segment on the straight line $y = 2$ bounded by the curve $y = 4x^2$.

- (a) $\frac{1}{\sqrt{2}}$
- (b) $\sqrt{2}$
- (c) $1 + \sqrt{2}$
- (d) $1 + \frac{1}{\sqrt{2}}$

[Ans: (b)]

Questions

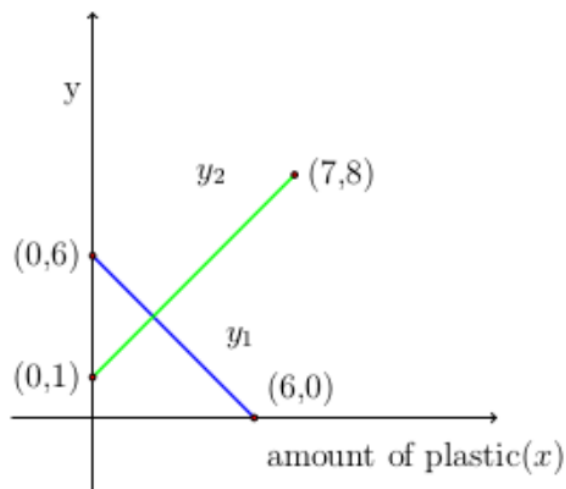
3. Mr. Mehta has two sons. Both sons send money to their father each month separately as $M_1(x) = (x - 2)^2$ and $M_2(x) = (x + 2)^2$ respectively. If x denotes the month, then choose the curve which best represents the total amount ($P(x)$) received by Mr. Mehta every month.



[Ans: (d)]

Questions

4. A civil engineer found that the durability d of the road she is laying depends on two functions y_1 and y_2 as follows: $d = ay_1y_2$ where $a > 0$. Functions y_1 and y_2 depend on the amount of plastic (x) mixed in bitumen, and their variations are shown in the graph given below. Find the values of functions y_1 and y_2 such that the durability of the road is maximum.



- (a) $\frac{5}{2}$ and $-\frac{7}{2}$
- (b) $\frac{7}{2}$ and $\frac{7}{2}$
- (c) $\frac{7}{2}$ and $-\frac{5}{2}$
- (d) $\frac{17}{2}$ and $\frac{5}{2}$
- (e) $\frac{17}{2}$ and $\frac{7}{2}$
- (f) Inadequate information about a .

[Ans: (b)]

Questions

5. Let A be the set of all points on the curve defined by the function $f_1(x) = x^2 - x - 42$ and let B be the set of all points on the curve f_2 defined by the reflection of the curve f_1 with respect to X -axis. If C is the set of all points on the axes then choose the correct option regarding the cardinality of set D where $D = (A \cap B) \cup (A \cap C) \cup (B \cap C)$.
- (a) infinite
 - (b) 8
 - (c) 4
 - (d) 6
 - (e) 2
 - (f) zero

[Ans: (c)]

6. Let $f_1(x) = x^2 - 25$. Let A be the set of all points inside the region by the curves representing $f_1(x)$ and its reflection $f_2(x)$ with respect to X -axis (excluding the points on curve). Choose the correct option.
- (a) The cardinality of A is 2.
 - (b) The cardinality of A is 4.
 - (c) Y -coordinates of the points in set A belong to the interval $(-25, 25)$
 - (d) Y -coordinates of the points in set A belong to the interval $[-25, 25]$.
 - (e) X -coordinates of the points in set A belong to the interval $[-5, 5]$.
 - (f) X -coordinates of the points in set A will be all real numbers because f_1 is a quadratic function

[Ans: (c)]

7. Choose the correct set of options regarding the function $f(x) = x^2 + 6x + 8$.
- (a) $y = -3$ is the axis of symmetry.
 - (b) -2 and -4 are the zeroes of the above function
 - (c) The maximum value of the above function is -1
 - (d) Slope of the function at $(-3, -1)$ is zero
 - (e) $2x + 6$ is the slope of this curve at any given x .
 - (f) The function is symmetric around $x = 3$

[Ans: (b), (d), (e)]

Questions

8. A quadratic function f is such that its value decreases over the interval $(-\infty, -2)$ and increases over the interval $(-2, \infty)$, and $f(0) = f(-4) = 23$. Then, f can be:

- (a) $-3x^2 - 12x + 23$
- (b) $3x^2 + 12x + 23$
- (c) $5(x - 2)^2 + 3$
- (d) $5(x + 2)^2 + 3$
- (e) $ax^2 + 4ax + 23, a > 0$
- (f) $ax^2 + 4ax + 23, a < 0$

[Ans: (b), (d), (e)]

9. Suppose one root of a quadratic equation of the form $ax^2 + bx + c = 0$, with $a, b, c \in \mathbb{R}$, is $2 + \sqrt{3}$. Then choose the correct set of options.

- (a) There can be infinitely many such quadratic equations.
- (b) There is no such quadratic equation
- (c) There is a unique quadratic equation satisfying the properties.
- (d) $x^2 - 4x + 1 = 0$ is one such quadratic equation
- (e) $x^2 - 2x - 3 = 0$ is one such quadratic equation

[Ans: (a), (d)]

10. A company's profits are known to be dependent on the months of a year. The profit pattern (in lakhs of Rupees) from January to December is $P(x) = -2x^2 + 25x$. Here, x represents the month number, starting from 1 (for January) and ending at 12 (for December). On this basis, choose the correct option.

- (a) The maximum profit in a month is Rs.78 lakhs
- (b) The maximum profit in a month is Rs.78.125 lakhs
- (c) The maximum profit in a month is Rs.77 lakhs.
- (d) The maximum profit is recorded in June
- (e) The profit in December is 144 lakhs
- (f) None of the above

[Ans: (a),(d)]

Questions

11. Raghav sells 2000 packets of bread for Rs. 20000 each day, and makes a profit of Rs. 4,000 per day. He finds that if the cost price increases by Rs. x per packet, he can increase the selling price by Rs. $2x$ per packet. However, when this price increase happens, he loses $200x$ of his customers. Choose the correct options.
- (a) For the maximum profit per day, cost price is Rs. 12 per packet
 - (b) For the maximum profit per day, cost price is Rs. 4 per packet.
 - (c) For the maximum profit per day, the sale price increases by Rs. 4 per packet
 - (d) For the maximum profit per day, Raghav will lose 400 customers
 - (e) The maximum difference in profit per day could be Rs. 3200
 - (f) The maximum difference in profit per day could be Rs. 7200

[Ans: (a), (e)]

12. A farmer has a wire of length 576 metres. He uses it to fence his rectangular field to protect it from animals. If he fences his field with four rounds of wire, and the field has the maximum area possible to accommodate such a fencing, what is the area (in square metres) of the field?

[Ans: 1296]

- 13-14. Consider the quadratic function $f(x) = x^2 - 2x - 8$. Two points P and Q are chosen on this curve such that they are 2 units away from the axis of symmetry. R is the point of intersection of axis of symmetry and the X -axis. And S is the vertex of the curve. Based on this information, answer the following:

13) What is the height of $\triangle PQR$ taking PQ as the base? [Ans: 5]

14) What is the height of $\triangle PQS$ taking PQ as the base? [Ans: 4]

15. What will be the value of parameter k , if the discriminant of equation $4x^2 + 9x + 10k = 0$ is 1?

(a) $\frac{82}{80}$

(b) $\frac{41}{80}$

(c) $\frac{1}{2}$

(d) $\frac{41}{160}$

(e) 1

(f) None of the above

[Ans: (c)]

Questions

16. A boat has a speed of 30 km/hr in still water. In flowing water, it covers a distance of 50 km in the direction of flow and comes back in the opposite direction. If it covers this total of 100 km in 10 hours, then what is the speed of flow of the water (in km/hr)?

- (a) $5 - 5\sqrt{37}$
- (b) $-10\sqrt{6}$
- (c) $10\sqrt{6}$
- (d) $20\sqrt{3}$
- (e) $-20\sqrt{3}$
- (f) 2

[Ans: (c)]

17. A stunt man performs a bike stunt between two houses of the same height as shown in Figure P-5.1. His bike (lowest part of the bike) makes an angle of θ at house A with the horizontal at the beginning of the stunt, follows a parabolic path and lands at house B with an angle of $(180 - \theta)$ with the horizontal.

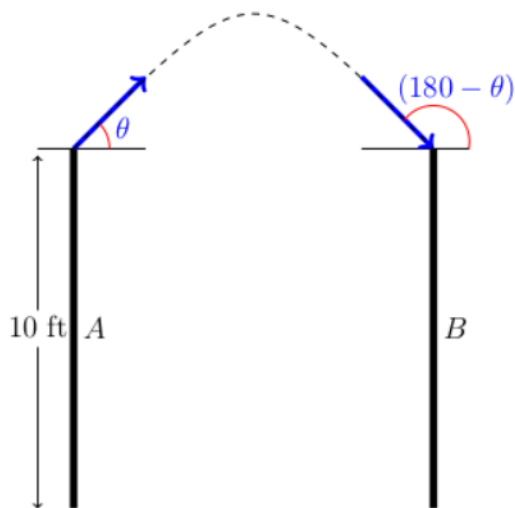


Figure P-5.1:

Questions

17. If the maximum height achieved by the bike is 12.5 ft from the ground and $\tan \theta = 1$, then find the distance between the two houses.
- (a) 1 ft
 - (b) 2.5 ft
 - (c) 5 ft
 - (d) 10 ft
 - (e) 15 ft
 - (f) 20 ft

[Ans: (d)]

18. Given that $f_1(x) = -x^2 - 6x$ and $f_2(x) = x^2 + 6x + 10$. Let $f(x)$ be a function such that the domain of $f(x)$ is $[\alpha, \beta]$, where $f_1(\alpha) = f_2(\alpha)$ and $f_1(\beta) = f_2(\beta)$, then choose the set of correct options.
- (a) Range of $f(x)$ is $[-1, 3]$
 - (b) Range of $f(x)$ is $[0, 5]$
 - (c) Domain of $f(x)$ is $[-5, 5]$
 - (d) Domain of $f(x)$ is $[-5, -1]$
 - (e) Inadequate information provided for finding range
 - (f) Inadequate information provided for finding domain

[Ans: (d), (e)]

19. If $f(x) = 2x^2 + (5 + k)x + 7$, $g(x) = 5x^2 + (3 + k)x + 1$, $h_1(x) = f(x) - g(x)$, and $h_2(x) = g(x) - f(x)$, then choose the set of correct options.
- (a) Roots for $h_1(x) = 0$ and roots for $h_2(x) = 0$ are real, distinct, and the roots are the same for $h_1(x) = 0$ and $h_2(x) = 0$
 - (b) Roots for $h_1(x) = 0$ and roots for $h_2(x) = 0$ are real and distinct but the roots are not the same for $h_1(x) = 0$ and $h_2(x) = 0$
 - (c) Sum of roots of quadratic equation $h_1(x) = 0$ will be $\frac{2}{3}$
 - (d) Product of roots of quadratic equation $h_2(x) = 0$ will be -2
 - (e) Axis of symmetry for both the functions $h_1(x)$ and $h_2(x)$ will be the same
 - (f) Vertex for both the functions $h_1(x)$ and $h_2(x)$ will be the same

[Ans: (a), (c), (d), (e)]

Questions

- 20-21. Vaishali wants to set up a small plate making machine in her village. Table P-5.1 shows the different costs involved in making the plates. Figure P-5.2 shows her survey regarding the demand (number of packets of plate) versus selling price of plate per packet (in ₹) per day.

Cost type	Cost
Electricity	₹1.5 per packet
Miscellaneous	₹6.5 per packet
Raw material	₹10 per packet

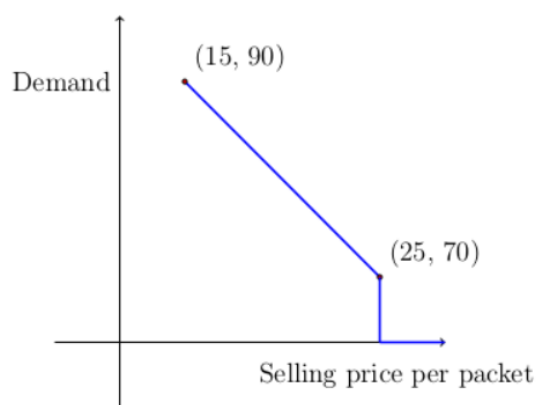


Figure P-5.2:

20) Choose the correct option which shows the profit obtained by Vaishali per day. Here, x is the selling price per packet.

- (a) $2(60 - x)$
- (b) $x(x - 18)$
- (c) $2(x - 18)(60 - x)$
- (d) $2(x + 18)(60 - x)$
- (e) Inadequate information

[Ans: (c)]

Questions

21. Choose the set of correct options.

- (a) Vaishali should sell a packet with minimum price of 18 so as not to incur any loss
- (b) Vaishali should sell a packet with minimum price of 12 so as not to incur any loss
- (c) To make maximum profit per day, the selling price per packet should be 39
- (d) To make maximum profit per day, the selling price per packet should be 25
- (e) Vaishali should sell a packet with maximum price of 60 so as not to incur any loss
- (f) Vaishali should sell a packet with maximum price of 25 so as not to incur any loss

[Ans: (a), (d), (f)]

22. What should be the price of plate per packet to make a profit of 490 per day? [Hint: $(x - 53)$ a factor of $2(-x^2 + 78x - 1325)$.] [Ans: 25]

23. What will be the value of $m + n$ if the sum of the roots and the product of the roots of equation $(5m + 5)x^2 - (4n + 3)x + 10 = 0$ are 3 and 2 respectively? [Ans: 3]

24. What will the sum of two positive integers be if the sum of their squares is 369 and the difference between them is 3? [Ans: 27]

END OF WEEK 3

AQ 4.1

Questions

1. Which of the following statements is(are) correct with respect to the polynomials?
 - (a) A polynomial with one, two, and three terms are called binomial, trinomial, and monomial respectively.
 - (b) A polynomial is an algebraic expression in which the only arithmetic is addition, subtraction, multiplication, and non-negative integer exponents of the variables.
 - (c) A polynomial in one variable can be represented as:

$$\sum_{m=0}^n a_m x^m \quad \text{where } m \text{ and } n \text{ are natural numbers.}$$

- (d) $t^{5/2} + t$ is a binomial.
- (e) $\sqrt{(x-1)^4}$ is a trinomial.
- (f) $\sqrt{(x-1)}$ is a binomial.

[Ans: (b), (c), (e)]

2. From the list of algebraic expressions:

$$x + \sqrt{\frac{1}{2}}, \sqrt{\frac{x}{2}} - 2x^2 \times x^8, \quad \sqrt{\frac{x^4}{2}}, \quad 0.001, \quad \frac{x^3 + 5x^2 - 8x}{x^2}, \quad 5x^{111} + x^{-9}, \quad 8x^2 + (-3)$$

Choose the correct option(s) from the following:

- (a) Total number of monomials are 2.
- (b) Total number of binomials are 2.
- (c) Total number of polynomials are 4.
- (d) $\frac{x^3 + 5x^2 - 8x}{x^2}$ is a trinomial.
- (e) $5x^{111} + x^{-9}$ is a binomial.

[Ans: (a), (b), (c)]

AQ 4.2

Questions

1. The degree of the 3rd term of a polynomial $5x^6 - 4x^5 + x^{0.5} - x^1 + 4$ is:

- (a) $\frac{1}{2}$ or 0.5
- (b) It is not a polynomial
- (c) 6
- (d) 13.5
- (e) 0

[Ans: (b)]

2. The degree of a polynomial $\sqrt{x^8} + \frac{5}{2}x^3 - \sqrt{2}$ is:

- (a) It is not a polynomial
- (b) 3
- (c) 8
- (d) 4

[Ans: (d)]

3. The degree of zero polynomial is:

- (a) 0
- (b) Can be a positive real number
- (c) 1
- (d) Not defined

[Ans: (d)]

Questions

4. Choose the correct options.

- (a) The degree of the polynomial is the largest degree of any one of the terms with or without zero coefficients.
- (b) Degree of any term in a polynomial is the sum of the degrees of the variables in that term.
- (c) Degree of the polynomial $\sqrt{10x^{16}y^{12}z^8}$ is 18.
- (d) The degree of the binomial $\frac{8x^4+\sqrt{8x^4}}{\sqrt{5x^4}}$ is 4.

[Ans: (b), (c)]

5. Choose the correct options with respect to polynomials.

- (a) The degree of a constant polynomial is 1.
- (b) The degree of the binomial $\sqrt{2x^4} \cdot y + 2y + 0$ is 3.
- (c) The equation of a straight line is an example of a linear polynomial, which is always a binomial.
- (d) The binomial $2x^2 + 0x + \sqrt{2}$ is a quadratic polynomial.
- (e) The cubic polynomial must be a trinomial.
- (f) Degree of a cubic monomial, x^2yz^0 , is 3.

[Ans: (b), (d), (f)]

AQ 4.3

Questions

1. Choose the correct option with respect to the domain and range of the polynomial function $p(x)$.
 - (a) Domain and range of $p(x)$ is $\mathbb{R}$.
 - (b) Domain of $p(x)$ depends on the function and the range is $\mathbb{R}$.
 - (c) Domain of $p(x)$ is $\mathbb{R}$ and the range depends on the function.
 - (d) Both range and domain of $p(x)$ depend on the function.

[Ans: (c)]

2. The non-constant polynomials $p(x)$ and $q(x)$ with degree m and n respectively, when added, the resultant polynomial is $r(x)$. Choose the correct option from the following.
 - (a) If $m < n$, then the degree of the polynomial $r(x)$ is n .
 - (b) If $m = n$, then the degree of the polynomial $r(x)$ is $m + n$.
 - (c) If $m > n$, then the degree of the polynomial $r(x)$ is $n - m$.
 - (d) If $m > n$, then the degree of the polynomial $r(x)$ is $m - n$.

[Ans: (a)]

3. The difference between the cubic polynomials $p(x) = \sum_{k=0}^3 a_k x^k$ and $q(x) = \sum_{j=0}^3 b_j x^j$, using subtraction algorithm, is: (Hint: $p(x) - q(x) = \sum_{k=0}^{m \vee n} (a_k - b_k) x^k$, where m or n denotes whichever is maximum.)

- (a) $(a_3 - b_3)x^3 + (a_2 - b_2)x^2 + (a_1 - b_1)x^1 + (a_0 - b_0)x^0$
- (b) $(a_3 + b_3)x^3 + (a_2 + b_3)x^1 + (a_1 + b_2)x^1 + (a_0 + b_0)x^0$
- (c) $(a_3 + b_3)x^3 - (a_2 + b_2)x^2 - (a_1 + b_1)x^1 - (a_0 + b_0)x^0$
- (d) $(a_3 - b_3)x^3 - (a_2 - b_2)x^2 - (a_1 - b_1)x^1 - (a_0 - b_0)x^0$

[Ans: (a)]

Questions

4. Choose the right option(s) with respect to the polynomials $p(x) = \sum_{k=0}^n a_k x^k$ and $q(x) = \sum_{j=0}^m b_j x^j$:

- (a) $p(x) + q(x) = \sum_{k=0}^{m \vee n} (a_k + b_k) x^k$
- (b) $p(x) + q(x) = \sum_{k=0}^{m \vee n} (a_k + b_k) x^{m+n}$
- (c) $p(x) - q(x) = \sum_{k=0}^{m \vee n} (a_k - b_k) x^{m-n}$
- (d) $p(x) - q(x) = \sum_{k=0}^{m \vee n} (a_k - b_k) x^k$
- (e) $p(x) + q(x) = \sum_{k=0}^{m \vee n} (a_k + b_k) x^k$
- (f) $p(x) - q(x) = \sum_{k=0}^{m \vee n} (a_k + b_k) x^k$

[Ans: (a), (d)]

5. Choose the correct option(s) with respect to the polynomial $p(x) = \sum_{k=0}^n a_k x^k$ and $q(x) = \sum_{j=0}^m a_j x^j$:

- (a) Sum of two distinct binomials is always binomial.
- (b) Difference of two distinct trinomials is always trinomial.
- (c) Sum or difference of a binomial and a trinomial can be trinomial.
- (d) A quadratic trinomial when added to itself, results in a quadratic trinomial.
- (e) A quadratic trinomial when subtracted from itself, results in a quadratic trinomial.

[Ans: (c), (d)]

AQ 4.4

Questions

1. If the polynomial $r(x)$ is the product of a degree two polynomial $p(x) = -8x^2 - 12x + 2$ and its slope, then the coefficient of the term with the highest power is:

- (a) -128
- (b) -8
- (c) 128
- (d) 24
- (e) -24

[Ans: (c)]

2. The product of the polynomials $p(x) = 2x^2 + 2x + 2$ and $q(x) = x^2 - x - 1$ is:

- (a) $2x^4 + 2x^2 - 4x + 2$
- (b) $4x^4 - 4x^2 - 4x - 4$
- (c) $4x^4 - 4$
- (d) $2x^4 - 2x^2 - 4x - 2$

[Ans: (d)]

3. Choose the correct option(s) with respect to the product of the polynomials $p(x) = \sum_{k=0}^n a_k x^k$ and $q(x) = \sum_{j=0}^m b_j x^j$:

[Ans: (c), (d)]

- ☐ $p(x) \times q(x) = \sum_{k=0}^{m+n} \sum_{j=0}^k (a_j b_{k-j}) x^{k-j}$
- ☐ $q(x) \times p(x) = \sum_{k=0}^{m+n} \sum_{j=0}^k (a_j b_{k-j}) x^{m+n}$
- ☐ $p(x) \times q(x) = \sum_{k=0}^{m+n} \sum_{j=0}^k (a_j b_{k-j}) x^k$
- ☐ $q(x) \times p(x) = \sum_{k=0}^{m+n} \sum_{j=0}^k (a_j b_{k-j}) x^k$

Figure 1

Questions

4. If $r(x) = p(x)q(x)$ where $p(x) = -2x^9 + 4x^5 + 2x^2 + x + 1$ and $q(x) = 8x^4 - 2x^2 + x$, then choose the correct option(s) from the following:
- (a) Degree of $r(x)$ is 13 and coefficient of x^{13} is -16 .
 - (b) Degree of polynomial $r(x)$ is 36 and it has a coefficient of zero.
 - (c) Coefficient of x^2 in $r(x)$ and $q(x)$ is -2 .
 - (d) Coefficient of x^8 in $r(x)$ is zero.
 - (e) Coefficient of x^9 in $r(x)$ and $p(x)$ is 32 and -2 , respectively.

[Ans: (a), (c), (d), (e)]

AQ 4.5

Questions

1. If $p(x) = x^5 - 4x^4 + 3x^3 + 3x^2 - 4$ and $q(x) = (x - 2)^2$, then $\frac{p(x)}{q(x)}$ is:

- (a) $x^3 + x + 1$
- (b) $x^4 - x^2 - 1$
- (c) $x^3 - x - 1$
- (d) $x^4 - x^2 + 1$

[Ans: (c)]

2. When a polynomial $q(x) = x^6 + 3x^4 + 3x^3 - 3x^2 - 10$ is divided by another polynomial $p(x) = (x - 2)^4(x + 2)^3$, then which of the following is correct?

- (a) $2x^2 - 2x + 5$
- (b) Division is not possible
- (c) $12x^2 - 10x + 10$
- (d) None of the above

[Ans: (b)]

3. Which of the following is(are) correct with respect to the division of a polynomial if the dividend and the divisor are $p(x) = \sum_{k=0}^n a_k x^k$ and $q(x) = \sum_{j=0}^m b_j x^j$ respectively, and they are not constant polynomials?

- (a) The division is possible when $n \geq m$
- (b) If $p(x)$ is a trinomial, then the quotient is always a trinomial
- (c) The quotient is always a polynomial
- (d) If $m = 6$ and $n = 4$, then division is possible

[Ans: (a), (c)]

AQ 4.6

Questions

1. If dividend is $p(x) = x^6 - 6x^5 + 17x^4 - 48x^3 + 76x^2 - 24x + 36$ and divisor is $g(x) = x^2 - 6x + 9$, then quotient $q(x)$ and remainder $r(x)$ will be:

- (a) $q(x) = x^4 + 8x^2 + 4$ and $r(x) = 0$
- (b) $q(x) = x^4 - 8x^2 + 4$ and $r(x) = x + 1$
- (c) $q(x) = x^4 + 8x^2 + 4$ and $r(x) = x + 1$
- (d) $q(x) = x^4 + 8x^2 - 4$ and $r(x) = 0$

[Ans: (a)]

2. What will be the dividend $p(x)$, if the divisor, the quotient, and the remainder are $g(x) = x^{10} + 2x + 2$, $q(x) = x^4 - 2$, and $r(x) = 2x - 4$ respectively?

- (a) $p(x) = x^{14} - 2x^{10} + 2x^5 + 2x^4 - 2x + 8$
- (b) $p(x) = x^{14} - 2x^{10} + 2x^5 + 2x^4 - 2x - 8$
- (c) $p(x) = x^{14} - 2x^{10} - 2x^5 + 2x^4 - 2x - 8$
- (d) $p(x) = x^{40} - 2x^{10} + 2x^5 + 2x^4 - 2x - 8$

[Ans: (b)]

3. Choose the correct option(s) with respect to the polynomials $p(x) = x^3 + 2x^2 - 11x - 12$ and $g(x) = x - 3$:

- (a) The remainder of the division of $p(x)$ by $g(x)$ is 0.
- (b) The division of $g(x)$ by $p(x)$ is not possible.
- (c) The remainder of the division of $g(x)$ by $p(x)$ is 0.
- (d) If $g_1(x) = x + 4$ and $g_2(x) = x + 1$, then the remainder is 0 when $p(x)$ is divided by $g_1(x)$ and $p(x)$ is divided by $g_2(x)$.

[Ans: (a), (b), (d)]

Questions

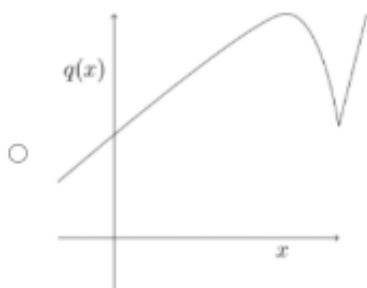
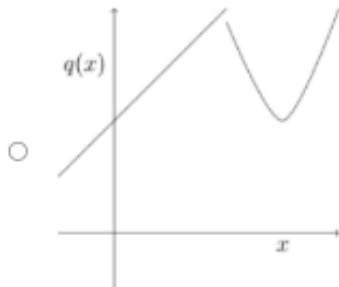
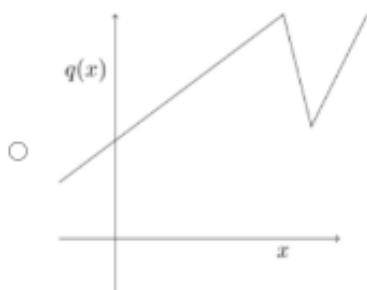
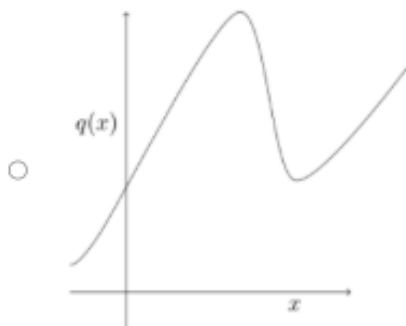
4. Choose the correct option(s) from the following if $f(x) = 10x^4 + 17x^3 - 62x^2 + 28x - 18$ and $g(x) = 2x^2 - x + 1$ are two polynomials, and $r(x)$ and $q(x)$ represent the remainder and quotient respectively:
- (a) Division of $f(x)$ by $g(x)$ is not possible.
 - (b) If $f(x)$ is divided by $q_1(x)$, where $q_1(x) = 5x^2 + 11x - 28$, then the quotient and remainder will be $2x^2 - x + 1$ and $-11x + 10$, respectively.
 - (c) $q(x) = 5x^2 + 11x - 28$ and $r(x) = -11x - 10$ then $f(x) = g(x) \times q(x) + r(x)$
 - (d) $q(x) = 5x^2 + 11x - 28$ and $r(x) = -11x + 10$ then $f(x) = g(x) \cdot q(x) + r(x)$

[Ans: (b), (d)]

AQ 4.7

Questions

1. Which of the following can be graph of a polynomial?



[Ans: (a)]

Questions

1. Ram follows a path according to the given graph as in the Figure AQ – 7.1 of a polynomial $q(x)$. He started his journey from his house and ends his journey to the market. Choose the correct set of options.

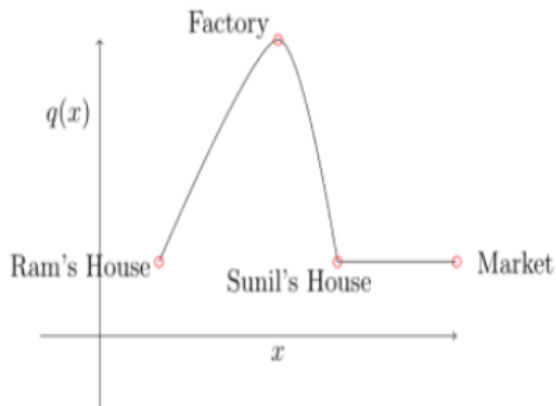


Figure AQ-7.1

- (a) The path from Ram's House to Sunil's House may represent a graph of a polynomial.
- (b) The whole path may represent a graph of a polynomial.
- (c) The path from factory to market may represent a graph of a polynomial.
- (d) The path from factory to Sunil's House may represent a graph of a polynomials.

[Ans: (a), (d)]

AQ 4.8

Questions

1. Let $p(x) = 5x^3 - 50x^2 + 155x - 150$ be a polynomial and $(x - 2)$ be a factor of $p(x)$. Which of the following is the complete factorized form of the given polynomial $p(x)$?

- (a) $(x - 5)(x - 2)(x - 3)$
- (b) $5(x - 3)(x - 2)(x - 5)$
- (c) $(5x + 25)(x - 2)(x - 3)$
- (d) $(5x - 15)(x - 2)(x + 5)$

[Ans: (b)]

2. Zeros of the polynomial $2x^3 - 18x^2 - 24x + 40$ is(are):

- (a) 7
- (b) 14
- (c) 10
- (d) 1
- (e) 2
- (f) -2

[Ans: (c), (d), (f)]

3. Let $p(x) = (x^2 + x + 1)(x + 1)x$, choose the set of correct options.

- (a) The monomial factor of $p(x)$ is 5.
- (b) The monomial factor of $p(x)$ is x .
- (c) The binomial factor of $p(x)$ is $(x + 1)$.
- (d) The binomial factor of $p(x)$ is $(x^2 + x + 1)$.
- (e) The trinomial factor of $p(x)$ is $(x^2 + x + 1)$.

[Ans: (b), (c), (e)]

Questions

4. Let $p(x) = x^3 + 2x^2 + 5x + 10$ be a polynomial and $\mathbf{S}$ be the set of x -intercepts, choose the set of correct options:

- (a) $\mathbf{S} = \phi$
- (b) $5 \in \mathbf{S}$
- (c) $-2 \in \mathbf{S}$
- (d) $-5 \in \mathbf{S}$
- (e) $2 \in \mathbf{S}$
- (f) The cardinality of the set $\mathbf{S}$ is 1.
- (g) The cardinality of the set $\mathbf{S}$ is 3.

[Ans: (c), (f)]

Let $p(x) = (x^2 + 5)(x^2 + 4)$ be a polynomial and $\mathbf{S}$ is the set of x -intercepts and $\mathbf{R}$ is the set of y -intercepts. Answer the following questions:

5. Find the cardinality of the set $\mathbf{S}$.

[Ans: 0]

6. Find the y -intercept.

[Ans: 20]

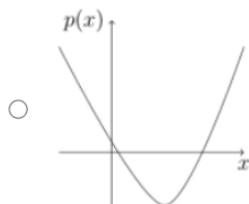
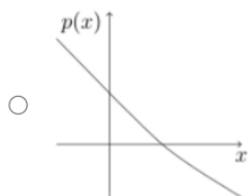
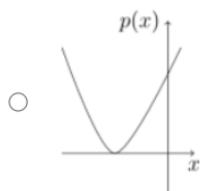
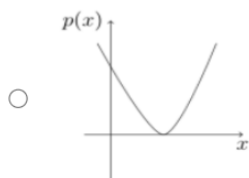
7. Find the cardinality of the set $\mathbf{R}$.

[Ans: 1]

AQ 4.9

Questions

1. If $p(x)$ be a polynomial function and 1 is a zero of the given polynomial with even multiplicity, then which of the following graph is the best representation of the graph of $p(x)$?



[Ans: (a)]

Questions

Answer question 2 and 3 from the *Figure AQ – 7.2* which represents the graph of $p(x)$

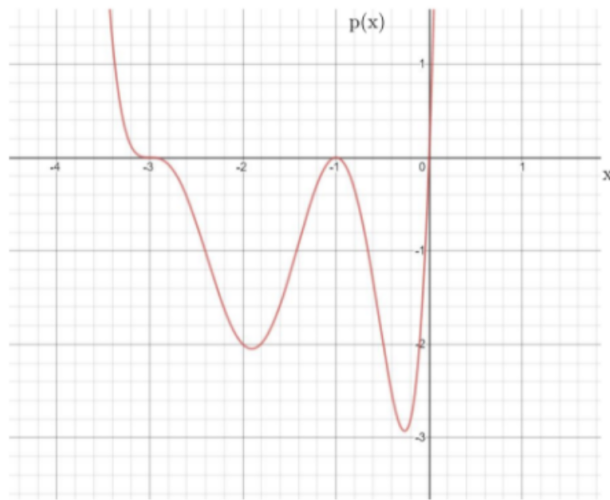


Figure AQ-7.2

2. Choose the set of correct options.

- (a) Zeros of the polynomial are $-3, -2, 1$
- (b) Zeros of the polynomial are $-3, -1, 0$
- (c) y -intercept of the polynomial is 1
- (d) y -intercept of the polynomial is 0

[Ans: (b), (d)]

3. Choose the correct options regarding the multiplicities of the zeros.

- (a) -3 with even multiplicity
- (b) -3 with odd multiplicity
- (c) 0 with even multiplicity
- (d) 0 with 1 multiplicity
- (e) -1 with even multiplicity
- (f) -1 with odd multiplicity

[Ans: (b), (d), (e)]

Questions

4. Let $p(x)$ be a polynomial and 7, 9, 2 are the only zeros of the polynomial with multiplicities 3, 1, and 4, respectively. Then the degree of the polynomial can be at least:

- (a) 7
- (b) 3
- (c) 8
- (d) 1
- (e) 4

[Ans: (c)]

5. Suppose $p(x) = (x + 4)^5(2x^2 + 4)$, choose the set of correct options.

- (a) 4 is the zero of the polynomial
- (b) -4 is the zero of the polynomial
- (c) Only one root is of multiplicity 5
- (d) Two roots are of multiplicity 5
- (e) There is no real root
- (f) The number of real roots is 2
- (g) The number of real roots is 1

[Ans: (b), (c), (g)]

AQ 4.10

Questions

Answer Questions 1 and 2 from the Figure AQ-7.3 which represents a graph of a polynomial of degree 9.

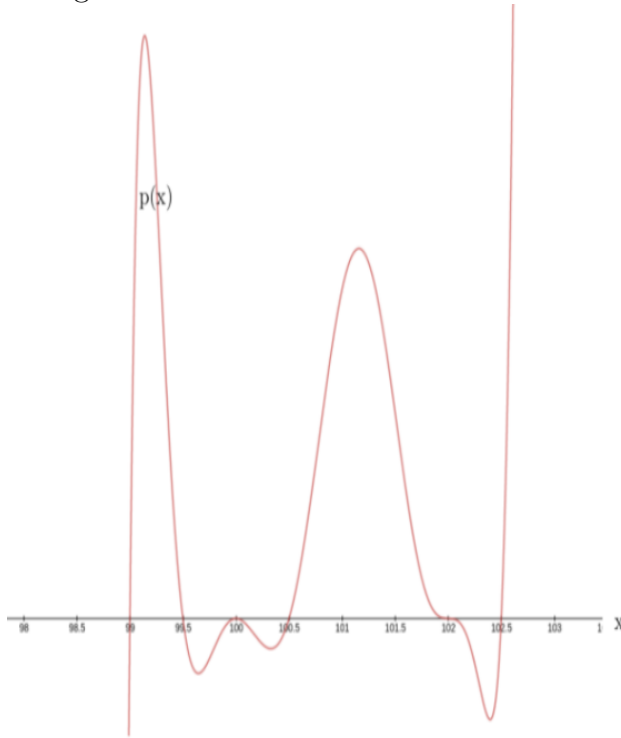


Figure AQ-7.3

1. The zeros of the polynomial are:
 - (a) 100, 99, 101, 99.5, 98, 100.5, 102.5
 - (b) 100, 101, 99, 99.5, 100.5, 102.5, 102
 - (c) 100, 99, 101, 99.5, 98, 100.5, 102.5, 99
 - (d) 100, 99, 99.5, 102, 100.5, 102.5

[Ans: (d)]

Questions

2. The most appropriate algebraic expression corresponding to the given graph of the polynomial is:

- (a) $5(x - 100)^2(x - 99)(x - 102)^2(x - 100.5)(x - 102.5)(x - 99.5)$
- (b) $a(x - 100)^2(x - 99)(x - 102)^3(x - 100.5)(x - 102.5)(x - 99.5)$ where $a > 0$, $a \in \mathbb{R}$
- (c) $(x - 100)^2(x - 102)^2(x - 100.5)(x - 102.5)(x - 99.5)$
- (d) $b(x - 100)^2(x - 99)(x - 102)^3(x - 100.5)(x - 102.5)$ where $b \neq 0$, $b \in \mathbb{R}$
- (e) None of the above

[Ans: (b)]

A function $f(x)$ is said to be even function if $f(-x) = f(x)$ and said to be odd function if $f(-x) = -f(x)$. Use this information for question 3 and 4.

3. Let $f(x)$ be a function whose graph is shown in *Figure AQ – 7.4*:

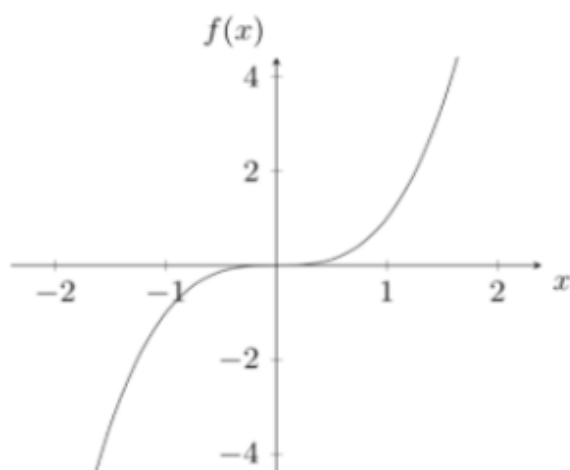


Figure AQ-7.4

Choose the correct option:

- (a) $f(x)$ is an even function.
- (b) $f(x)$ is an odd function.
- (c) $f(x)$ is neither even nor odd.
- (d) $f(x)$ is both even and odd.

[Ans: (b)]

Questions

4. Let $f(x)$ be a function whose graph is shown in Figure AQ-7.5:

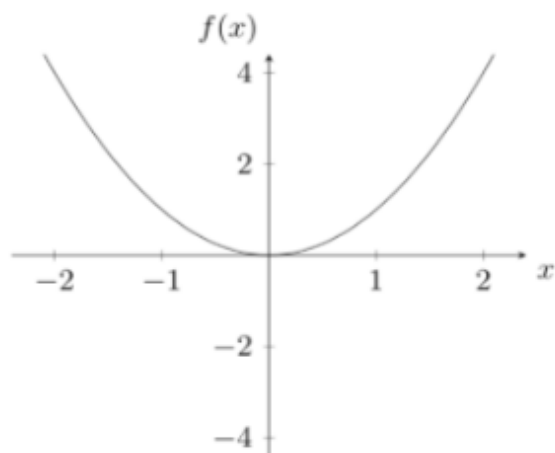


Figure AQ-7.5

Choose the correct option:

- (a) $f(x)$ is an even function.
- (b) $f(x)$ is an odd function.
- (c) $f(x)$ is neither even nor odd.
- (d) $f(x)$ is both even and odd.

[Ans: (a)]

AQ 4.11

Questions

1. If $p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ be a polynomial, then the end behavior of the polynomial depends:
 - (a) Only on the coefficient a_{n-1}
 - (b) Only on the coefficients a_n and a_{n-1}
 - (c) Only on the coefficient a_n and the degree of the polynomial
 - (d) Only on the degree of the polynomial
 - (e) On all coefficients in the polynomial
 - (f) On all coefficients and degree of the polynomial

[Ans: (c)]

Let $p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ be the polynomial, Use this for questions 2-5

2. If $a_n > 0$ and n is even, then which of the following is(are) true?
 - (a) $x \rightarrow \infty \implies p(x) \rightarrow -\infty$
 - (b) $x \rightarrow \infty \implies p(x) \rightarrow \infty$
 - (c) $x \rightarrow -\infty \implies p(x) \rightarrow -\infty$
 - (d) $x \rightarrow -\infty \implies p(x) \rightarrow \infty$

[Ans: (b), (d)]

3. If $a_n > 0$ and n is odd, then which of the following is(are) true?
 - (a) $x \rightarrow -\infty \implies p(x) \rightarrow -\infty$
 - (b) $x \rightarrow -\infty \implies p(x) \rightarrow \infty$
 - (c) $x \rightarrow \infty \implies p(x) \rightarrow -\infty$
 - (d) $x \rightarrow \infty \implies p(x) \rightarrow \infty$

[Ans: (a), (d)]

Questions

4. If $a_n < 0$ and n is odd, then which of the following is(are) true?

- (a) $x \rightarrow -\infty \implies p(x) \rightarrow \infty$
- (b) $x \rightarrow -\infty \implies p(x) \rightarrow -\infty$
- (c) $x \rightarrow \infty \implies p(x) \rightarrow \infty$
- (d) $x \rightarrow \infty \implies p(x) \rightarrow -\infty$

[Ans: (a), (d)]

5. If $a_n < 0$ and n is even, then which of the following is(are) true?

- (a) $x \rightarrow \infty \implies p(x) \rightarrow -\infty$
- (b) $x \rightarrow -\infty \implies p(x) \rightarrow -\infty$
- (c) $x \rightarrow -\infty \implies p(x) \rightarrow \infty$
- (d) $x \rightarrow \infty \implies p(x) \rightarrow \infty$

[Ans: (a), (b)]

6. If $p(x) = (-5)^3x^3 + 2x^2 + x + 1$ be a polynomial, then which of the following is(are) true?

- (a) $x \rightarrow \infty \implies p(x) \rightarrow -\infty$
- (b) $x \rightarrow -\infty \implies p(x) \rightarrow -\infty$
- (c) $x \rightarrow -\infty \implies p(x) \rightarrow \infty$
- (d) $x \rightarrow \infty \implies p(x) \rightarrow \infty$

[Ans: (a), (c)]

7. If $p(x) = 7x^n + 9x^5 + 1$ be a polynomial and $x \rightarrow -\infty, \implies p(x) \rightarrow -\infty$, then which of the following is(are) true?

- (a) $n > 5$ is even
- (b) $n > 5$ is odd
- (c) n is irrational
- (d) $n = 0$

[Ans: (b), (d)]

AQ 4.12

Questions

1. If $p(x) = 3x^{10} + 5x + 2$ be a polynomial, then the maximum number of turning points of the polynomial $p(x)$ can be:

- (a) 10
- (b) 1
- (c) 9
- (d) 0

[Ans: (c)]

2. If the graph of a polynomial $p(x)$ is shown in Figure AQ-7.6, then the number of turning points of the polynomial $p(x)$ is:

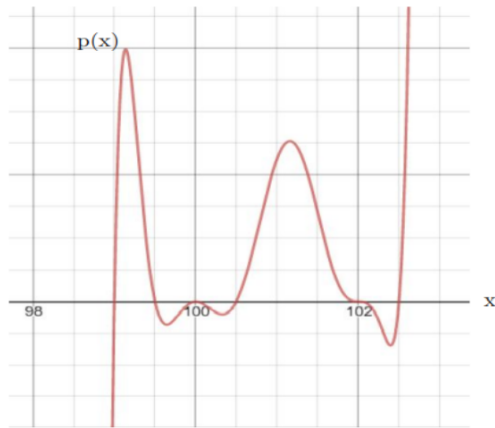


Figure AQ -7.6

- (a) 7
- (b) 6
- (c) 9
- (d) 0
- (e) 5

[Ans: (b)]

Questions

3. If the graph of a polynomial $p(x)$ is shown in Figure AQ-7.7, then the minimum possible degree of the polynomial $p(x)$ is:

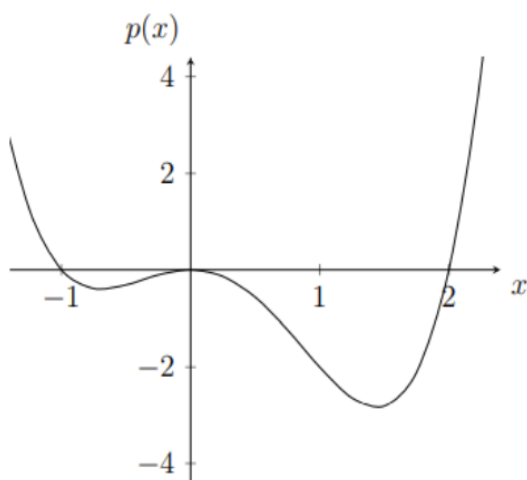


Figure AQ - 7.7

- (a) 3
- (b) 4
- (c) 2
- (d) 1

[Ans: (b)]

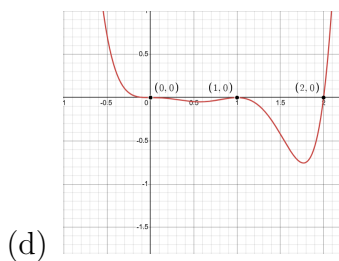
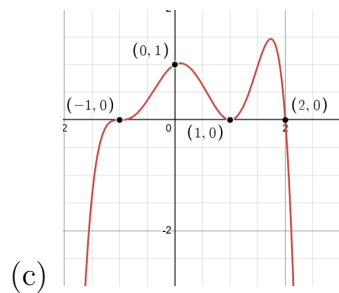
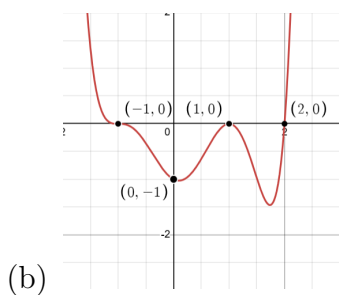
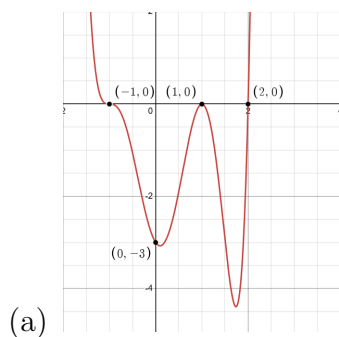
4. Find the number of turning points of the polynomial $P(x) = x^3$.

[Ans: 0]

AQ 4.13

Questions

- Which of the following represents the graph of the polynomial $p(x) = \frac{1}{2}(x-1)^2(x+1)^3(x-2)$?



[Ans: (b)]

Questions

2. If $p(x)$ is a polynomial function and $p(3)$ and $p(5)$ have opposite signs to each other, then which of the following is true?
- (a) There exists, at least two values a and b such that $3 < a, b < 5$ and $p(a) = 0$ and $p(b) = 0$
 - (b) There does not exist any value a such that $3 < a < 5$ and $p(a) = 0$
 - (c) There exists, at least one value a such that $3 < a < 5$
 - (d) There exists, at least one value a such that $3 > a$ and $p(a) = 0$

[Ans: (c)]

3. The algebraic expression for the polynomial of the graph as shown in *Figure AQ-7.8* is:

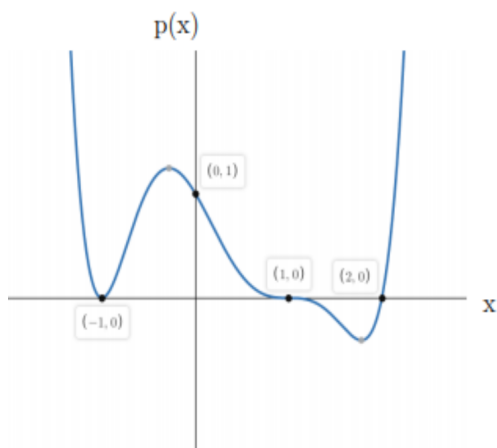


Figure AQ- 7.8

- (a) $\frac{1}{2}(x-1)^3(x-2)(x+1)^2$
- (b) $\frac{3}{4}(x-1)^3(x-2)(1+x)^2$
- (c) $\frac{3}{4}(x-1)^2(x-2)(x+1)$
- (d) $\frac{1}{2}(x-1)^3(x-2)(x+1)$

[Ans: (a)]

Questions

4. Given three polynomials are:

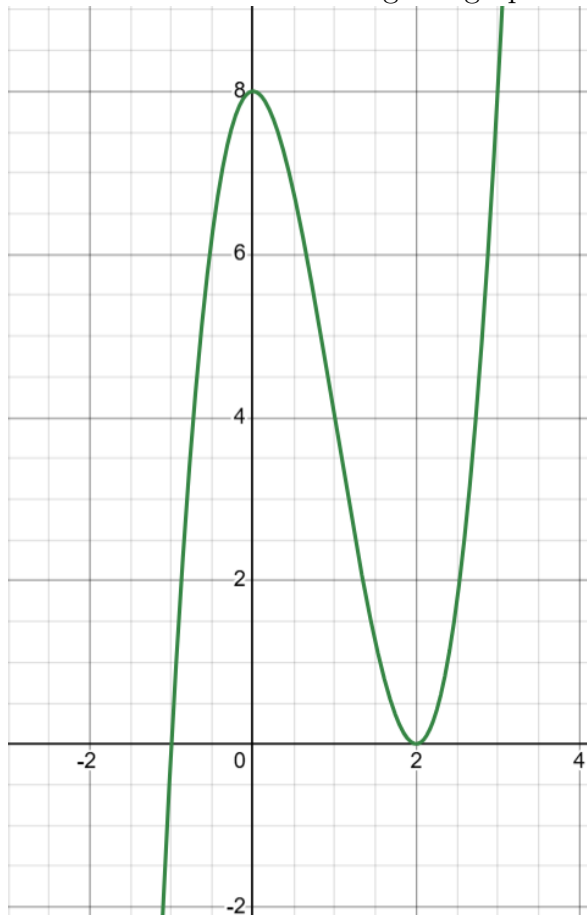
$$p_1(x) = (-x^2 + x^4)^2, \quad p_2(x) = -x^3 + 5x, \quad p_3(x) = 9x^3 + 7x,$$

choose the set of correct options.

- (a) p_1, p_2 are even polynomials
- (b) p_3 is an odd polynomial
- (c) p_1 is an even polynomial
- (d) p_3 is an even polynomial
- (e) p_1, p_2 are odd polynomials

[Ans: (b), (c)]

1. Stretch factor of the below given graph is



[Ans: 2]

PA 4

Questions

1-2. (Use the following data for Question 1 and Question 2 only). Let x be the number of years since the year 2000 (i.e., $x = 0$ denotes the year 2000). The total amount of profit (in ₹) on books in a shop is given by the function $T(x) = 5x^3 + 3x + 1$. The shop sells books of four languages English, Bengali, Hindi, and Tamil. The profits from selling English and Bengali books are given by $E(x) = 3x^3 - 5x^2 + x$ and $B(x) = x^2 + 4x + 5$ respectively. The profit from selling Hindi and Tamil books are found to be the same.

1. Which of the following polynomial functions represents the profit from selling Tamil books?

- (a) $2x^3 + 4x^2 - 2x - 4$
- (b) $x^3 - 2x^2 - x + 2$
- (c) $x^3 + 2x^2 - x - 2$
- (d) $2x^3 - 4x^2 - 2x + 4$

[Ans: (c)]

2. In which year was the profit from Hindi books zero?

- (a) 2001
- (b) 2002
- (c) 2004
- (d) 2010

[Ans: (a)]

3. Find the quadratic polynomial which when divided by x , $x - 1$ and $x + 1$ gives the remainders 7, 14 and 8 respectively.

- (a) $4x^2 - 3x + 7$
- (b) $x^2 + 7x + 7$
- (c) $7x^2 + x + 7$
- (d) $4x^2 + 3x + 7$

[Ans: (d)]

Questions

4. Box A has length x unit, breadth $(x + 1)$ unit, and height $(x + 2)$ unit. Box B has length $(x + 1)$ unit, breadth $(x + 1)$ unit, and height $(x + 2)$ unit. There are two more boxes C and D of cubic shape with side x unit. The total volume of A and B is y cubic unit more than the total volume of C and D . Find y in terms of x .

- (a) $x^3 + 7x^2 + 7x + 2$
- (b) $7x^2 + 7x + 2$
- (c) $7x^2 - 7x - 2$
- (d) $x^3 + 7x^2 - 7x - 2$

[Ans: (b)]

5. The population of a bacteria culture in laboratory conditions is known to be a function of time of the form $p(t) = at^5 + bt^2 + c$, where p represents the population (in lakhs) and t represents the time (in minutes). Suppose a student conducts an experiment to determine the coefficients a, b and c in the formula and obtains the following data: $p(0) = 3$ $p(1) = 5$ $p(2) = 39$ Which of the following options is correct?

- (a) $p(t) = 3t^5 - t^2 + 3$
- (b) $p(t) = 4t^5 - 2t^2 + 3$
- (c) $p(t) = t^5 + t^2 + 3$
- (d) $p(t) = 39t^5 + 5t^2 + 3$

[Ans: (c)]

6. If the polynomials $x^3 + ax^2 + 5x + 7$ and $x^3 + 2x^2 + 3x + 2a$ leave the same remainder when divided by $(x - 2)$, then the value of a is:

- (a) $\frac{3}{2}$
- (b) $-\frac{3}{2}$
- (c) $\frac{5}{2}$
- (d) $-\frac{5}{2}$

[Ans: (b)]

Questions

7. Let $r(x)$, be a polynomial function which is obtained as the remainder after dividing the polynomial $2x^3 + x^2 - 5$ by the polynomial $2x - 3$. Choose the correct option which represents the polynomial $r(x)$ most appropriately.

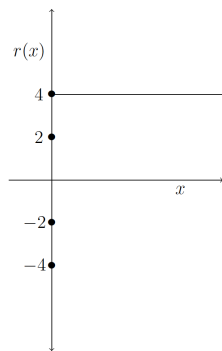


Fig P-6.2

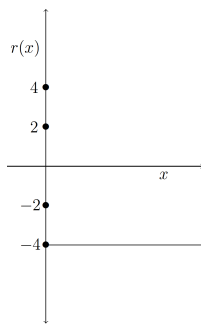


Fig P-6.3

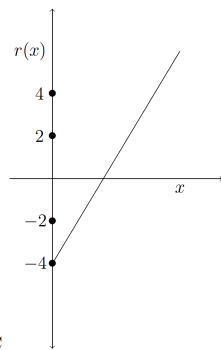


Fig P-6.4

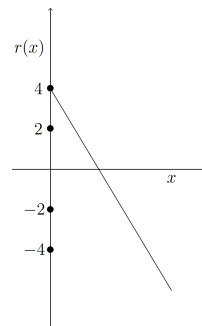


Fig P-6.5

[Ans: (a) [fig P6.2]]

- 8-9. (Use the following data for Question 8 and Question 9 only). By dividing a polynomial $p(x)$ with another polynomial $q(x)$ we get $h(x)$ as the quotient and $r(x) (\neq 0)$ as the remainder.

8. The maximum degree of $r(x)$ can be,

- (a) $\deg p(x)$
- (b) $\deg p(x) - 1$
- (c) $\deg q(x)$
- (d) $\deg q(x) - 1$

[Ans: (d)]

9. If $\deg p(x) < \deg q(x)$, then choose the set of correct answers:

- (a) $h(x) = 0$
- (b) $\deg h(x) = \deg q(x)$
- (c) $\deg r(x) = \deg q(x)$
- (d) $\deg r(x) = \deg p(x)$

[Ans: (a), (d)]

Questions

- 10-11. (Use the following data for Question 10 and Question 11 only). An open box can be made from a piece of cardboard of length $7x$ unit and breadth $5x$ unit, by cutting squares of side x unit out of the corners of the rectangular cardboard, then folding up the sides as shown in the Figure P-6.1.

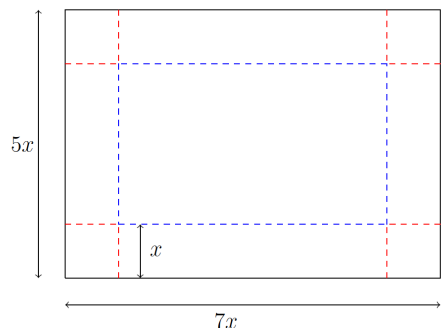


Figure P-6.1

10. What will be the coefficient of x^3 in the polynomial representing the volume of the box? [Ans: 15]
11. What will be the coefficient of x^2 in the polynomial representing the volume of the box? [Ans: 0]
12. What should be subtracted from the polynomial $P(x) = 6x^4 + 5x^3 + 4x - 4$ to make it divisible by $2x^2 + x - 1$?
- (a) $4x$
 - (b) $4x - 3$
 - (c) $6x - 3$
 - (d) $2x - 3$

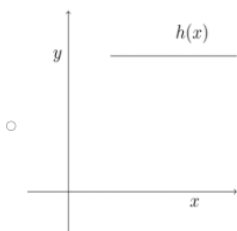
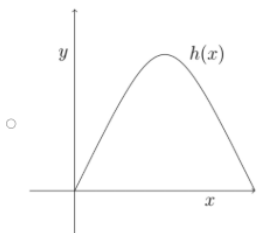
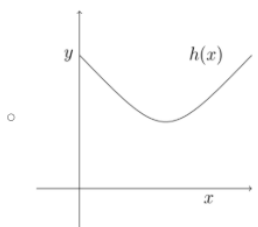
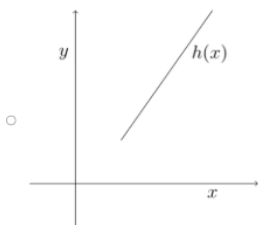
[Ans: (b)]

Questions

13. Table A-6.1 provides the information regarding some polynomials. Which is the most suitable (not exact) representation of $h(x)$, where $h(x)$ is known to be a polynomial in x , and if $h(x) = \frac{P(x)Q(x)-R(x)S(x)+S(x)P(x)}{P(x)+P(x)Q^2(x)}$?

Polynomial	Degree	Condition
$P(x)$	m	$m > 0$
$Q(x)$	n	$m > 2n > 0$
$R(x)$	k	$k = m - n$
$S(x)$	t	$t = 2n$

Table A-6.1



[Ans: (d)]

Questions

14-16. Use the following information to solve questions 14-16.

A manufacturing company produces three type of products A , B , and C from one raw material in a single continuous process. This process generates total solid wastes (W) (in kg) as $W(r) = -0.0001r^3 + 0.1r^2 + r$, where r is the amount of raw material used in kg. If instead, the company uses three different batch-processes (one batch process for one product) to produce the above products, then the amount of waste generated because of products A , B , and C are given as $W_A = -0.00001r^4 + 0.015r^3$, $W_B = -0.005r^3 + 0.05r^2$ and $W_C = 0.05r^2$ respectively.

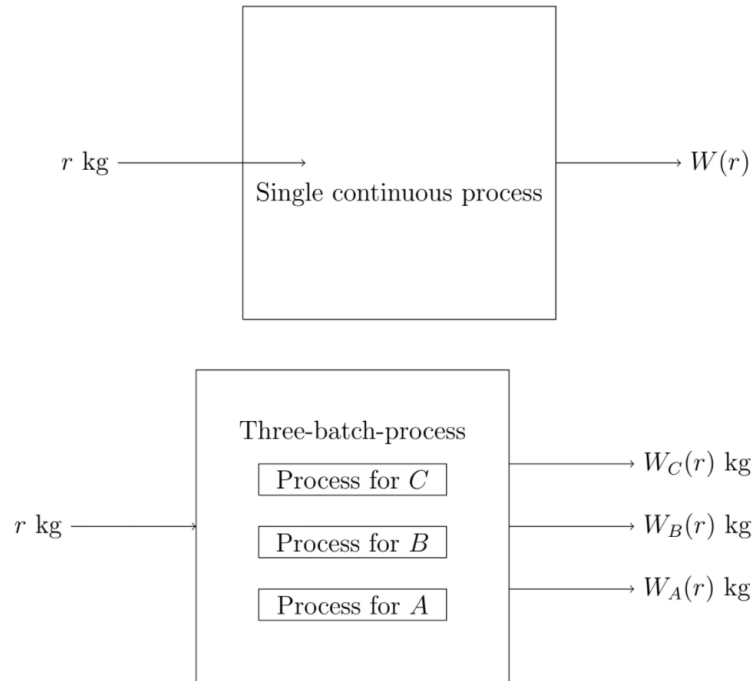


Figure A-5.1

14. What is the total amount of waste generated because of the three different batch-processes?
- (a) $-0.00001r^4 + 0.01r^3 + 1.5r^2$
 - (b) $-0.00001r^4 + 0.015r^3 + 1.5r^2$
 - (c) $-0.00001r^4 + 0.01r^3 + 0.1r^2$
 - (d) $-0.00001r^4 + 0.01r^3 + 0.5r^2$

[Ans: (c)]

Questions

1. 15. What is the ratio of the total waste generated by the three-batch-processes with respect to the single continuous process?
- (a) $-0.01r$
 - (b) $-0.1r$
 - (c) $0.1r$
 - (d) $0.01r$

[Ans: (c)]

16. Let the company wastes Rs. 5,000 in waste treatment when it uses the single continuous process by consuming 100 kg of raw material. If instead of continuous process the company uses the three-batch-processes, then how much extra amount (in Rs.) will the company have to pay for waste treatment with respect to the continuous process?
- (a) 50,000
 - (b) 500
 - (c) 45,000
 - (d) 5,000
 - (e) 4,000

[Ans: (c)]

17. Given that $P(x)$ and $Q(x)$ are two non zero polynomials of degrees m and n respectively. Let $f(x) = P(x)+Q(x)$, $g(x) = P(x)Q(x)$, and $h(x) = P(x) \left\{ P(x)Q(x) + \frac{P(x)}{Q(x)} \right\}$. If $h(x)$ is known to be a polynomial in x , then choose the set of correct options.
- (a) The degree of $f(x)$ is $m + n$.
 - (b) The degree of $g(x)$ is $m + n$.
 - (c) The degree of $f(x)$ is $\max\{m, n\}$ if $m \neq n$, where $\max\{m, n\}$ represents the maximum value from m and n .
 - (d) The degree of $h(x)$ is m^3 .
 - (e) The degree of $h(x)$ is n^3 .
 - (f) The degree of $h(x)$ is $2m + n$.

[Ans: (b), (c), (f)]

Questions

18. Given a polynomial $P(x) = (2x+5)(1-3x)(x^2+3x+1)$, then choose the set of correct options.

- (a) Coefficient of x^5 is 0
- (b) Coefficient of x^3 is -18
- (c) Degree of P is 4
- (d) Coefficient of x^3 is -13
- (e) Degree of P is 7
- (f) All of the above

[Ans: (a), (c)]

19. A sheet $ABCD$ of dimensions 10 ft x 3 ft is shown in Figure A-6.1. A box is made by removing two squares of equal dimensions $AEFG$ and $DHIJ$ and two rectangles of equal dimensions $BKLM$ and $CNOP$ respectively.

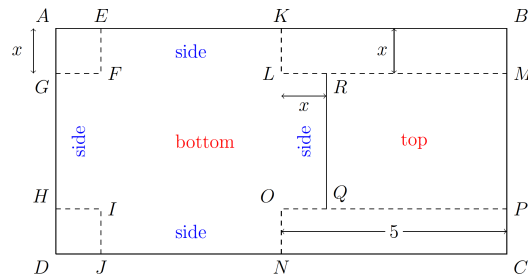


Figure A-6.1

Choose the set of correct options.

- (a) The volume of the box is $2x^2 - 23x + 30$
- (b) The volume of the box is $2x^3 - 13x^2 + 15x$
- (c) If $x = 0.5$, then the volume of the box is 5.625 cubic ft.
- (d) To create the box, value of x should always be greater than 0 but less than 1.5.

[Ans: (b), (d)]

Questions

20. A curious student created a performance profile of his favourite cricketer as $R = -x^5 + 6x^4 - 30x^3 + 80x^2 + 70x + c$, where R is the total cumulative runs scored by the cricketer in x matches. He picked three starting values shown in Table A-5.2 and tried to find the value of c . If he uses Sum Squared Error method, then what will be the value of c ?

No. of matches	Total cumulative score
1	120
2	285
3	361

[Ans: -2]

21. What is the minimum value of x -coordinate for the points of intersection of functions $f(x) = -x^5 + 5x^4 - 7x - 2$ and $g(x) = -x^5 + 5x^4 - x^2 - 2$? [Ans: 0]

END OF WEEK 4

AQ 5.1

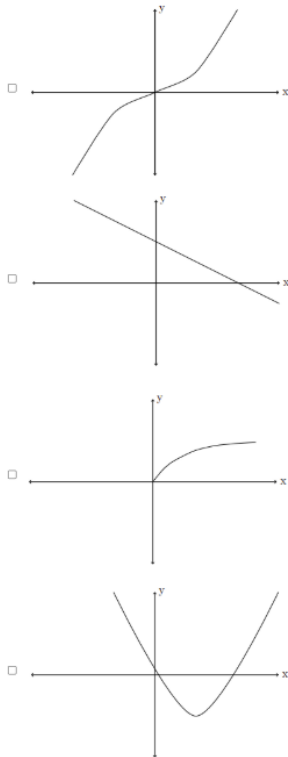
Questions

1. Which of the following statements is(are) TRUE?

- (a) The Vertical line test is used to find whether the given function is one-to-one or not.
- (b) The Horizontal line test is used to find whether the given function is one-to-one or not.
- (c) If for one value of x in domain gives more than one $f(x)$, then f is one-to-one function.
- (d) If for more than one value of x in domain gives one $f(x)$, then f is one-to-one function.

[Ans: (b)]

2. Which of the following graph fails the Horizontal line test?



[Ans: (d)]

Questions

3. Which of the following statements is(are) TRUE?

- (a) One-to-one functions never fail the Horizontal line test.
- (b) One-to-one functions may sometime fail the Horizontal line test.
- (c) No function should fail the Horizontal line test.
- (d) There are some functions that fail the Vertical line test.

[Ans: (a)]

4. Which of the following statements is(are) FALSE?

- (a) A function $f : X \rightarrow Y$ is called one-to-one if, $f(x_1) = f(x_2) \in Y$, then $x_1 = x_2$.
- (b) One-to-one functions are not always reversible on their range.
- (c) If a function fails the Horizontal line test, then it is not reversible.
- (d) A function $f : X \rightarrow Y$ is called one-to-one if, $f(x_1) \neq f(x_2) \in Y$, then $x_1 \neq x_2$.

[Ans: (b), (d)]

5. Suppose $f : [0, \infty) \rightarrow R$. Which of the following is not a function?

- (a) $f(x) = 4x + 3$
- (b) $f(x) = \pm\sqrt{x+1}$
- (c) $f(x) = x^2 + 4x + 30$
- (d) $f(x) = 100$

[Ans: (b)]

AQ 5.2

Questions

1. Which of the following statements is(are) TRUE?

- (a) If any horizontal line intersects the graph of a function f in at most one point, then f is one-to-one.
- (b) If any horizontal line intersects the graph of a function f in at least one point, then f is one-to-one.
- (c) If f is a decreasing function, then f is not one-to-one.
- (d) If f is an increasing function, then f is one-to-one.

[Ans: (b), (d)]

2. If $f : \mathbb{R} \rightarrow \mathbb{R}$, then which of the following functions is(are) one-to-one?

- (a) $f(x) = x + 1 - 20$
- (b) $f(x) = x^2 + 4x$
- (c) $f(x) = x^3 + 15$
- (d) $f(x) = x^3 - 5x^2 + 2x + 8$

[Ans: (c)]

3. Which of the following functions is(are) decreasing?

- (a) $f(x) = 4x^2 + 4x + 1$, for all $x \in \mathbb{R}$
- (b) $f(x) = 7 - 3x$, for all $x \in \mathbb{R}$
- (c) $f(x) = -2x^3$, for all $x \in \mathbb{R}$
- (d) $f(x) = \frac{1}{x}$, for all $x \in (\mathbb{R} \setminus \{0\})$

[Ans: (b), (c)]

4. Which of the following statements is(are) INCORRECT?

- (a) The function $f(x) = \frac{4x+3}{3x-5}$, for all $x \in (\mathbb{R} \setminus \{\frac{5}{3}\})$ is one-to-one.
- (b) The function $f(x) = mx + c$, for all $x \in (\mathbb{R} \setminus \{0\})$ and $c, m \in \mathbb{R}$ is one-to-one.
- (c) A quadratic function having two distinct roots is always one-to-one.
- (d) A quadratic function having equal roots can be one-to-one.

[Ans: (c), (d)]

AQ 5.3

Questions

1. Which of the following options is(are) TRUE?

- (a) $a^s \cdot a^t = a^{s+t}$, for $s, t \in \mathbb{R}$ and $a \in \mathbb{R}$
- (b) $(a^s)^t = a^{st}$, for $s, t \in \mathbb{R}$ and $a > 0$
- (c) $(ab)^s = a^s b^s$, for $s \in \mathbb{R}$ and $a, b > 0$
- (d) $a^s \times b^s = (a + b)^s$, for $s \in \mathbb{R}$ and $a, b > 0$
- (e) $a^s \cdot a^t = a^{s+t}$, for $s, t \in \mathbb{R}$ and $a > 0$

[Ans: (c),(e)]

2. Which of the following equations is(are) CORRECT?

- (a) $2^3 \times 2^4 = 2^{12}$
- (b) $2^3 \cdot 2^4 = 2^7$
- (c) $3^5 + 9^3 = 9^{11}$
- (d) $7^4 + 5^1 = 35^9$

[Ans: (b), (c)]

3. Simplify the expression $(2^2 \times 3^3)^5 \times 4^7 \times 5^3 \times (8 \times 25^3)^2$.

- (a) 60^{15}
- (b) 120^{15}
- (c) 30^{30}
- (d) 30^{15}

[Ans: (a)]

4. Which of the following statements is(are) INCORRECT?

- (a) Every exponential function is a one-to-one function.
- (b) 0^0 is undefined.
- (c) $a^0 = 1$, for all $a \in R$.
- (d) a is the exponent in the algebraic expression a^r .

[Ans: (c), (d)]

5. Which of the following options is not an exponential function?

(a) $f(x) = 3^{x/2}$

(b) $f(x) = x^{9/4}$

(c) $f(x) = \frac{15}{8^x}$

(d) $f(x) = 20 \times 6^x$

[Ans: (b)]

AQ 5.4

Questions

1. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = 3^x + 20$. Which of the following is the domain and range of the function f respectively?

- (a) $\mathbb{R}$ and $(0, \infty)$
- (b) $\mathbb{R}$ and $\mathbb{R}$
- (c) $(0, \infty)$ and $(20, \infty)$
- (d) $\mathbb{R}$ and $(20, \infty)$

[Ans: (d)]

2. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = 2^x - 32$. Which of the following is(are) true about $f(x)$?

- (a) x -intercept is $(5, 0)$.
- (b) y -intercept is $(0, -32)$.
- (c) $y = -31$ is the horizontal asymptote.
- (d) $y = -32$ is the horizontal asymptote.

[Ans: (a), (d)]

3. Which of the following statements is(are) INCORRECT?

- (a) $f(x) = 8^{-x}$ is a decreasing function.
- (b) Every $f(x) = a^x$, $a > 1$ has same properties as $g(x) = 6174^x$.
- (c) $f(x) = 20^{-x}$ is an increasing function.
- (d) Every $f(x) = a^x$, $a < 1$ has same properties as $g(x) = 1729^x$.

[Ans: (c), (d)]

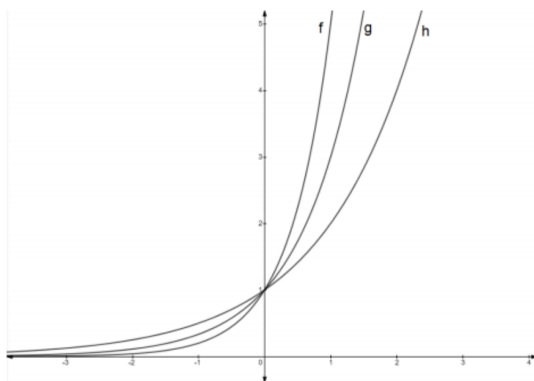
4. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = \frac{1}{7^x} + 15$. Which of the following is true about $f(x)$?

- (a) End behaviour of $f(x)$ as $x \rightarrow -\infty$ is 0.
- (b) End behaviour of $f(x)$ as $x \rightarrow +\infty$ is 15.
- (c) End behaviour of $f(x)$ as $x \rightarrow -\infty$ is 15.
- (d) End behaviour of $f(x)$ as $x \rightarrow +\infty$ is 0.

[Ans: (b)]

5. Observe Figure 1 and choose the correct set of options:

- (a) The curves f, g, h may represent $2^x, 3^x, 5^x$.
- (b) The y -intercepts of f, g, h are 2, 3, 5 respectively.
- (c) The curves f, g, h may represent $5^x, 3^x, 2^x$.
- (d) The x -intercepts of f, g, h are 1, 1, 1 respectively.



[Ans: (c)]

AQ 5.5

Questions

1. Suppose that a stock's price continues to increase at the rate of 5% per year. If the value of one share of this stock is \$1000 at present, then find the value S of one share of this stock two years from now. (Use the formula $S = a(1+r)^n$, where r is the increase rate per year, n is the number of years from now, a is the present value of one share of the stock, and S is the value of one share of the stock after n years from now.)

- (a) 1157.625
- (b) 1000
- (c) 1050
- (d) 1102.5

[Ans: (d)]

2. Which of the following options yield the value "e"?

- (a) The slope of the tangent line of the curve $f(x) = e^x$ at $x = 1$
- (b) The slope of the tangent line of the curve $f(x) = e^x + c$ at $x = 1$
- (c) The area under the curve $f(x) = 2^x$, from $(-\infty, 1)$
- (d) The area under the curve $f(x) = e^x$ from $(-\infty, 1)$

[Ans: (a), (d)]

3. Which of the following statements is(are) CORRECT?

- (a) If $n \rightarrow \infty$, then $(1 + \frac{1}{n})^n \rightarrow e$
- (b) If $n \rightarrow \infty$, then $(1 + \frac{1}{n})^n = e$
- (c) Every $f(x) = a^x$, $a < 1$, has same properties as $g(x) = e^x$
- (d) Every $f(x) = a^x$, $a > 1$, has same properties as $g(x) = e^x$

[Ans: (b), (d)]

4. In 1990, the population of blue whales in the world is approximately 3200. Let $F(t)$ be the approximate population of blue whales in the world after t years since 1990 ($t = 0$ corresponds to 1990), defined as

$$F(t) = 3200e^{-0.15t}.$$

What is the approximate population of blue whales in the world in the year 2020?

- (a) 360
- (b) 36
- (c) 12
- (d) 160

[Ans: (b)]

5. In 1990, the population of blue whales in the world is approximately 3200. Let $F(t)$ be the approximate population of blue whales in the world after t years since 1990 ($t = 0$ corresponds to 1990), defined as

$$F(t) = 3200e^{-0.15t}.$$

In which year will the population of blue whales approximately be 160? (HINT: Check from the given options)

- (a) 2000
- (b) 2005
- (c) 2010
- (d) 2020

[Ans: (c)]

6. Suppose the population of bacteria in a culture is growing exponentially. At today 2:00 pm, 100 bacteria were present and by 5 : 00 pm, 448 bacteria were present. Find an exponential function

$$y = ae^{kt},$$

that models this growth, where t is the number of hours since 2 : 00 pm ($t = 0$ corresponds to 2 : 00 pm). Here, y is the population of bacteria at time t , and a, k are arbitrary constants. (HINT: Check from given options)

- (a) $y = 100e^{0.5t}$
- (b) $y = 448e^{0.54t}$
- (c) $y = 100e^{2t}$
- (d) $y = 448e^{2t}$

[Ans: (a)]

AQ 5.6

Questions

1. Let $f(x)$ and $g(x)$ be two functions. Which of the following options is(are) INCORRECT?

- (a) $(f \circ g)(x) = f(g(x))$
- (b) $(g \circ f)(x) = g(f(x))$
- (c) $(f \circ g)(x) = g(f(x))$
- (d) $(g \circ f)(x) = f(g(x))$

[Ans: (c), (d)]

2. Suppose $f(x)$ and $g(x)$ are well-defined functions. Which of the following statements is(are) CORRECT?

- (a) For any given functions $f(x)$ and $g(x)$, $(f \circ g)(x) = (g \circ f)(x)$.
- (b) The domain of a composite function $(f \circ g)(x)$ is always a subset of the domain of the function $g(x)$.
- (c) The domain of a composite function $(f \circ g)(x)$ is always equal to the domain of the function $g(x)$.
- (d) The range of a composite function $(f \circ g)(x)$ is always a subset of the range of the function $f(x)$.

[Ans: (b), (d)]

3. Suppose $f(x) = 8x$, $g(x) = 50 - 8x$, $h(x) = 50$, and $k(x) = 50 - 64x$ are functions. Which of the following options is true?

- (a) $(f \circ g)(x) = k(x)$
- (b) $(g \circ f)(x) = k(x)$
- (c) $(g \circ f)(x) = h(x)$
- (d) $(f \circ g)(x) = h(x)$

[Ans: (b)]

AQ 5.7

Questions

1. If $f(x) = 10(x - 5)^2 + 100x - 225$ and $g(x) = \sqrt{2x + 5}$, then the composite functions $f(g(x))$ and $g(f(x))$ are respectively:

- (a) $20x + 75$ and $\sqrt{20x^2 + 5}$
- (b) $\sqrt{20x + 55}$ and $20x^2 + 25$
- (c) $20x + 75$ and $20x^2 + 55$
- (d) $20x + 75$ and $\sqrt{20x^2 + 55}$

[Ans: (d)]

2. Suppose $f(x) = 3x + 10$ and $g(x) = \sqrt{x + 11}$ are two functions. Find the value of $f(g(5))$.

[Ans: (22)]

3. Suppose $f(x) = 3x + 10$ and $g(x) = \sqrt{x + 11}$ are two functions. Find the value of $g(f(5))$.

[Ans: (6)]

4. Which of the following pairs of functions $f(x)$ and $g(x)$ satisfies the equation $f(g(x)) = g(f(x))$?

- (a) $f(x) = 2x$ and $g(x) = 10x + 3$
- (b) $f(x) = 7x + 6$ and $g(x) = 4x + 3$
- (c) $f(x) = e^x$ and $g(x) = x$
- (d) $f(x) = 4e^x$ and $g(x) = 4x$

[Ans: (b), (c)]

5. If $f(x) = 2\sqrt{x + 9}$, $g(x) = 8x^2$, and $h(x) = 5x + 7$, then which of the following options are CORRECT?

- (a) $f(g(0)) < g(f(0))$
- (b) $h(f(0)) > f(h(0))$
- (c) $f(g(x)) < g(f(x))$, for all $x \in (-\infty, \infty)$
- (d) $f(h(x)) < h(f(x))$, for all $x \in [0, \infty)$
- (e) $h(g(1)) > g(h(1))$

[Ans: (a), (b), (d)]

AQ 5.8

Questions

1. If $f(x) = \frac{x+5}{x-4}$ and $g(x) = \frac{1}{x+1}$, then find $(f \circ g)(x)$.

- (a) $\frac{x-4}{2x+1}$
- (b) $\frac{2x+1}{x-4}$
- (c) $\frac{-5x-6}{4x+3}$
- (d) $\frac{5x+6}{4x-3}$

[Ans: (c)]

2. If $f(x) = \frac{x+5}{x-4}$ and $g(x) = \frac{1}{x+1}$, then find the domain of the function $(f \circ g)(x)$.

- (a) $\mathbb{R} \setminus \{-1, -\frac{3}{4}, 4\}$
- (b) $\mathbb{R} \setminus \{-1, -\frac{3}{4}\}$
- (c) $\mathbb{R} \setminus \{-\frac{3}{4}, 4\}$
- (d) None of these

[Ans: (b)]

3. Which of the following statements is(are) CORRECT?

- (a) If there exists $x \in \mathbb{R}$ that is not in the domain of a function f , then that x will not be in the domain of some composite function $f \circ g$.
- (b) The domain of a composite function $f(g(x))$ is the set of all x such that x is in the domain of a function f and $f(x)$ is in the domain of a function g .
- (c) If there exists $x \in \mathbb{R}$ that is not in the domain of a function g , then that x will not be in the domain of some composite function $f(g(x))$.
- (d) The domain of a composite function $f(g(x))$ is the set of all x such that x is in the domain of a function g and $g(x)$ is in the domain of a function f .

[Ans: (c),(d)]

4. If the domain of a composite function $f(g)$ is $(-2, +\infty)$ and $f(x) = \frac{1}{x^2}$, then which of the following options can be $g(x)$?

(a) $\frac{\sqrt{x+2}}{10}$

(b) $\frac{\sqrt{x+2}}{5x}$

(c) $\frac{3}{\sqrt{x+2}}$

(d) $\sqrt{2-x}$

[Ans: (a), (c)]

AQ 5.9

Questions

1. Which of the following options is(are) CORRECT for a reversible function f ?
 - (a) For any function f , $f^{-1}(x) = \frac{1}{f(x)}$
 - (b) The domain of a function f is always equal to the range of f^{-1} function.
 - (c) For any $x = a$ in the domain of a function f , if $(a, f(a))$ is on the graph of f , then $(f(a), a)$ is on the graph of f^{-1} .
 - (d) For any function f , $(f \circ f^{-1})(x) = x$.

[Ans: (b), (c), (d)]

2. If $f(x) = \sqrt{x-5}$, then the domain and range of the function f^{-1} are respectively:
 - (a) $[0, \infty)$ and $[5, \infty)$
 - (b) $[5, \infty)$ and $[0, \infty)$
 - (c) $\mathbb{R}$ and $[5, \infty)$
 - (d) $[5, \infty)$ and $\mathbb{R}$

[Ans: (a)]

3. Which of the following functions satisfy the condition $f(x) = f^{-1}(x)$?
 - (a) $f(x) = \frac{1}{x^3}$
 - (b) $f(x) = \frac{1}{x}$
 - (c) $f(x) = \frac{2x-1}{3x-2}$
 - (d) $f(x) = 9 - 5x$
 - (e) $f(x) = -x$

[Ans: (b), (c), (e)]

4. Which of the following statements is INCORRECT?

- (a) The inverse function $g(x)$ of a function $f(x) = \frac{1}{x^3}$ is $g(x) = \frac{1}{x^3}$.
- (b) If $f(x) = \frac{1}{x-11}$ and $g(x) = 11 - \frac{1}{x}$, then $f(x) = g^{-1}(x)$ for all $x \in \mathbb{R}$.
- (c) For any function f , $(f^{-1}f)(x) = x$, for all x in the domain of the function f .
- (d) A function f is symmetric to function f^{-1} about the line $y = x$.

[Ans: (b)]

5. If $f(x) = x^3 - 5$, $g(x) = x^{1/3} + 5$, $h(x) = (x - 5)^3$, and $k(x) = (x + 5)^{1/3}$ are functions, then which of the following options is(are) true?

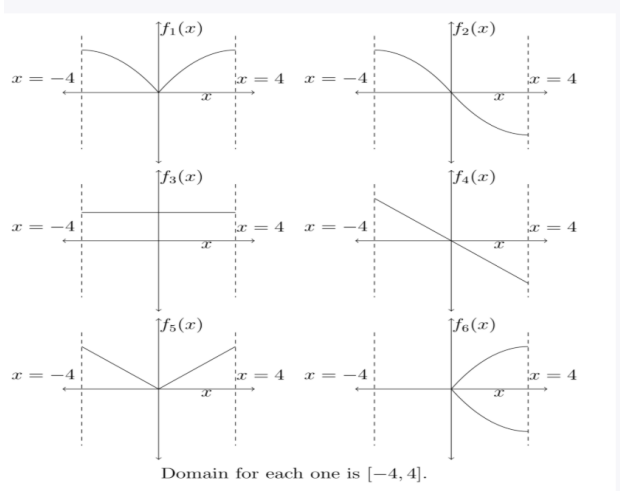
- (a) $k(x) = f^{-1}(x)$, for all $x \in \mathbb{R}$
- (b) $f(x) = h^{-1}(x)$, for all $x \in \mathbb{R}$
- (c) $g(x) = k^{-1}(x)$, for all $x > 0$
- (d) $h(x) = g^{-1}(x)$, for all $x > 0$

[Ans: (a),(d)]

PA 5

Questions

Answer the questions 1, 2, and 3 based on the given graphs.



1. Choose the correct option.

- (a) f_3 is not a function.
- (b) f_6 is not a function.
- (c) f_5 is not a function.
- (d) All of the above are functions.

[Ans: (b)]

2. Choose the correct option.

- (a) f_1 and f_3 are one-one functions in the given domain.
- (b) f_2 and f_4 are one-one functions in the given domain.
- (c) f_3 and f_5 are one-one functions in the given domain.
- (d) f_5 is one-one function in the given domain.

[Ans: (b)]

3. Choose the correct option.

- (a) f_1 and f_5 are strictly increasing functions in the given domain.
- (b) f_2 and f_4 are strictly decreasing functions in the given domain.
- (c) f_4 and f_5 are strictly decreasing functions in the given domain.

(d) f_5 is strictly increasing function in the given domain.

[Ans: (b)]

Use the below information for questions 4 and 5.

Let N_0 be the number of atoms of a radioactive material at the initial stage i.e., at time $t = 0$, and $N(t)$ be the number of atoms of the same radioactive material at a given time t , which is given by the equation:

$$N(t) = N_0 e^{-\lambda t},$$

where λ is the decay constant

4. If at time t_1 , the number of atoms reduces to half of N_0 , and at time t_2 , the number of atoms reduces to one-fourth of N_0 , then which one of the following equations is correct?

- (a) $\frac{t_1}{t_2} = 2$
- (b) $\frac{t_2}{t_1} = 2$
- (c) $e^{\lambda(t_2 - t_1)} = 2$
- (d) $e^{\lambda(t_1 - t_2)} = 2$

[Ans: (c)]

5. If $N_{\frac{1}{\lambda}}$ is the number of atoms at time $t = \frac{1}{\lambda}$, then what will be the ratio of N_0 to $N_{\frac{1}{\lambda}}$?

- (a) $1 : e$
- (b) $e : 1$
- (c) $1 : e^{-\lambda}$
- (d) $1 : e^{\lambda}$

[Ans: (b)]

6. Selvi deposits P in a bank A, which provides an interest rate of 10

- (a) $M_A(x)$ is a one-one function of x , for $x \in (0, 10)$.
- (b) $M_B(y)$ is a one-one function of y .
- (c) $M_A(12) = P \times 1.1^{12}$.
- (d) $M_A(12) = 0$.
- (e) $M_A(x)$ is a strictly increasing function for $x \in (0, 10)$.
- (f) $M_B(y)$ is a decreasing function of y .
- (g) $M_B(n) = (P \times 1.1^{10}) \times (1.125)^n$.
- (h) $M_B(n) = (P \times 1.1^n) \times (1.125)^{10}$.

[Ans: (a, b, d, e, g)]

Use the below information for questions 7 and 8.

There are two offers in a shop. In the first offer, the discount in total payable amount is $M(n)\%$ if the number of products bought at a time is n . The second offer involves a discount of ₹1000 on the total payable amount. If Geeta shops for ₹15,000, then answer the following questions:

7. If the total payable amounts after applying the first and second offers (one at a time) are represented by the functions $f(n)$ and $g(n)$ respectively, and the total payable amount after applying both the offers together is represented by $T(n)$, then choose the set of correct options.
- (a) $f(n) = (100 - M(n)) \times 15000$ and $g(n) = 14000$
 - (b) $f(n) = (100 - M(n)) \times 1500$ and $g(n) = (100 - M(n)) \times 15000 - 1000$
 - (c) $f(n) = (100 - M(n)) \times 150$ and $g(n) = 14000$
 - (d) $T(n) = (100 - M(n)) \times 15000$ is the total payable amount when the first offer is applied after the second.
 - (e) $T(n) = (100 - M(n)) \times 140$ is the total payable amount when the first offer is applied after the second.
 - (f) $T(n) = (100 - M(n)) \times 150 - 1000$ is the total payable amount when the second offer is applied after the first.

[Ans: (c, e, f)]

8. If Geeta is allowed to use the offer in any sequence and $M(n) = -n^2 + 18n - 72$, where $n \in \{6, 7, 8, 9\}$, then choose the set of correct options which minimizes the total payable amount.
- (a) Total payable amount is same irrespective of the order in which the offers are applied.
 - (b) She should choose offer one and then offer two i.e., $f(g(M(n)))$.
 - (c) She should choose offer two and then offer one i.e., $g(f(M(n)))$.
 - (d) If she chooses offer one and then offer two, the minimum payable amount will be ₹12650.

[Ans: (b, d)]

Use the below information for questions 9-13.

Given two real-valued functions $f(x) = \frac{5x+9}{2x}$, $g(y) = \sqrt{y^2 - 9}$, and $h(x) = f(g(x))$

9. if the domain of $f(x)$ and $g(x)$ are $(-\infty, m) \cup (m, \infty)$ and $\mathbb{R} \setminus (-n, n)$ respectively, then find the value of $m + n$.

[Ans: (3)]

10. If the range of $f(x)$ and $g(x)$ are $(-\infty, m) \cup (m, \infty)$ and $[n, \infty)$ respectively, then find the value of $2(m + n)$.

[Ans: (5)]

11. If the domain of $h(x)$ is $(-\infty, -3) \cup (m, \infty)$, then find the value of m .

[Ans: (3)]

12. If the domain of $f^{-1}(x)$ is $(-\infty, m) \cup (m, \infty)$, then find the value of $2m$.

[Ans: (5)]

13. If $f^{-1}(5) = 9/m$, then find the value of m .

[Ans: (5)]

END OF WEEK 5

AQ 6.1

Questions

1. For which point on the Cartesian plane, will the equations $x = \log_2 y = \log_{0.2} y$ be valid?

- (a) (1, 1)
- (b) (0, 1)
- (c) (2, 1)
- (d) (0, 2)

[Ans: (b)]

2. The domain of $f(x) = \log_2(2 - x)$ is:

- (a) $(-\infty, 1)$
- (b) $(-\infty, 2)$
- (c) $(\infty, 1]$
- (d) $(\infty, 2]$

[Ans: (b)]

3. Consider the function $f(x) = \log_{10}((\log_{10} x)^2 - 8 \log_{10} x + 15)$. What is the domain of $f(x)$?

- (a) $(0, 10^{-5}) \cup (10^{-3}, \infty)$
- (b) $(0, 10^{-5}] \cup [10^{-3}, \infty)$
- (c) $(0, 10^3) \cup (10^5, \infty)$
- (d) $(0, 10^3] \cup [10^5, \infty)$

[Ans: (c)]

4. The set of all x for which there are no functions of the form $f(x) = \log_{\frac{x-1}{x+1}}\left(\frac{x^2+1}{x^2-1}\right)$ is:

- (a) $(-\infty, 0) \cup (0, \infty)$
- (b) $[-1, 1]$
- (c) $(-1, 1)$
- (d) $(-\infty, -1] \cup [1, \infty)$

[Ans: (b)]

5. What is the domain of the function $f(x) = \log(-(xe^{3x} - 27x))$?

- (a) $(0, \ln 27)$
- (b) $(0, \ln 2)$
- (c) $(0, \ln 3)$
- (d) $(0, \ln 3 - \ln 2)$

[Ans: (c)]

AQ 6.2

Questions

1. The only difference between the graphs of $f(x) = \log_2 x$ and $g(x) = \log_2(x - 1)$ will be:
 - (a) The ordinate values of $g(x)$ will be 1 more than the ordinate values of $f(x)$.
 - (b) The abscissa values of $g(x)$ will be 1 more than the abscissa values of $f(x)$.
 - (c) Both abscissa and ordinate values of $g(x)$ will be 1 more than the abscissa and ordinate values of $f(x)$, respectively.
 - (d) There will be no difference.

[Ans: (b)]

2. Which of the following statements is(are) true for the function $f(x) = \ln |x|$?
 - (a) f is an increasing function.
 - (b) The straight line $x = 0$ is an asymptote of f .
 - (c) The inverse of f is $e^{|x|}$.
 - (d) The domain of f is $(-\infty, 0) \cup (0, \infty)$.

[Ans: (b), (d)]

3. Consider the function $f(x) = \log_2(\log_2(\log_2(x)))$. Which of the following is(are) true about f ?
 - (a) f is one-to-one.
 - (b) The domain of f is $(0, \infty)$.
 - (c) f has an inverse.
 - (d) The straight line $x = 8$ is an asymptote of f .

[Ans: (a), (c)]

AQ 6.3

Questions

1. If the domain of $f(x)$ is $(-3, 1)$, then the domain of $f(\ln x)$ is:

- (a) (e^{-1}, e^3)
- (b) $(0, \infty)$
- (c) $(1, \infty)$
- (d) (e^{-3}, e^1)

[Ans: (d)]

2. Consider the function $f(x) = \ln(e^x - e^{-x})$. Find the correct options:

- (a) f is a one-one function.
- (b) The domain of f is $\mathbb{R}$.
- (c) f is an increasing function.
- (d) None of the above.

[Ans: (a), (c)]

3. Solve for x in $3^{x+2} = 243$:

- (a) 4
- (b) 3
- (c) 2
- (d) 1

[Ans: (b)]

4. Solve for x in $e^{(4^x - 2^x - 12)} = 1$:

- (a) -1
- (b) 2
- (c) 1
- (d) -2

[Ans: (b)]

AQ 6.4

Questions

1. For $0 < a < 1$ and $M > 0$, $\log_a(M^2) = ?$

- (a) 1
- (b) $2\log_a M$
- (c) $2\log_M a$
- (d) $\log_a 3M$

[Ans: (b)]

2. Solve for x : $\log_{12}(x - 2) + \log_{12}(x + 2) = 1$

- (a) $x = -4$
- (b) $x = 4$
- (c) $x = 2$
- (d) $x = -2$

[Ans: (b)]

3. The number of solutions of $\ln\left(\frac{x^2}{4}\right) - \ln(x - 1) = \log_5 5$ is:

- (a) 0
- (b) 1
- (c) 2
- (d) 3

[Ans: (c)]

AQ 6.5

Questions

1. How many digits are there in 15^{72} ?

- (a) 49
- (b) 50
- (c) 57
- (d) 58

[Ans: (d)]

2. Suppose a certain amount of money M is invested in a mutual fund at an annual rate of interest of 5%. How long (approximately) does it take to triple the initial investment, assuming interest is compounded thrice a year?

- (a) 11 years
- (b) 14 years
- (c) 19 years
- (d) 22 years

[Ans: (d)]

AQ 6.6

Questions

1. If $\ln(x^5) = (\ln x)^2$, then the value of $\ln x$ is:

- (a) 2
- (b) 5
- (c) -5
- (d) -2

[Ans: (b)]

2. The number of solutions of $\log_8(x^3 - 1) = \log_2(x - 1)$ is:

- (a) 0
- (b) 1
- (c) 3
- (d) ∞

[Ans: (a)]

AQ 6.7

Questions

1. If the value of $\left(\frac{1}{\log_3 \pi} + \frac{1}{\log_4 \pi}\right)$ is m , then which of the following options are true for m ?

- (a) $m \geq 3.178$
- (b) $2 < m < 2.2$
- (c) $m \leq 2$
- (d) $m < \log_2 2$

[Ans: (b)]

2. Consider the function $f(x) = \log_{10}(x - \lfloor x \rfloor)$, where $\lfloor x \rfloor$ is the greatest integer less than or equal to x , and $D \subseteq \mathbb{R}$ is the set of points at which f is defined. Which of the following options are correct?

- (a) The set D is infinite.
- (b) The cardinality of the set $\mathbb{R} \setminus D$ is infinite.
- (c) The graph of $f(x)$ has an infinite number of vertical asymptotes.
- (d) f is an invertible function on D .
- (e) The range of the function f is $\mathbb{R}$.

[Ans: (a), (b), (c)]

3. Suppose x and y are positive real numbers. If $\log_x(2) + \log_y(8) = 0$, then which of the following options are true for x and y ?

- (a) $x = 2, y = 1/8$
- (b) $xy^3 = 1$
- (c) $x = 27, y = 1/3$
- (d) $x^3y = 1$

[Ans: (a), (d)]

PA 6

Questions

1. If $b > 0$ and $4 \log_x b + 9 \log_{b^5 x} b = 1$, then the possible value(s) of x is(are):

- (a) b^{10}
- (b) b^9
- (c) b^{-2}
- (d) b^5
- (e) b^4

[Ans: (a), (c)]

2. George deposits ₹2L in a bank that compounds quarterly at the rate of 20 percent per year. How long will it take to increase his money to 16 times the principal amount (in years)?

- (a) $\frac{\ln 16}{4}$
- (b) $\frac{\ln 16}{4 \ln \frac{21}{20}}$
- (c) $\frac{\ln 2}{\ln \frac{21}{20}}$
- (d) $\log_{\frac{21}{20}} 2$
- (e) $\frac{\ln 2^{\frac{1}{4}}}{\ln \frac{21}{20}}$

[Ans: (b), (c), (d)]

3. Choose the set of correct options.

- (a) $\log_5 2$ is a rational number.
- (b) If $0 < b < 1$ and $0 < x < 1$ then $\log_b x < 0$.
- (c) If $\log_3(\log_5 x) = 1$, then $x = 125$.
- (d) If $0 < b < 1$, $0 < x < 1$, and $x > b$, then $\log_b x > 1$.
- (e) If $0 < b < 1$ and $0 < x < y$ then $\log_b x > \log_b y$.

[Ans: (c), (e)]

4. Suppose that two types of insects are found in a pond. Their growth in number after t seconds is given by the equations:

$$f(t) = 5^{3t-2} \quad \text{and} \quad h(t) = 3^{2t+1} \quad (t \neq 0).$$

For what value of t will both insects be of the same number in the pond?

- (a) $\frac{\ln 3 + 2 \ln 5}{3 \ln 5 - 2 \ln 3}$
- (b) $\frac{\ln 75}{\ln \frac{125}{9}}$
- (c) $\log_{\frac{125}{9}} 75$
- (d) $\frac{\ln 5 + 2 \ln 3}{3 \ln 3 - 2 \ln 5}$

[Ans: (a), (b), (c)]

5. Which of the following is/are true?

- (a) If m and n are positive real numbers, then $m^{\log(n)} = n^{\log(m)}$.
- (b) $\log_5(123456789999999999)$ is an irrational number.
- (c) The function $f(x) = \log_{10}(x^2 + x + 1)$ is one-to-one on the interval $(-0.5, +\infty)$.
- (d) None of the above.

[Ans: (a), (c)]

6. Which of the following is/are true?

- (a) Suppose D is an arbitrary subset of $\mathbb{R}$ and f is a one-one function on D . $\log f(x)$, whenever defined, is also a one-one function on D .
- (b) $(14!)^{\frac{1}{14}} < (15!)^{\frac{1}{15}}$, where "!" denotes the factorial function, and for a non-negative integer n , the value of $n!$ is $n \times (n-1) \times \cdots \times 2 \times 1$.
- (c) $f(x) = 2^x + 3^x + \cdots + 100^x$ is one-one function on $\mathbb{R}$.
- (d) There exists a function $f(x)$ on $\mathbb{R}$, such that $\log(f(x)) \geq 100$ for all $x \in \mathbb{R}$.

[Ans: (a), (b), (c), (d)]

7. If $\log_{\sqrt{2}}(x+4) - \log_2(\frac{1}{2}x+2) = 1$, then x is:

- (a) -3
- (b) 1
- (c) -4
- (d) 5

[Ans: (a)]

8. Seismologists use the Richter scale to measure and report the magnitude of earthquakes as given by the equation:

$$R = \ln I - \ln I_0,$$

where I is the intensity of an earthquake with respect to a minimal or reference intensity I_0 ($I = cI_0$, where c is a constant). The reference intensity is the smallest earth

movement that can be recorded on a seismograph. If an earthquake in city A recorded a magnitude of 8.0 in the Richter scale and the intensity of the earthquake in city B is the reference intensity, then what is the ratio of intensity of earthquake in city A with respect to city B?

- (a) $e^0 : 1$
- (b) $e^1 : 2$
- (c) $e^8 : 1$
- (d) $e^5 : 1$
- (e) $e^8 : 2$

[Ans: (c)]

9. Suppose that the number of bacteria present in a loaf of rotten bread after t minutes is given by the equation:

$$G(t) = G_0 3^{kt},$$

where G_0 represents the number of bacteria at $t = 0$, k is a constant (Given $\ln 730 = 6.59$ and $\ln 3 = 1.09$). If the initial number of bacteria is 1000 and it takes 1 min to increase to 9000, then how long (in minutes) would it take for the bacteria count to grow to 730000 (integer value of t)?

- (a) 2
- (b) 1
- (c) 3
- (d) 6

[Ans: (c)]

10. (Answer the questions 10 and 11 based on below information.)

Let c_A and c_B be the luminosity (luminous efficacy) of the bulbs A and B , respectively. The bulb A is $f(x)$ times brighter than the bulb B . If $f(x) = 3^{x^2+1}$ (i.e., $c_A = f(x) \times c_B$), where x is the difference of the magnitude of supply voltage between the bulb A and the bulb B .

If the bulb A is 10 times brighter than the bulb B , then the difference of the magnitude of supply voltage between the two bulbs is:

- (a) $\sqrt{\log_3 5 - 1}$
- (b) $\sqrt{\log_3 10}$
- (c) $\sqrt{\frac{\ln 10}{\ln 3}}$
- (d) $\sqrt{\log_3 \frac{10}{3}}$

[Ans: (d)]

11. If 4 volts and 3 volts are the supply voltages for the bulbs A and B , respectively, then how many times the bulb A is brighter than the bulb B ?

[Ans: (9)]

12. Find the number of values of x satisfying the equation:

$$(5x)^{\log_{(5x)} \frac{1}{5}} (6x^3 - 36x^2 + 66x - 35)^{\frac{1}{5}} = 1$$

[Ans: (3)]

END OF WEEK 6

AQ 7.1

Questions

1. Which of the following may represent a function $f : \mathbb{R} \rightarrow \mathbb{R}$?

- (a) figure(a)
- (b) figure(b)
- (c) figure(c)
- (d) figure(d)

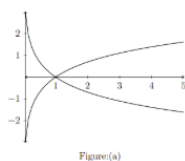


Figure: (a)

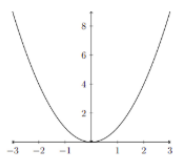


Figure: (b)

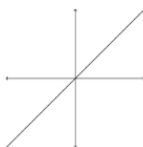


Figure: (c)

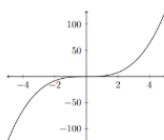


Figure: (d)

[Ans: (b),(c),(d)]

2. Which of the following represents a linear function?

- (a) $f(x) = x + 5$
- (b) $f(x) = 2x^2 + 5x + 10$
- (c) $f(x) = \ln(x^2 + 5)$
- (d) $f(x) = x$

(e) $f(x) = \frac{x}{5} + 3$

[Ans: (a), (d), (e)]

3. Which of the following represents a quadratic function?

(a) $f(x) = 7x + 10$

(b) $f(x) = 3x^2 + 9x + 199$

(c) $f(x) = \ln(2x^2 + 10)$

(d) $f(x) = e^2x$

(e) $f(x) = -x^2 + 1$

[Ans: (b), (e)]

4. Consider the function $f(x) = \frac{x-1}{x^2-5x+6}$. Then the number of points at which the function is not defined is:

[Ans: (2)]

5. Which of the following collections are sets?

(a) Collection of irrational numbers.

(b) Collection of rich people in India.

(c) Collection of movies directed by Rituparno Ghosh.

(d) Collection of novels written by George Orwell.

(e) Collection of good actors and actresses in India.

(f) Collection of good badminton players in India.

[Ans: (a), (c), (d)]

6. Which of the following option(s) is(are) true?

(a) All quadratic functions are polynomial functions.

(b) All polynomial functions are quadratic functions.

(c) All linear functions are polynomial functions.

(d) All quadratic functions are linear functions.

[Ans: (a), (c)]

7. Assume the following Figure M2W1AQ1 represents the graph of a function $f : \mathbb{R} \rightarrow \mathbb{R}$.

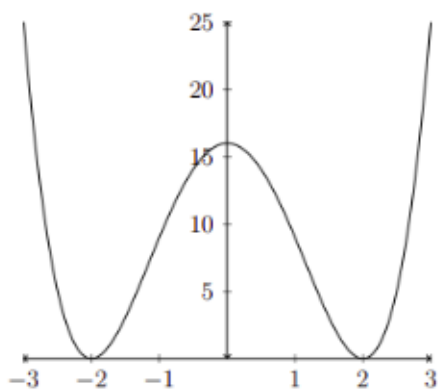


Figure M2W1AQ 1

Consider the following statements about the above graph:

- (a) In the interval $(-\infty, 2]$, the function is increasing.
- (b) In the interval $[-2, 2]$, the function is decreasing.
- (c) In the interval $[-2, 0]$, the function is increasing.
- (d) In the interval $[0, 2]$, the function is decreasing.
- (e) In the interval $[0, \infty)$, the function is increasing.
- (f) In the interval $[2, 3]$, the function is increasing.

Find the number of correct statements:

[Ans: (3)]

8. In the following Figure M2W1AQ2, C_1 , C_2 , and C_3 represent curves.

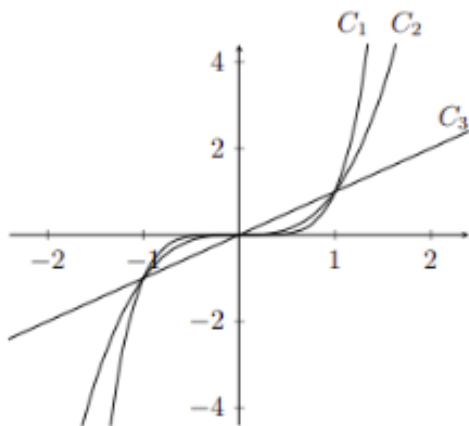


Figure M2W1AQ 2

Which of the following option(s) is(are) true?

- (a) Among these three curves, C_1 is the fastest growth curve.
- (b) Among these three curves, C_2 is the fastest growth curve.
- (c) Among these three curves, C_3 is the fastest growth curve.
- (d) C_1 is a faster growing curve than C_2 .
- (e) C_3 is a faster growing curve than C_2 .
- (f) C_2 is a faster growing curve than C_3 .

[Ans: (a), (d), (f)]

9. Which of the following option(s) is(are) true?

- (a) Any linear function is always an increasing function.
- (b) Any quadratic function is always an increasing function.
- (c) e^x is always an increasing function.
- (d) $\ln x$ is always an increasing function.

[Ans: (c), (d)]

10. Consider the following statements:

- (a) $f(x) = mx + c$ is a one-one function from $\mathbb{R}$ to $\mathbb{R}$, where m is an arbitrary non-zero integer.
- (b) $f(x) = x^2 + bx + c$ is an onto function from $\mathbb{R}$ to $\mathbb{R}$, where b and c are arbitrary real numbers.
- (c) $\frac{1}{e^x}$ is an unbounded function on $\mathbb{R}$.
- (d) The function $g(x) = \frac{1}{1+e^x}$ gives a bijection from $\mathbb{R}$ to $(0, 1)$.

Find the number of correct statements:

[Ans: (3)]

11. Consider the function $f(x) = x + \frac{1}{x}$. Which of the following options are correct?

- (a) f is an injective function, if the domain of f is $\mathbb{R} \setminus \{0\}$.
- (b) f is an injective function, if the domain of f is $\mathbb{N}$.
- (c) f is an injective function, if the domain of f is $\mathbb{Q} \setminus \{0\}$.
- (d) f is an injective function, if the domain of f is $\mathbb{Z} \setminus \{0\}$.
- (e) f is a surjective function, if the domain of f is $\mathbb{N}$ and codomain of f is $\mathbb{Q}$.
- (f) f is a surjective function, if the domain of f is $\mathbb{Z} \setminus \{0\}$ and codomain of f is $\mathbb{Q}$.

[Ans: (b), (d)]

AQ 7.2

Questions

1. Which of the following option(s) is(are) true?

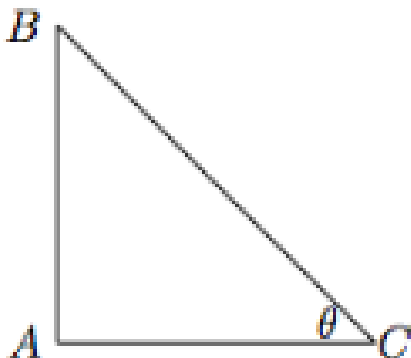


Figure M2W1AQ 3

- (a) $\sin \theta = \frac{AB}{AC}$
- (b) $\cos \theta = \frac{AB}{AC}$
- (c) $\tan \theta = \frac{AB}{BC}$
- (d) $\sin \theta = \frac{AC}{BC}$
- (e) $\cos \theta = \frac{AC}{BC}$
- (f) $\csc \theta = \frac{BC}{AB}$
- (g) $\sec \theta = \frac{BC}{AC}$
- (h) $\cot \theta = \frac{AB}{AC}$

[Ans:(c),(e),(f),(g)]

Consider two functions $f(x) = x^2 + 2x + 3$ and $g(x) = x + 1$, where $x \neq -1$. Compute f and g and answer the questions 2 and 3.

- 2. The value of $(3f/g)(2)$ is: [Ans:(11)]
- 3. The value of $(fg)(0)$ is: [Ans:(6)]
- 4. Suppose $\sin(x + y) = \sin x \cos y + \cos x \sin y$, where $0 < x, y < \frac{\pi}{2}$, then which of the following is(are) true?

- (a) $\sin 2x = 2 \sin x \cos x$
- (b) $\sin 2x = 2 \sin x \sin x$
- (c) $\sin 3x = 3 \sin x - 4 \sin^3 x$
- (d) $\sin 3x = \sin 2x \cos x - \cos 2x \sin x$

[Ans:(a),(c)]

5. Suppose $\cos(x + y) = \cos x \cos y - \sin x \sin y$, where $0 < x, y < \frac{\pi}{2}$, then which of the following is(are) true?

- (a) $\cos 2x = \cos^2 x - \sin^2 x$
- (b) $\cos 2x = \sin^2 x - \cos^2 x$
- (c) $\cos 3x = \cos 2x \cos x - \sin 2x \sin x$
- (d) $\cos 3x = 4 \cos^3 x - 3 \cos x$

Then, the number of correct statements is:

[Ans:(a),(c),(d)]

6. Consider the following statements:

$\sin(x)$ is a bounded function on $\mathbb{R}$.

$\cos(x)$ is an unbounded function on $\mathbb{R}$.

$\tan(x)$ is an unbounded function on its domain.

$\frac{1}{2+\cos(x)}$ is an unbounded function on its domain.

Then the number of correct statements is:

[Ans:(2)]

7. Which of the following is(are) true?

- (a) If $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{Q} \rightarrow \mathbb{R}$, then $g \circ f$ is well-defined.
- (b) If $f : \mathbb{N} \rightarrow \mathbb{Z}$ and $g : \mathbb{Q} \rightarrow \mathbb{R}$, then $g \circ f$ is well-defined.
- (c) If $f : \mathbb{R} \rightarrow (0, 1)$ and $g : (0, 1] \rightarrow \mathbb{R}$, then $g \circ f$ is well-defined.
- (d) If $f : \mathbb{Z} \rightarrow [0, 1]$ and $g : (0, 1) \rightarrow \mathbb{Q}$, then $g \circ f$ is well-defined.

[Ans:(b),(c)]

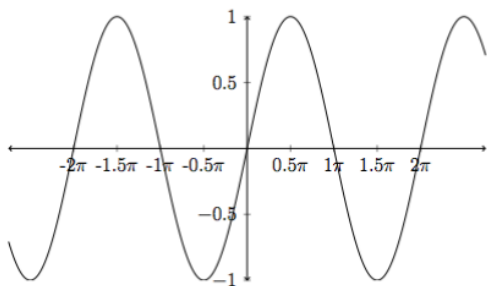


Figure M2W1AQ 4

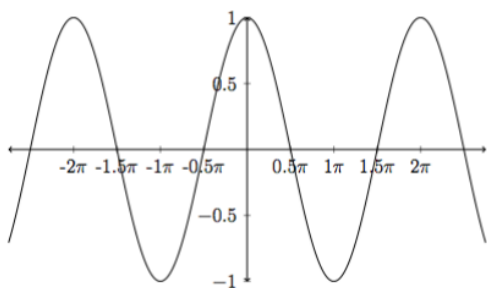


Figure M2W1AQ 5

Figure M2W1AQ 4 shows the graph of $\sin$ and Figure M2W1AQ 5 shows the graph of $\cos$. Use the information to answer the questions 8,9,10

8. Let $0 < \theta < \frac{\pi}{2}$, which of the following option(s) is(are) true?

Hint: Shift the curves in the figures by $\frac{\pi}{2}$ units.

- (Opt1on 1) $\sin\left(\frac{\pi}{2} + \theta\right) = \cos \theta$
 (Opt2on 2) $\sin\left(\frac{\pi}{2} + \theta\right) = -\cos \theta$
 (Opt3on 3) $\sin\left(\frac{\pi}{2} - \theta\right) = -\cos \theta$
 (Opt4on 4) $\sin\left(\frac{\pi}{2} - \theta\right) = \cos \theta$

[Ans:(a),(d)]

9. Let $0 < \theta < \frac{\pi}{4}$, which of the following option(s) is(are) true?

Hint: Shift the curves in the figures by $\frac{\pi}{2}$ units.

- (Opt1on 1) $\cos\left(\theta + \frac{\pi}{2}\right) = -\sin \theta$
 (Opt2on 2) $\cos\left(\theta + \frac{\pi}{2}\right) = \sin \theta$
 (Opt3on 3) $\cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta$
 (Opt4on 4) $\cos\left(\frac{\pi}{2} - \theta\right) = -\sin \theta$

[Ans:(b),(c)]

10. Let $0 < \theta < \frac{\pi}{2}$, which of the following option(s) is(are) true?

Hint: Shift the curves in the figures by π units.

(a) $\sin(\pi + \theta) = -\sin \theta$

(b) $\sin(\pi + \theta) = \sin \theta$

(c) $\sin(\pi - \theta) = \sin \theta$

(d) $\sin(\pi - \theta) = -\sin \theta$

(e) $\sin(\pi + \theta) = \cos \theta$

(f) $\sin(\pi + \theta) = \cos \theta$

[Ans:(a),(c)]

11. Let $f(x) = \frac{x}{x+a}$, where $x > 0$ and $a > 0$. If $f(f(x)) = \frac{x}{3x+4}$, then find the value of a .

[Ans:(2)]

AQ 7.3

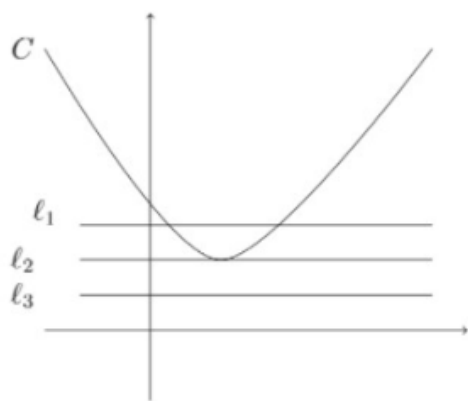
Questions

1. Consider a function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = x^2 + 1$. Which of the following options represents the graph of the function?

- (a) $\{(x, x^2 + 1) \mid x \in \mathbb{R}\}$
- (b) $\{(x, x^2) \mid x \in \mathbb{R}\}$
- (c) $\{(x, 1) \mid x \in \mathbb{R}\}$
- (d) The graph of f is a subset of $\mathbb{Z} \times \mathbb{R}$.
- (e) The graph of f is a subset of $\mathbb{R} \times \mathbb{R}$.

[Ans:(a),(e)]

2. Let C be a curve in the following Figure M2W1AQ 6.



Which of the following statements are correct about the curve C ?

- (a) ℓ_1 is the tangent line at some point to the curve C .
- (b) ℓ_2 is the tangent line at some point to the curve C .
- (c) ℓ_3 is not a tangent line at some point to the curve C .
- (d) None of the above.

[Ans:(b),(c)]

3. Let C be a curve in the following Figure M2W1AQ 7.

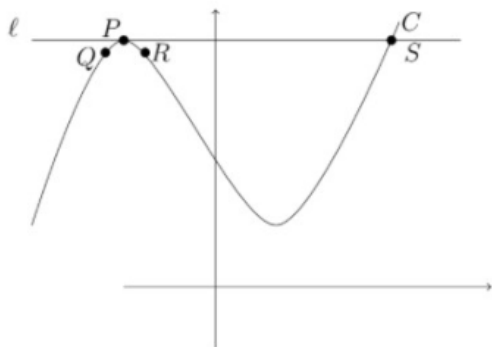


Figure M2W1AQ 7

Which of the following statements are correct about the curve C ?

- (a) ℓ_1 is the tangent line to the curve C at P .
- (b) ℓ_1 is the tangent line to the curve C at Q .
- (c) ℓ_1 is the tangent line to the curve C at R .
- (d) ℓ_1 is not a tangent line to the curve C at S .

[Ans:(a),(d)]

4. Let C be a curve and three straight lines ℓ_1 , ℓ_2 and ℓ_3 pass through the point P in the following Figure M2W1AQ 8.

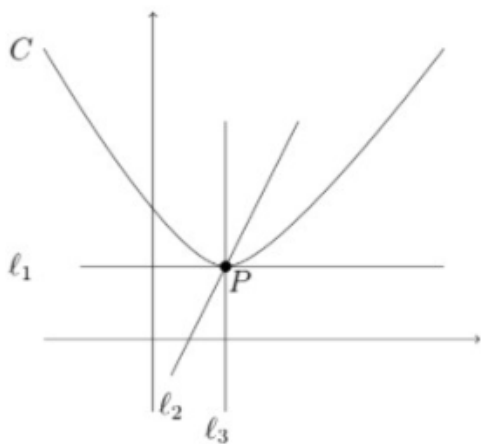


Figure M2W1AQ 8

Which of the following statements are correct about the curve C ?

- (a) ℓ_3 is not a tangent line to the curve C at P .
- (b) ℓ_2 is the tangent line to the curve C at P .
- (c) ℓ_1 is the tangent line to the curve C at P .

(d) None of the above.

[Ans:(a),(c)]

5. Which of the following options represents the graph of a quadratic function?

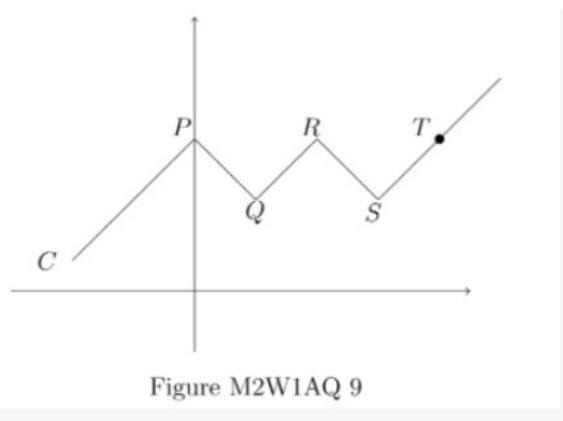
- (a) $\{(x, x^3 + 2x + 1) \mid x \in \mathbb{R}\}$
- (b) $\{(x, x^2 + 4x + 5) \mid x \in \mathbb{R}\}$
- (c) $\{(x, x^4 + 5x^2 + 3) \mid x \in \mathbb{R}\}$
- (d) $\{(x, 9x^2) \mid x \in \mathbb{R}\}$

[Ans:(b),(d)]

6. Let $f : A \rightarrow B$ be a function, where A and B are subsets of $\mathbb{R}$ and the cardinalities of the sets A and B are 4 and 4, respectively. What is the cardinality of $\Gamma(f)$, the graph of f ?

[Ans: (4)]

Let C represent a curve in the Figure M2W1AQ 9. Answer the questions 7 and 8 using the curve C .



7. Let C represent a curve in the Figure M2W1AQ 9.

Which of the following option(s) is(are) true?

- (a) At point P , there is no tangent.
- (b) At point Q , there is no tangent.
- (c) At point P , there are an infinite number of tangents.
- (d) None of the above.

[Ans: (a), (b)]

8. Let A be the set of points where the curve C has no tangents. Find a lower bound for the cardinality of the set A . [Ans: (4)]
9. Let $f : A \rightarrow B$ be a function, where A and B are finite subsets of $\mathbb{R}$. Which of the following options are correct?
- (a) The cardinality of $\Gamma(f)$, the graph of f , is the same as the cardinality of $A \times B$.
 - (b) The cardinality of $\Gamma(f)$, the graph of f , is the same as the cardinality of A .
 - (c) The cardinality of $\Gamma(f)$, the graph of f , is the same as the cardinality of B .
 - (d) The cardinality of $\Gamma(f)$, the graph of f , is less than or equal to the cardinality of $A \times B$.

[Ans: (b),(d)]

AQ 7.4

Questions

1. Consider the following functions, where $\mathbb{R}$ is the set of real numbers, $\mathbb{N}$ is the set of natural numbers, $\mathbb{Z}$ is the set of integer numbers, $\mathbb{R}^+$ is the set of positive real numbers, and $\mathbb{R}^-$ is the set of negative real numbers:

- (a) $f_1 : \mathbb{R} \rightarrow \mathbb{R}$
- (b) $f_2 : \mathbb{N} \rightarrow \mathbb{R}$
- (c) $f_3 : \mathbb{R}^- \rightarrow \mathbb{R}$
- (d) $f_4 : \mathbb{R}^+ \rightarrow \mathbb{R}$
- (e) $f_5 : \mathbb{N} \rightarrow \mathbb{Z}$

Which of the following represents a sequence?

[Ans: (b),(e)]

2. Let (a_n) be a sequence defined as $a_n = \frac{3n^2+5n+2}{6n^2+2n+1}$. The limit of the sequence (a_n) is:

[Ans: (0.5)]

3. Consider a sequence $(1, 2, 3, 4, 5, 6, \dots)$ that is $a_n = n$. Which of the followings are subsequence of a_n ?

- (a) $(2, 2, 2, 2, 2, \dots)$, that is $b_n = 2$ for all n .
- (b) $(2, 4, 6, 8, 10, \dots)$, that is $b_n = 2n$ for all n .
- (c) $(10, 4, 6, 8, 10, 12, \dots)$, that is $b_1 = 10$ and $b_n = 2n$ for all $n > 2$.
- (d) $(3, 5, 7, 9, 11, \dots)$, that is $b_n = 2n + 1$ for all n .

[Ans: (b), (d)]

4. Consider a sequence $\{n^2\}$. Which of the following options are subsequences of the given sequence?

- (a) $\{(n+5)^2\}$
- (b) $\{(2n+1)^2\}$
- (c) $\{(2n+1)^3\}$
- (d) $\{n^3\}$
- (e) $\{n^2\}$
- (f) $\{(2n)^2\}$
- (g) $\{(\frac{1}{2}n+1)^2\}$

[Ans: (a), (b), (e), (f)]

5. Consider a sequence $(1, -1, 1, -1, \dots)$, that is $a_n = 1$ if n is odd and $a_n = -1$ if n is even. Which of the following options are correct?

- (a) Limit of $\{a_n\}$ is 1.
- (b) Limit of $\{a_n\}$ is -1.
- (c) Limit of $\{a_n\}$ is both 1 and -1.
- (d) $(1, 1, 1, 1, \dots)$, that is $b_n = 1$ for all n , is a subsequence of a_n .
- (e) Limit of the sequence a_n does not exist.

[Ans: (d), (e)]

6. Let $\{a_n\}$ be a sequence converging to the limit 1. Then $\lim_{n \rightarrow \infty} a_{3n}$ is: [Ans: (1)]

7. Consider the sequence $\{a_n\}$, defined as $a_n = \sqrt{2n+1} - \sqrt{2n}$. Which of the following option(s) is(are) true?

- (a) $\{a_n\}$ is convergent.
- (b) Limit of $\{a_n\}$ is 0.
- (c) Limit of $\{a_n\}$ is 1.
- (d) Limit of $\{a_n\}$ is $+\infty$.

[Ans: (a), (b)]

8. Let $\{a_n\}$ be a sequence defined by $a_n = n^{\frac{1}{n}}$, has limit 1. Then which of the following option(s) is(are) true?

- (a) $\lim_{n \rightarrow \infty} \ln(n^{\frac{1}{n}}) = 0$
- (b) $\lim_{n \rightarrow \infty} \ln(1 + n^{\frac{1}{n}}) = 0$
- (c) $\lim_{n \rightarrow \infty} \frac{\ln 2 - \frac{1}{n}}{\ln(1 + n^{\frac{1}{n}})} = 1$
- (d) $\lim_{n \rightarrow \infty} (4n^{\frac{3}{n}} - 1) = 3$

[Ans: (a),(b),(d)]

9. If $\{a_n\}$ and $\{b_n\}$ be two sequences such that $|a_n| \leq |b_n|$ for all $n \geq m$, where $m \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} b_n = 0$, then find the limit of a_n . [Ans: (0)]

10. Consider a sequence $(-1, \frac{1}{2}, -\frac{1}{3}, \frac{1}{4}, -\frac{1}{5}, \dots)$, that is $a_n = \frac{(-1)^n}{n}$. The limit of sequence $\{a_n\}$ is: [Ans: (0)]

11. Which of the following option(s) is(are) true?

- (a) $\{\frac{3n-5}{4n+2}\}$ is a decreasing sequence.
- (b) $\{-\frac{1}{6n-5}\}$ is an increasing sequence.
- (c) $\{\sqrt{n+1} - \sqrt{n}\}$ is an increasing sequence.
- (d) $\{\sqrt{2n}\}$ is an increasing sequence.

[Ans: (b), (d)]

AQ 7.5

Questions

1. Define a function $f(x) = \frac{x}{x^2-5x+6}$. Then $\lim_{x \rightarrow 1} f(x)$ is:

[Ans: 0.5]

2. Define a function $f(x) = (-x^2 + 2)(x^3 + 3)$. Then $\lim_{x \rightarrow 1} f(x)$ is:

[Ans: 4]

3. Define a function

$$f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases}$$

Which of the following option(s) is(are) true?

- (a) $\lim_{x \rightarrow 2} f(x) = -1$
- (b) $\lim_{x \rightarrow 0} f(x) = 1$
- (c) $\lim_{x \rightarrow -0.5} f(x) = -1$
- (d) Left limit at 0 i.e., $\lim_{x \rightarrow 0^-} f(x) = -1$
- (e) Right limit at 0 i.e., $\lim_{x \rightarrow 0^+} f(x) = -1$
- (f) Right limit at 0 i.e., $\lim_{x \rightarrow 0^+} f(x)$ does not exist.
- (g) Limit of the function at 0 does not exist.
- (h) Limit of the function does not exist at any real number.

[Ans: (c),(d),(g)]

4. Define a function

$$f(x) = \begin{cases} x & \text{if } x \leq 2 \\ 5 & \text{if } x > 2 \end{cases}$$

Which of the following option(s) is(are) true?

- (a) Limit exists at $x = 2$.
- (b) Right limit exists at $x = 2$.
- (c) Left limit at $x = 2$ is 2.
- (d) Right limit at $x = 2$ is -1.

[Ans: (b),(c)]

5. Consider the following graph of a function in the Figure M2W1AQ 10, where a bullet point represents the point included in the line segment and a circle represents the point does not included in the line segment.

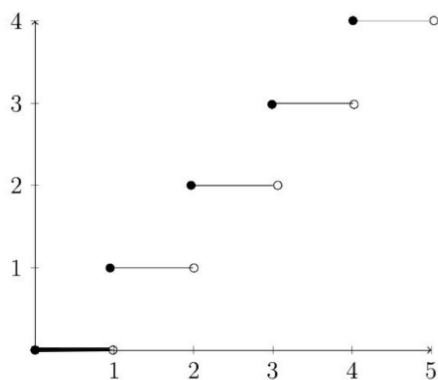


Figure M2W1AQ 10

Which of the following is(are) true?

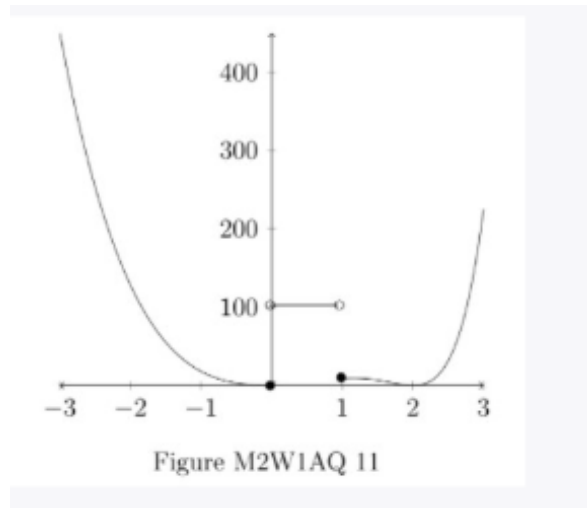
- (a) Left limit exists at $x = 1$.
- (b) Left limit at $x = 3$ is 2.
- (c) Right limit at $x = 2$ is 2.
- (d) Limit exists at $x = 3$.
- (e) Limit exists at every point $x \in (2, 3)$.

[Ans:(a),(b),(c),(e)]

6. Consider the function $f(x) = \frac{x^2}{|x|}$. Then $\lim_{x \rightarrow 0} f(x)$ is:

[Ans: 0]

Consider the following graph of a function in the Figure M2W1AQ 11, where bullet point represents the point included in the curve and circle represents the point does not included in the line segment. Answer the following 2 questions.



7. Which of the following is(are) true?

- (a) Left limit at $x = 0$ is 0.
- (b) Limit exists at $x = 1$.
- (c) Limit exists at $x = 2$.
- (d) Limit exists at $x = 0$.
- (e) Limit exists at $x = \frac{1}{2}$ which is 100.
- (f) Limit exists at $x = 0$ which is 0.

[Ans: (a),(c),(e)]

8. Which of the following is(are) true?

- (a) Function is increasing in the interval $(1, 3)$.
- (b) Function is increasing in the interval $(1, 2)$.
- (c) Function is decreasing in the interval $(-3, 0)$.
- (d) Function is decreasing in the interval $(2, 3)$.

[Ans: (b),(c)]

9. Which of the following option(s) is(are) true?

- (a) $\lim_{x \rightarrow -1} \frac{x^2 - 6x - 7}{x^2 + 3x + 2} = -8$
- (b) $\lim_{x \rightarrow 0} \frac{x^2 - 6x - 7}{x^2 + 3x + 2} = -8$

$$(c) \lim_{x \rightarrow 2} (x^3 + 4x^2 - 6x - 7) = 5$$

$$(d) \lim_{x \rightarrow 3} \frac{x^2 - 6x + 9}{x - 3} = 1$$

[Ans: (a),(c)]

10. Which of the following option(s) is(are) true?

$$(a) \lim_{x \rightarrow +\infty} \frac{1}{x} = 0$$

$$(b) \lim_{x \rightarrow +\infty} \frac{x^2}{1+x} = 1$$

$$(c) \lim_{x \rightarrow -\infty} \frac{1+x}{x^2} = 0$$

$$(d) \lim_{x \rightarrow +\infty} \frac{1+x+x^2}{5x^2+1} = \frac{1}{5}$$

$$(e) \lim_{x \rightarrow +\infty} \frac{x^{2021} + x^{2020} + \dots + x + 1}{x^{2021} + 2021x^{2020} + \dots + 2021} = 2021$$

[Ans:(a),(c),(d)]

PA 7

Questions

1. The graph of some function is drawn below. Choose the set of correct statements about it:

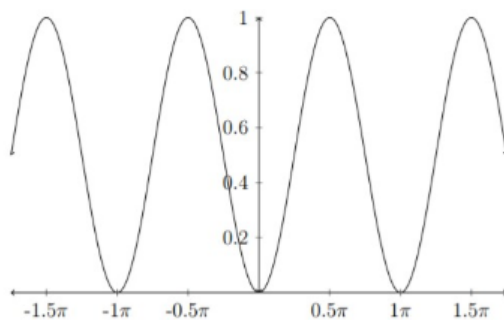


Figure: M2W1G1

- (a) Limit of the function as x tends to 0 is 1.
- (b) Limit of the function as x tends to 0 is 0.
- (c) Limit of the function as x tends to 0 is undefined.
- (d) There is a (unique) tangent at the point $x = \pi$, but not at $x = -\pi$.
- (e) There is a (unique) tangent at $x = \pi$, as well as at $x = -\pi$.
- (f) The given function is monotonically increasing in the interval $[-0.5\pi, 0]$.
- (g) The given function is monotonically decreasing in the interval $[-0.5\pi, 0]$.

[Ans: (b, e, g)]

2. Consider the sequence $\{a_n\}$ given by $a_n = n^{1/n}$, which is known to converge to 1. What will be the limit of the sequence $\{b_n\}$ given by $b_n = n^{1/n} + 12n^{1/n} - 1$?

[Ans: (12)]

3. Let $\{a_n\}$ be a sequence defined as $a_n = \frac{17n+16}{17n+17}$. What will be the limit of the sequence $\{b_n\}$ defined by $b_n = 4913a_n^2 + 289a_n + \frac{31n^2}{n^3+1}$?

[Ans: (5202)]

4. Let $\{a_n\}$ and $\{b_n\}$ be two sequences of real numbers. Consider the following statements:

- (a) If $\{a_n\}$ and $\{b_n\}$ both converge to some non-zero real number, then $\{a_nb_n\}$ also converges to some non-zero real number.
- (b) $\{a_nb_n\}$ may converge even if neither $\{a_n\}$ nor $\{b_n\}$ converge.

- (c) A constant sequence, i.e., $a_i = c$ for some real number c , for all $i \in \mathbb{N}$ is always convergent and it converges to c .

How many statements are correct among the above?

[Ans: (3)]

5. Recall $n! = n(n-1)(n-2) \cdots 3 \cdot 2 \cdot 1$. Define the sequence $\{a_n\}$ by $a_n = \frac{n}{(n!)^{\frac{1}{n}}}$. Note that $\lim a_n = e$. Find the value of:

$$\lim_{n \rightarrow \infty} \frac{2n+1}{((2n+1)!)^{\frac{1}{2n+1}}} + \lim_{n \rightarrow \infty} \frac{3n}{((3n)!)^{\frac{1}{3n}}}.$$

[Ans: (2)]

6. Consider the following sequences $\{a_n\}$, $\{b_n\}$, and $\{c_n\}$:

$$a_n = \frac{1}{n^2}, \quad b_n = 0, \quad c_n = 1 - \frac{1}{n}.$$

How many among the three sequences given above are subsequences of the sequence $\frac{1}{n}$?

[Ans: (1)]

7. Depending on the graphs given below, predict how many of the following curves do not have a (unique) tangent at the origin (i.e., $(0,0)$)? **Note:** Enter an integer value.

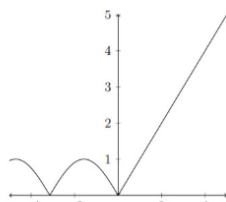


Figure: Curve 1

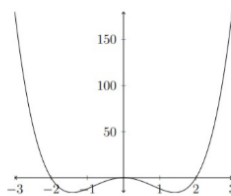


Figure: Curve 2

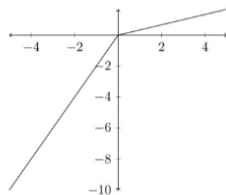


Figure: Curve 3

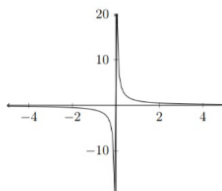


Figure: Curve 4

[Ans: (3)]

(Answer questions 8, 9, and 10 using the below information.)

Suppose there are three schemes available for renting a studio room as given in the

table below:

Schemes	Rent (in ₹) for t hours
Scheme A	$p_1(t) = 100\lfloor t \rfloor + 200$
Scheme B	$p_2(t) = 100\lceil t \rceil + 200$
Scheme C	$p_3(t) = 100t + 200$

where $\lfloor t \rfloor$ denotes the largest integer less than or equal to t , and $\lceil t \rceil$ denotes the smallest integer greater than or equal to t .

8. If Rana wants to book the studio for 2.6 hours and chooses the scheme to avail the studio at minimum cost, then find out the cost he has to pay to book the studio for 2.6 hours.

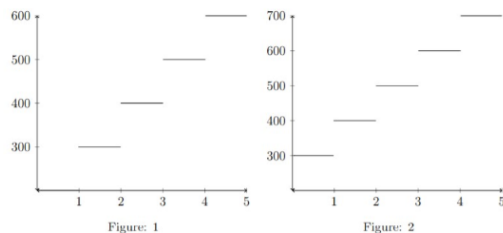
[Ans: (400)]

9. Choose the set of correct options:

- (a) The tangent to the function $p_1(t)$ which represents Scheme A does not exist at 1.
- (b) The tangent to the function $p_1(t)$ which represents Scheme A does not exist at a , for any natural number a .
- (c) The tangent to the function $p_3(t)$ which represents Scheme C does not exist at 1.
- (d) The tangent to the function $p_2(t)$ which represents Scheme B does not exist at a , for any natural number a .

[Ans: (a, b, d)]

10. Consider the following graphs where X -axis denotes the time in hours and Y -axis denotes the cost for rent of the studio:



Enter the figure number which represents Scheme B.

[Ans: (2)]

END OF WEEK 7—————

AQ 8.1

Questions

1. If $\lim_{x \rightarrow a} f(x) = L$, which of the following statements must be true?

- (a) f is defined at a .
- (b) $\lim_{x \rightarrow a^-} f(x) = L$.
- (c) $f(a) = L$.
- (d) $\lim_{x \rightarrow a} f(x) = L$.
- (e) f is continuous at a .

[Ans: (b),(d)]

2. Let f be a function and the Figure M2W2AQ1 represent the graph of function f . The solid points denote the value of the function at the points, and the values denoted by the hollow points are not taken by the functions.

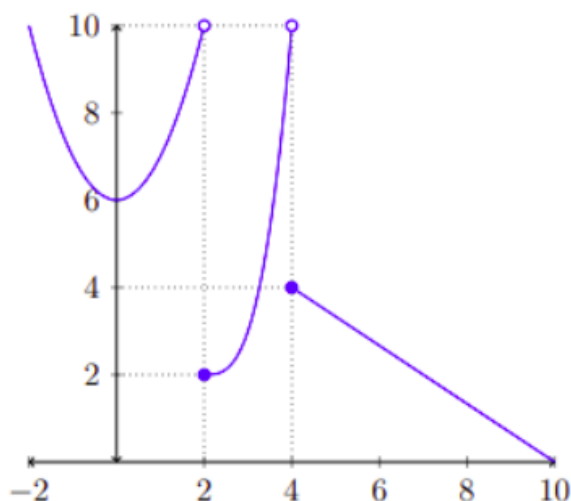


Figure: M2W2AQ1

Choose the set of correct options.

- (a) $\lim_{x \rightarrow 2^-} f(x) = 10$
- (b) $\lim_{x \rightarrow 2^+} f(x) = 2$
- (c) $\lim_{x \rightarrow 4^-} f(x) = 4$
- (d) $\lim_{x \rightarrow 4^+} f(x) = 4$
- (e) f is continuous at $x = 2$.
- (f) f is continuous at $x = 6$.

[Ans: (a),(b),(d),(f)]

3. Consider the following function:

$$f(x) = \begin{cases} \frac{x}{(x+1)(x+2)} & \text{if } x \geq 1 \\ 1/6 & \text{if } x < 1 \end{cases}$$

The number of discontinuity of f is

[Ans: (0)]

4. Define a function f as follows:

$$f(x) = \begin{cases} ax & \text{if } x \geq 1 \\ bx^2 & \text{if } x < 1 \end{cases}$$

where a and b are two arbitrary real numbers.

- (a) f is always continuous.
- (b) f is continuous if and only if $a = b = 0$.
- (c) f is continuous if $a = b = 0$.
- (d) f is continuous if $a = b$.

[Ans: (c),(d)]

5. Which of the following option(s) is(are) true?

- (a) There exists a continuous function f from $[1, 5]$ to $[1, 7]$ such that $f(3) = \pi$.
- (b) There exists a continuous function f from $[1, 10]$ to $[2, 8]$ such that $f(1) = 3$ and $f(10) = 5$.
- (c) There exists a non-constant continuous function f from $\mathbb{R}$ to $\mathbb{R}$ which is bounded.
- (d) There exists an unbounded continuous function from $(2, 3]$ to $\mathbb{R}$.

[Ans: (a),(b), (c), (d)]

6. Consider a function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(cx) = cf(x)$ for all $c, x \in \mathbb{R}$. Which of the following option(s) is(are) correct?

- (a) $f(x + y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$.
- (b) f is not continuous in $\mathbb{R}$.
- (c) f is continuous in $\mathbb{R}$.
- (d) $\lim_{x \rightarrow a} f(x)$ exists for all $a \in \mathbb{R}$, but f is not continuous in $\mathbb{R}$.

[Ans: (a, c)]

7. Define a function f as follows:

$$f(x) = \begin{cases} \frac{x}{\tan^{-1}(2x)} & \text{if } x > 0 \\ b & \text{if } x = 0 \\ \frac{\sin(ax)}{x} & \text{if } x < 0, \end{cases}$$

If f is continuous then the value of $a + b$ is

[Ans: (1)]

8. Define a function f as follows:

$$f(x) = \begin{cases} \frac{1}{e^{\frac{1}{x}} + 1} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Which of the following option(s) is(are) true?

- (a) $\lim_{x \rightarrow 0^-} f(x) \neq \lim_{x \rightarrow 0^+} f(x)$
- (b) $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x)$
- (c) f is a bounded function on $\mathbb{R}$.
- (d) f is continuous at $x = 0$.

[Ans: (a), (c)]

AQ 8.2

Questions

1. Choose the set of correct options.

- (a) If $\lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h}$ exists, then f is differentiable at a .
- (b) A function f may be differentiable at a point a , even if it is not continuous at a .
- (c) If a function is differentiable at a point a , then it must be continuous at a .
- (d) There can exist some continuous functions which are not differentiable at some points in the domain.

[Ans: (a, c, d)]

2. Which of the following options showing step wise solution to check whether a function is differentiable or not are true?

- (a) Checking whether a constant function $f(x) = c$ is differentiable at any real number a or not: $\lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h} = \lim_{h \rightarrow 0} \frac{c+h-c}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = 1$.
- (b) Checking whether $f(x) = x - c$ is differentiable at a for some real number a or not: $\lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h} = \lim_{h \rightarrow 0} \frac{a+h-c-(a-c)}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = 1$.
- (c) Checking whether $f(x) = x^2$ is differentiable at any real number a or not: $\lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h} = \lim_{h \rightarrow 0} \frac{(a+h)^2-a^2}{h} = \lim_{h \rightarrow 0} \frac{2ah+h^2}{h} = 0$.
- (d) Checking whether $f(x) = e^x$ is differentiable at any real number a or not: $\lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h} = \lim_{h \rightarrow 0} \frac{e^{a+h}-e^a}{h} = \lim_{h \rightarrow 0} \frac{e^a(e^h-1)}{h} = e^a \lim_{h \rightarrow 0} \frac{e^h-1}{h} = e^a \cdot 1 = e^a$.

[Ans: (b, d)]

3. Consider the function defined as follows:

$$f(x) = \begin{cases} \lfloor x \rfloor & \text{if } x \geq 0 \\ \lceil x \rceil & \text{if } x < 0 \end{cases}$$

Choose the set of correct options:

- (a) f is differentiable at $x = 0$.
- (b) f is differentiable at $x = 1$.
- (c) f is differentiable at $x = -1$.
- (d) f is differentiable at $x = 1.5$.

[Ans: (a, d)]

4. Consider the graphs given below:

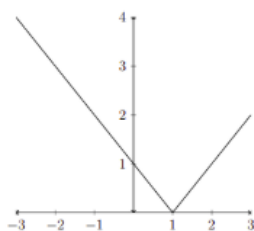


Figure: f_1

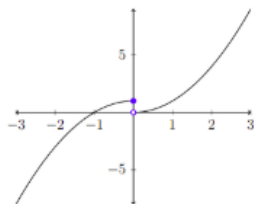


Figure: f_2

Which of the following options are correct?

- (a) f_1 is continuous and differentiable at each real number.
- (b) f_1 is not differentiable at 1.
- (c) f_2 is not continuous at 0.
- (d) f_2 is differentiable in the interval $[1, 2]$.

[Ans: (b, c, d)]

5. The following curve shown in Figure M2W2AQ2 represents the function $f : \mathbb{R} \rightarrow \mathbb{R}$, such that $f(x) = x^{\frac{1}{3}}$.

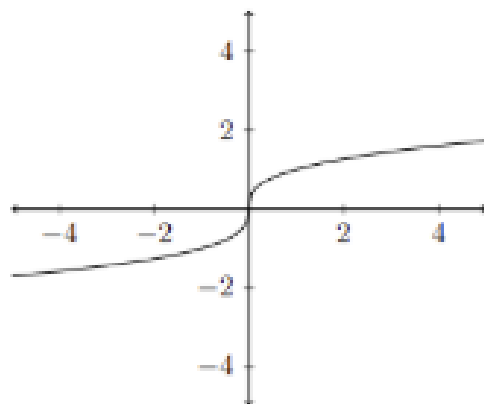


Figure: M2W2AQ2

Which of the following options is(are) correct?

- (a) f is continuous on $\mathbb{R}$.
- (b) f is differentiable everywhere on $\mathbb{R}$.
- (c) f is nowhere differentiable on $\mathbb{R}$.
- (d) f is not differentiable at 0.

[Ans: (a, d)]

6. Consider the function $f(x) = x|x|$. Which of the following options are correct?

- (a) f is not continuous on $\mathbb{R}$.
- (b) f is continuous on $\mathbb{R}$.
- (c) f is continuous on $\mathbb{R}$ and not differentiable at 0.
- (d) f is both continuous and differentiable on $\mathbb{R}$.

[Ans: (b, d)]

7. Choose the set of correct options.

- (a) There exists a function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that f is not differentiable exactly on the set of natural numbers
- (b) Inverse of a differentiable function is differentiable.
- (c) Let $f(x) = x^2$ for x rational and $f(x) = 0$ for x irrational. Then f is differentiable at $x = 0$.
- (d) None of the above

[Ans: (a, c)]

8. Consider the function $f(x) = |\sin x|$. Then f is

- (a) periodic with period π .
- (b) everywhere continuous and differentiable.
- (c) everywhere continuous and not differentiable at $n\pi$, where $n \in \mathbb{Z}$.
- (d) neither continuous nor differentiable at $n\pi$, where $n \in \mathbb{Z}$.

[Ans: (a, c)]

9. Consider Figure M2W2AQ3, which represents some function f , to choose the correct options from the following:

- (a) There are some points in the interval $[-5, 5]$, at which the derivative of f is 0.
- (b) There are some points in the interval $[-5, 5]$, at which f is not differentiable.
- (c) f is not differentiable at 0.
- (d) There are at least 2 points where the derivative of f are non-zero but the same.

[Ans: (a, d)]

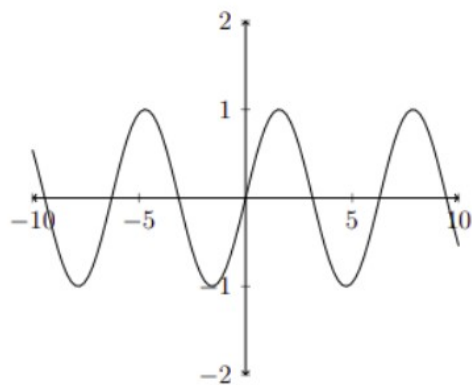


Figure: M2W2AQ3

Figure 2: Your caption

10. Define a function f as follows:

$$f(x) = \begin{cases} kx^2 + l & \text{if } x \leq 1 \\ lx^2 + kx + m & \text{if } x > 1 \end{cases}$$

If f is continuous and differentiable at $x = 1$ then the value of $k - 2l + m$ is [Ans: (0)]

AQ 8.3

Questions

1. Which of the following statements are correct?

- (a) Derivative of $e^{x^2 \cos x}$ is $e^x(2x \cos x - x^2 \sin x)$.
- (b) Derivative of $e^{x \sin x}$ is $e^{x \sin x}(\sin x + x \cos x)$.
- (c) Derivative of $e^{x \sin x}$ is $e^{x \sin x}(\sin x - x \cos x)$.
- (d) Derivative of $e^{x^2 \cos x}$ is $e^{x^2 \cos x}(2x \cos x - x^2 \sin x)$.

[Ans: (b, d)]

2. $\lim_{x \rightarrow 0^+} \frac{\sin x}{\sqrt{x}}$ is

[Ans: (0)]

3. Consider a function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = x|x|$. Which of the following option(s) is(are) true?

- (a) f is not differentiable at any point of $\mathbb{R}$.
- (b) f is differentiable on $\mathbb{R}$.
- (c) $f'(x) = \begin{cases} 2x & \text{if } x > 0 \\ -2x & \text{if } x < 0 \end{cases}$
- (d) $f'(x) = \begin{cases} -2x & \text{if } x > 0 \\ 2x & \text{if } x < 0 \end{cases}$
- (e) $f'(x) = 2x$
- (f) $f'(x) = \begin{cases} 2x & \text{if } x > 0 \\ -2x & \text{if } x < 0, \end{cases}$ and f is not differentiable at 0.

[Ans: (b, c)]

4. If $f(x) = \sqrt{9 - x^2}$, then find out the value of $\sqrt{8} \times \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1}$.

[Ans: (-1)]

5. The value of $\lim_{x \rightarrow +\infty} x e^{-x}$ is

[Ans: (0)]

6. Consider the following statements:

- Derivative of $f(x) = \log_x 5$, where $x > 0$ and $x \neq 1$ is $\frac{-\log_x 5 \log_x e}{x^2}$.
- Derivative of $f(x) = \log_x 5$, where $x > 0$ and $x \neq 1$ is $\frac{-\log_x 5 \log_x e}{x}$.
- Derivative of $f(x) = \log_e x^2$ is $\frac{-2}{x^2}$.
- Derivative of $f(x) = \log_e x^2$ is $\frac{2}{x}$.

Then the number of correct statements is

[Ans: (2)]

7. In which of the following, one can apply L'Hospital's rule to evaluate the limits?

(a) $\lim_{x \rightarrow +\infty} \frac{x}{x + \sin x}$

(b) $\lim_{x \rightarrow 0} \frac{\sin^2 x}{1 - \cos(2x)}$

(c) $\lim_{x \rightarrow +\infty} \frac{7 + \ln x}{x^3 + 6}$

(d) $\lim_{x \rightarrow 0^+} \frac{1}{x}$

[Ans: (b, c)]

8. If $f(x + y) = f(x)f(y)$ for all $x, y \in \mathbb{R}$ and $f(9) = 6, f'(0) = 4$, then $f'(9)$ is

[Ans: (24)]

9. Let f and g be two distinct functions from $\mathbb{R}$ to $\mathbb{R}$. Which of the following statements are true?

(a) If fg is differentiable, then both f and g are differentiable.

(b) Assume that $g(x) \neq 0$ for all $x \in \mathbb{R}$. If $\frac{f}{g}$ is differentiable, then both f and g are differentiable.

(c) If f is an even differentiable function, then f' is an odd function.

(d) If f is an odd differentiable function, then f' is an even function.

[Ans: (c, d)]

10. The value of $\lim_{x \rightarrow 0^+} \cot(x) \sin(4x)$ is

[Ans: (4)]

AQ 8.4

Questions

1. Suppose m_1 , m_2 and m_3 denote the slopes of the tangents of the curve represented by the function $f(x) = x^3 + 3x$, at the points $(-1, f(-1))$, $(0, f(0))$, and $(1, f(1))$, respectively. The value of $m_1 + m_2 + m_3$ is [Ans: (15)]
2. Which of the following statements are correct?
 - (a) Tangent to the parabola $(y - k) = a(x - h)^2$ at the vertex is $x = h$.
 - (b) Tangent to the parabola $(y - k) = a(x - h)^2$ at the vertex is $y = k$.
 - (c) Tangent to the parabola $y = ax^2$ at (a, a^3) is $y = a^2(2x - a)$.
 - (d) Tangent to the parabola $y = ax^2$ at (a, a^3) is $y = 2ax + a^3$.[Ans: (b, c)]
3. Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = ax^2 + bx + c$. Which of the following statements are correct?
 - (a) The slope of the linear approximation $L_f(x)$ at $(m, f(m))$ is $2am + b$.
 - (b) The slope of the linear approximation $L_f(x)$ at $(m, f(m))$ is $2ax + b$.
 - (c) The equation of the linear approximation $L_f(x)$ at $(m, f(m))$ is $y = (2am + b)x + c - am^2$.
 - (d) The equation of the linear approximation $L_f(x)$ at $(m, f(m))$ is $y = (2am + b)x + c$.

[Ans: (a, c)]

4. Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = e^x \sin x$. Which of the following expression represents the linear approximation $L_f(x)$ at $x = \frac{\pi}{2}$?
- (a) The slope of the linear approximation $L_f(x)$ at $x = \frac{\pi}{2}$ is $-e^{\frac{\pi}{2}}$.
 - (b) The slope of the linear approximation $L_f(x)$ at $x = \frac{\pi}{2}$ is $e^{\frac{\pi}{2}}$.
 - (c) The equation of the linear approximation $L_f(x)$ at $x = \frac{\pi}{2}$ is $L_f(x) = -e^{\frac{\pi}{2}}x + e^{\frac{\pi}{2}}(1 + \frac{\pi}{2})$.
 - (d) The equation of the linear approximation $L_f(x)$ at $x = \frac{\pi}{2}$ is $L_f(x) = e^{\frac{\pi}{2}}x + e^{\frac{\pi}{2}}(1 - \frac{\pi}{2})$.

[Ans: (b, d)]

5. Let f be a differentiable function at $x = 1$. The tangent line to the curve represented by the function f at the point $(1, 0)$ passes through the point $(5, 8)$. What will be the value of $f'(1)$? [Ans: (2)]
6. Suppose the tangent of the curve represented by a function f at the point $(1, f(1))$ is given by the equation $y = 3x + 2$. What is the value of $f(1)$? [Ans: (5)]
7. Using the linear approximation of $f(x) = \sqrt{x}$ at $x = 4$, the approximate value of $\sqrt{4.4}$ is [Ans: (2.1)]
8. Using the linear approximation of $f(x) = (1 + x)^4$ at $x = 0$, the approximate value of $(1.01)^4$ is [Ans: (1.04)]

PA 8

Questions

1. Based on the given Table M2W2P1, choose the options which represent the correct matching of a given function with its tangent at $(0,0)$ and its graph.

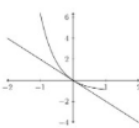
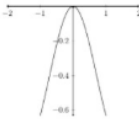
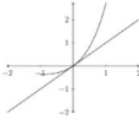
	Function (Column A)		It's tangent at $(0,0)$ (Column B)		Graph (Column C)
i)	$f(x) = xe^x$	a)	$y = -2x$	1)	
ii)	$f(x) = e^{-2x} - 1$	b)	$y = x$	2)	
iii)	$f(x) = e^{-x^2} - 1$	c)	$y = 0$	3)	

Table: M2W2P1

- (a) (i) $\rightarrow$ (b) $\rightarrow$ (3)
 (b) (ii) $\rightarrow$ (c) $\rightarrow$ (2)
 (c) (iii) $\rightarrow$ (c) $\rightarrow$ (2)
 (d) (iii) $\rightarrow$ (b) $\rightarrow$ (3)
 (e) (ii) $\rightarrow$ (a) $\rightarrow$ (1)
 (f) (i) $\rightarrow$ (a) $\rightarrow$ (1)

[Ans: (a,c,e)]

2. Choose the set of correct options:

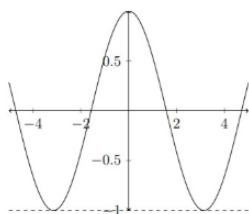


Figure: Curve 1

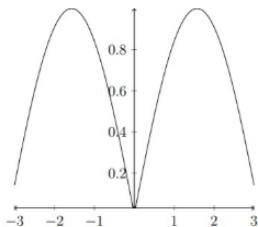


Figure: Curve 2

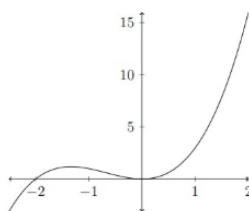


Figure: Curve 3

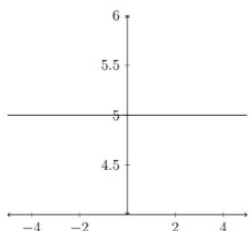


Figure: Curve 4

- (a) There are at least two points on Curve 1, where the derivatives of the function corresponding to Curve 1 are equal.
- (b) At the origin, the derivative of the function corresponding to Curve 2 does not exist.
- (c) The derivative of the function corresponding to Curve 3, at the origin and at the point $(-2, 0)$, are the same.
- (d) The derivative of the function corresponding to Curve 4 does not exist at any point.

[Ans: (a, b)]

3. Let f be a function and Figure M2W2P1 represent the graph of function f . The solid points denote the value of the function at the points, and the values denoted by hollow points are not taken by the functions. Choose the set of correct options:

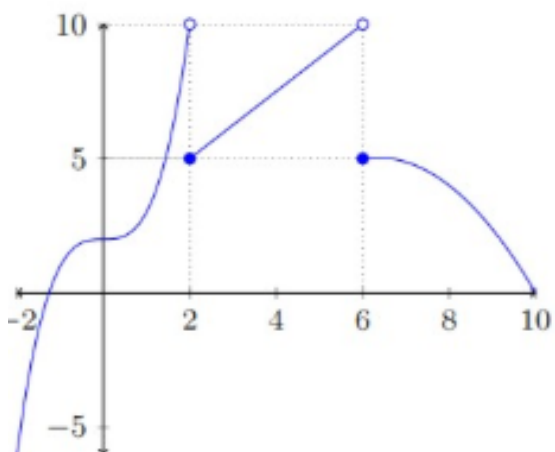


Figure: M2W2P1

- (a) $\lim_{t \rightarrow 2^-} f(t) = 10$
- (b) $\lim_{t \rightarrow 2^+} f(t) = 5$
- (c) $\lim_{t \rightarrow 6^-} f(t) = 10$
- (d) $\lim_{t \rightarrow 6^+} f(t) = 10$
- (e) f is continuous at $x = 2$
- (f) f is continuous at $x = 4$

[Ans: (a, b, c, f)]

4. Define a function f as follows:

$$f(x) = \begin{cases} x^3 & x > 1, \\ x^2 & 0 < x \leq 1, \\ x & x < 0, \\ 0 & x = 0. \end{cases}$$

Choose the set of correct options:

- (a) f is continuous but not differentiable at $x = 1$.
- (b) f is both continuous and differentiable at $x = 1$.
- (c) f is continuous but not differentiable at $x = 0$.
- (d) f is both continuous and differentiable at $x = 0$.
- (e) f is not continuous at $x = 0$.
- (f) f is not continuous at $x = 1$.

[Ans: (a, c)]

5. Let f and g be two real-valued functions defined as:

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = e^x - 1$$

$$g : \mathbb{R} \rightarrow \mathbb{R}, \quad g(x) = x$$

Choose the set of correct options:

- (a) The linear function $ex - 1$ is the best linear approximation of the function $f(x)$ at the point $x = 1$.
- (b) In this case, $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow 0} f(x)}{\lim_{x \rightarrow 0} g(x)}$.
- (c) In this case, $(f + g)(x)$ (where $(f + g)(x)$ is defined by $f(x) + g(x)$) is continuous at $x = 0$.
- (d) $\lim_{x \rightarrow 0} f(x)g(x) = 0$.

[Ans: (a, c, d)]

(Use the information given below to answer Questions 6,7 and 8.)

The profit of Company A with respect to time (in months) is given by the The profit of Company A with respect to time (in months) is given by the function $f(t)$ (in lakhs) as follows:

$$f(t) = \begin{cases} \frac{(t-2)^{n-1}}{t-3} & \text{if } 0 \leq t < 3, \\ [t] & \text{if } t \geq 3. \end{cases}$$

for some integer n .

The profit of Company B with respect to time (in months) is given by the function $g(t)$ (in lakhs) as follows:

$$g(t) = \begin{cases} \frac{t^3-3^3}{t-3} & \text{if } 0 \leq t < 3, \\ 3t^m & \text{if } t \geq 3, \end{cases} \quad \text{for some integer } m.$$

6. If the functions $f(t)$ and $g(t)$ denoting the profits of Company A and Company B, respectively, are known to be continuous at $t = 3$, then which of the following options is(are) true?
- (a) $n = 3$
 - (b) $n = 2$
 - (c) $m = 2$
 - (d) $m = 3$

[Ans: (a, c)]

7. Assuming g to be continuous at $t = 3$, choose the set of correct options:

- (a) $\lim_{t \rightarrow 3^-} \frac{g(t) - g(3)}{t - 3} = 18$
- (b) $\lim_{t \rightarrow 3^+} \frac{g(t) - g(3)}{t - 3} = 18$
- (c) $\lim_{t \rightarrow 3^+} \frac{g(t) - g(3)}{t - 3} = 9$
- (d) $\lim_{t \rightarrow 3^-} \frac{g(t) - g(3)}{t - 3} = 9$
- (e) g is not differentiable at $t = 3$.
- (f) g is differentiable at $t = 3$.

[Ans: (b, d, e)]

8. Let $L_f(t)$ be the best linear approximation of the function $f(t)$ at the point $t = 1$. Assuming f to be continuous at $t = 3$, then find the value of $L_f(2)$.

[Ans: (0)]

9. Let f be a differentiable function at $x = -9$. The tangent line to the curve represented by the function f at the point $(-9, 2)$ passes through the point $(8, 5)$. What will be the value of $f'(-9)$?

[Ans: Range: (0.17, 0.19)]

10. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^2$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $g(x) = x - 5$. Find the value of $(fg)'(17) - (f \circ g)'(17)$, where $f \circ g(x) = f(g(x))$ and $fg(x) = f(x)g(x)$.

[Ans: 673]

END OF WEEK 8

AQ 9.1 Level 1:

1. Let $f(x)$ be a function and P be a turning point of the function $f(x)$. Then which of the following option(s) is (are) true for the function $f(x)$ at the point P ?
 - (a) The function $f(x)$ may change from increasing to decreasing at the point P
 - (b) The function $f(x)$ may change from decreasing to increasing at the point P
 - (c) The function $f(x)$ remains the same
 - (d) None of the above

[Ans: (a),(b)]

2. Let $f(x)$ be a function and the graph of the function is given in Figure M2W3AQ1.

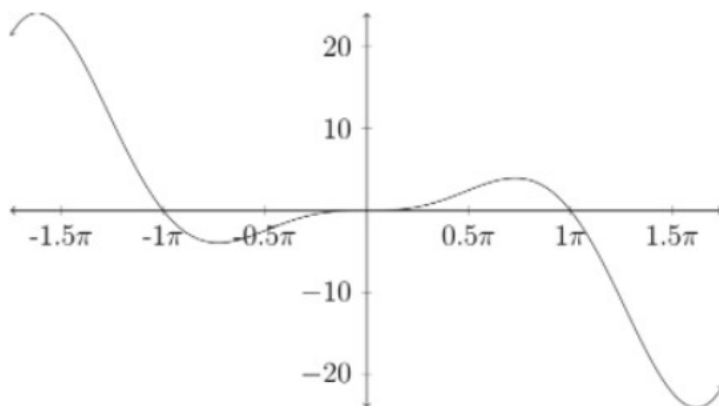


Figure: M2W3AQ1

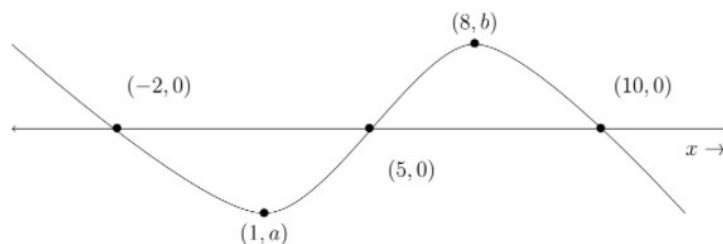
In the given domain as shown in the Figure M2W3AQ1, which of the following option is true?

- (a) The number of local maxima for the function $f(x)$ is 2.
 - (b) The number of local minima for the function $f(x)$ is 3.
 - (c) The function $f(x)$ has a local minimum at the origin.
 - (d) The function $f(x)$ has a local maximum at the origin.
- [Ans: (a)]
3. Let $f(x)$ be a function. If the function $f(x)$ has a local minimum at the point $x = 2$ and a local maximum at the point $x = 5$, then which of the following option(s) is(are) true?
 - (a) Slope of the tangent at the point $x = 2$ is 2.
 - (b) Slope of the tangent at the point $x = 5$ is 0.

- (c) Slope of the tangent at the point $x = 2$ is 0.
- (d) Slope of the tangent at the point $x = 5$ is 5.

[Ans: (b),(c)]

4. A function $f(x)$ represented by the curve is shown in Figure M2W3AQ2, where $a, b \in \mathbb{R}$.



M2W3AQ2

Choose the set of correct options.

- (a) If $x \in (2, 6)$, then $f(x)$ is an increasing function.
- (b) If $x \in (8, 10)$, then $f(x)$ is a decreasing function.
- (c) If $x \in (2, 6)$, then $f'(x) \leq 0$.
- (d) If $x \in (-2, 5)$, then $f'(x) \leq 0$.
- (e) The equation $f'(x) = 0$ has at least two solutions

[Ans: (a),(b),(e)]

5. Suppose $f(x)$ is a polynomial function. If $f'(x)$ intersects X -axis at 10 distinct points, then the degree of $f(x)$ is at least p . Find the value of p .

[Ans: 11]

AQ 9.1 Level 2:

1. Let $f(x)$ be a differentiable function defined as $f(x) = \frac{x^4}{2} - \frac{13x^3}{3} + 11x^2 - 8x$. Which of the following option(s) is (are) true?

- (a) The number of critical points are 3.
- (b) The number of critical points are 4.
- (c) The point $x = 4$ is a critical point.
- (d) The point $x = 1$ is a critical point.
- (e) The point $x = 2$ is a critical point.

[Ans: (a),(c),(e)]

2. Let $f(x) = \sin x + \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x$, $x \in [0, \pi]$. Assume $f(x)$ does not have a saddle point in the interval. Which of the following may represent the graph of the function?

[Hint: Differentiate the function $f(x)$ and then use the formula $\cos x + \cos y = 2 \cos \frac{x+y}{2} \cos \frac{x-y}{2}$ on $\cos x$ and $\cos 3x$ and then solve the equation $f'(x) = 0$ for critical points. Note: if $\cos x = 0$, then $x = \frac{\pi}{2}, \frac{3\pi}{2}$ and if $\cos x = -\frac{1}{2}$, then $x = \frac{2\pi}{3}$]

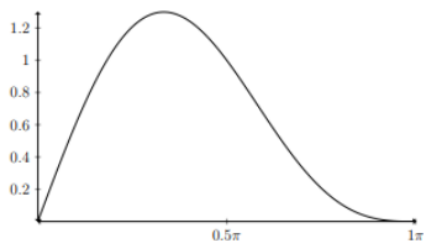


Figure: Curve 1

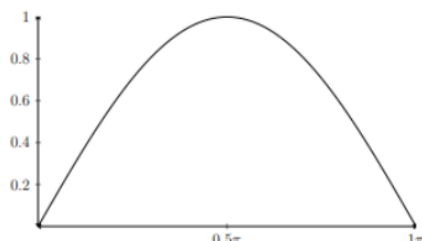


Figure: Curve 2

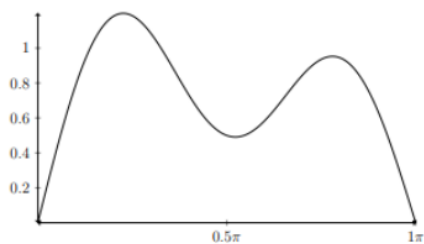


Figure: Curve 3

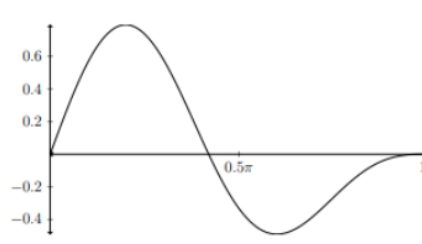


Figure: Curve 4

- (a) Curve 1
- (b) Curve 2
- (c) Curve 3
- (d) Curve 4

[Ans: (c)]

3. Find the number of distinct solutions for the equation $f'(x) = 0$, if $f(x) = \frac{1}{3}x^3 + x^2 + x$.
[Ans: 1]

4. Match the graphs of the functions $f(x)$ in column A with the graphs of their derivative functions $f'(x)$ in column B in Table M2W3AQ1.

[Hint: Observe that if a is a point on the X -axis at which a function $f(x)$ has a local maximum or a local minimum, then a is a root of $f'(x) = 0$.]

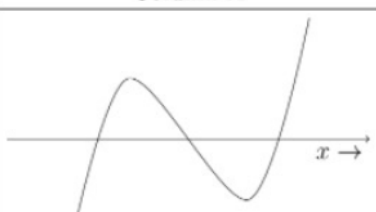
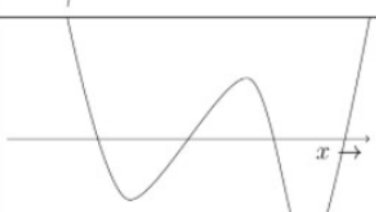
	Column A		Column B
a)		i)	
b)		ii)	

Table M2W3AQ1

- (a) $a \rightarrow \text{ii}, b \rightarrow \text{i}$
 (b) $a \rightarrow \text{i}, b \rightarrow \text{ii}$
 (c) None of the above.

[Ans: (a)]

5. Suppose $f(x) = \sin(x)$. Suppose there are k critical points and n zeros of $f(x)$ in the interval $(-10, 10)$. Find the value of $n - k$.
[Ans: 1]

AQ 9.2 Level 1:

1. Let A be the set of local minima and B be the set of local maxima for the function $f(x) = \frac{1}{4}x^4 - 2x^3 + \frac{11}{2}x^2 - 6x$. Choose the correct option.

- (a) $A = \{1, 2\}, B = \{3\}$
- (b) $A = \{3\}, B = \{1, 2\}$
- (c) $A = \{2, 3\}, B = \{1\}$
- (d) $A = \{1, 3\}, B = \{2\}$

[Ans: (d)]

2. The minimum value of the polynomial $P(x) = (x - \alpha)(x - \beta)$ occurs at

- (a) $x = \frac{\alpha - \beta}{2}$
- (b) $x = \alpha - \beta$
- (c) $x = \frac{\alpha + \beta}{2}$
- (d) $x = \alpha + \beta$

[Ans: (c)]

3. If $x + y = 50$, then find the maximum value of $2xy$.

[Ans: 1250]

4. Consider a rectangle of length l and width w such that the perimeter of the rectangle is 100 units. The maximum possible area of the rectangle is

- (a) 1250 square units
- (b) 625 square units
- (c) 1000 square units
- (d) 500 square units

[Ans: (b)]

5. Suppose $f(x) = 2x^3 - 3x^2 - 36x + 5$ and $f(x)$ has a local minimum at $x = k$. Find the value of k .

[Ans: 3]

AQ 9.2 Level 2:

1. Consider the function defined as follows: $f(x) = \begin{cases} -x^2 + 2x + 3 & \text{if } 0 \leq x \leq 50 \\ x^3 + 3 & \text{if } -50 \leq x < 0. \end{cases}$

Which of the following options are correct?

- (a) 1 is a local maximum.
- (b) -50 is the global minimum.
- (c) 0 is the global maximum.
- (d) 50 is the global minimum.

[Ans: (a),(b)]

2. Consider the function $f(x) = \frac{n}{x} + \frac{x}{n}$ from $(0, \infty)$ to R , where n is a positive integer.

Which of the following option(s) is(are) true?

- (a) n is a local maximum.
- (b) 1 is a local maximum.
- (c) 1 is a global maximum.
- (d) n is a local minimum.
- (e) n is a global minimum.

[Ans: (d),(e)]

3. Let $f(x) = \sin x + \cos x + 5$. Which of the following is the maximum value of the function in the interval $[0, \pi]$

- (a) $\sqrt{2}$
- (b) 5
- (c) $\sqrt{2} + 5$
- (d) 2

[Ans: (c)]

4. Let $f(x) = \frac{x^5}{5} - \frac{3x^4}{2} + \frac{11x^3}{3} - 3x^2$.

Which of the following option(s) is(are) true about the function $f(x)$?

- (a) The function $f(x)$ is increasing in the interval $(0, 1)$.
- (b) The function $f(x)$ is decreasing in the interval $(2, 3)$.
- (c) The function $f(x)$ is increasing in the interval $(1, 2)$.

(d) The function $f(x)$ is increasing in the interval $(3, \infty)$.

(e) The function $f(x)$ is decreasing in the interval $[1, 3]$.

[Ans: (b),(c),(d)]

5. What is the maximum value of the function $f(x) = x^3 - 2x^2 - 3x + 5$ in the interval $[0, 3]$? [Ans: 5]

AQ 9.3 Level 1:

1. Let A be a square with each side of length 6 units. Which of the following option(s) is (are) true?
 - (a) The maximum number of non-overlapping squares that can be drawn inside A such that each square has length 2 units is 9.
 - (b) The maximum number of non-overlapping squares that can be drawn inside A such that each square has length 1 unit is 35.
 - (c) The maximum number of non-overlapping rectangles that can be drawn inside A such that each square has length 2 units and breadth 3 units is 6.
 - (d) The maximum number of non-overlapping rectangles that can be drawn inside A such that each square has length 2 units and breadth 3 units is 5.

[Ans: (a),(c)]

2. Consider the parallelogram as shown in Figure M2W3AQ3. If $\theta = 60^\circ$ and perimeter of the parallelogram is 10 units then which of the following option(s) is(are) true?

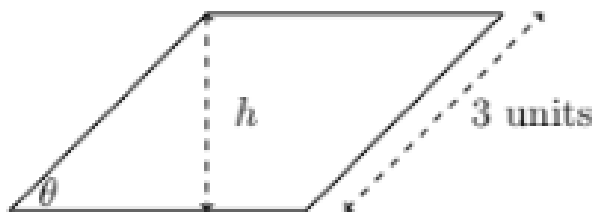


Figure M2W3AQ3

- (a) Height $h = \frac{3\sqrt{3}}{2}$ units.
- (b) Height $h = \frac{\sqrt{3}}{2}$ units.
- (c) Area of the parallelogram is $3\sqrt{3}$ square units.
- (d) Area of the parallelogram is $\sqrt{3}$ square units.

[Ans: (a),(c)]

3. Consider the regular hexagon in Figure M2W3AQ4 and triangles which are made inside the hexagon. If the length of each side of the hexagon is 4 units, then find the area of the hexagon.

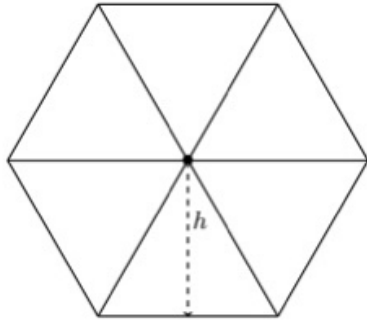


Figure M2W3AQ4

- (a) 24 square units.
- (b) $24\sqrt{3}$ square units.
- (c) 36 square units.
- (d) $36\sqrt{3}$ square units.

[Ans: (b)]

4. Let C_1 and C_2 be two circles where the radius are 2 units and 3 units, respectively. Which of the following option(s) is (are) true?

- (a) Area of $C_1 = 4\pi$ square units.
- (b) Area of $C_2 = 4\pi$ square units.
- (c) Area of $C_1 = 9\pi$ square units.
- (d) Area of $C_2 = 9\pi$ square units.

[Ans: (a),(d)]

5. How many number of unit squares required to make a rectangle of length 6 units and breadth 5 units?

[Ans: 30]

AQ 9.3 Level 2:

1. (Use the below information for question 1 and 2)

Consider the following two trapeziums $ACDB$ and $PQRS$ in Figure M2W3AQ5 and Figure M2W3AQ6 Respectively. In trapezium $ACDB$, $AB = 4$ units, $CD = 2$ units and height $h_1 = 2$ units. In trapezium $PQRS$, $\theta_1 = 60^\circ$, $\theta_2 = 45^\circ$, $PT = TU = 3$ units.

Find the area of the trapezium $ACDB$. [Hint: Area of a trapezium = $\frac{1}{2} \times h \times (a + b)$, where a and b are the lengths of the parallel of sides, and h is the height, i.e., the distance between the parallel sides.]

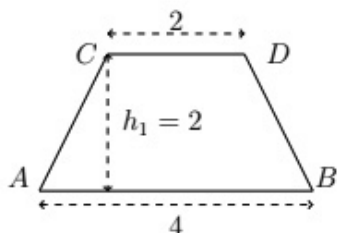


Figure M2W3AQ5

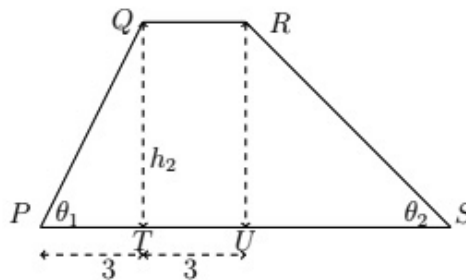


Figure M2W3AQ6

[Ans: 6]

2. Which of the option(s) is(are) true?
- (a) $h_2 = 3$ units
 - (b) $h_2 = \sqrt{3}$ units
 - (c) $US = 3\sqrt{3}$ units
 - (d) Area of $PQRS = \frac{27(\sqrt{3}+1)}{2}$ square units
 - (e) Area of $PQRS = \frac{27}{2}$ square units

[Ans: (c),(d)]

3. Let A be a triangle where all sides of the triangle are equal. If the length of a side of the triangle is 6 units, then area of the triangle is
- (a) $\sqrt{3}$ square units.
 - (b) $3\sqrt{3}$ square units.
 - (c) $9\sqrt{3}$ square units.
 - (d) 9 square units.

[Ans: (c)]

4. Suppose there are infinitely many circles $C_1, C_2, \dots, C_n, \dots$. The sequence $\{r_n\}$ of the radii of these circles is defined as $r_n = \frac{2n-1}{2n+2}$ units. Now as $n \rightarrow \infty$, the radius of the circles is increasing so areas of the circles are also increasing. (for example if $n = 1$, then the radius of the circle C_1 is $\frac{1}{4}$ units so area of circle(C_1) = $\frac{\pi}{16}$ square units, and if $n = 2$, then the radius of the circle C_2 is $\frac{3}{6} = \frac{1}{2} > \frac{1}{4}$ units and area of circle(C_2) = $\frac{\pi}{4} > \frac{\pi}{16}$ square units and so on). Which of the following option(s) is(are) true about the circles $C_1, C_2, \dots, C_n, \dots$

[Hint: Find the limit of the sequence $\frac{2n-1}{2n+2}$ and since the sequence is increasing so the radius of any circle is less than the limit of the sequence.]

- (a) Area of the biggest circle $\leq \pi$.
- (b) Area of the biggest circle $> \pi$.
- (c) Area of the smallest circle is $\frac{\pi}{16}$ square units.
- (d) Area of the smallest circle is π square units.

[Ans: (a),(c)]

5. A circle C is inscribed in a square S_1 and then a smaller square S_2 is inscribed in the circle C . Suppose the ratio of the area of the square S_1 to the area of the square S_2 is $m : n$, where $m, n \in \mathbb{Z}$ and $\gcd(m, n) = 1$. Find the value of $m + n$. [Ans: 3]

AQ 9.4 Level 1:

1. Consider the closed interval $[3, 9]$. Let $P = \{x_0, x_1, \dots, x_n\}$ be an ordered set, where $n \in \mathbb{N}$. Which of the following option(s) is (are) true regarding partitions of the interval $[3, 9]$?
 - (a) If $x_0 = 3$ and $n = 1$ (i.e., $x_1 = 9$), then P is a partition of the interval $[3, 9]$.
 - (b) If $x_0 = 3$ and $x_n = 9$ and $3 < x_i < 9$ with $x_i < x_{i+1}$ for $i = 1, 2, \dots, n-1$, then P is a partition of the interval $[3, 9]$.
 - (c) If $x_0 = 3$ and $x_n = 9$ and $3 < x_i < 9$ for $i = 2, 3, \dots, n-1$, and $x_1 = 2$, then P is a partition of the interval $[3, 9]$.
 - (d) If S is the collection of all partitions of the interval $[3, 9]$, then cardinality of the set S is finite.

[Ans: (a),(b)]

2. Consider an interval $[0, 10]$ and the following partitions $P_1 : 0 = x_0 < x_1 < \dots < x_{10} = 10$, where $x_i = x_{i-1} + 1, i = 1, 2, \dots, 10$. $P_2 : 0 = x_0 < x_1 < \dots < x_5 = 10$, where $x_i = x_{i-1} + 2, i = 1, 2, \dots, 5$. $P_3 : 0 = x_0 < x_1 < x_2 < x_3 = 10$, where $x_1 = 3, x_2 = 5, x_3 = 10$.

If $\Delta x_i = x_i - x_{i-1}$ and for a partition P_k , define $\|P_k\| = \max_i \{\Delta x_i\}$, then which of the following option(s) is (are) true?

- (a) For P_1 , $\Delta x_i = 1$ for all i
- (b) For P_2 , $\Delta x_i = 2$ for all i
- (c) For P_3 , $\Delta x_1 = 5$
- (d) For P_3 , $\Delta x_3 = 5$
- (e) $\|P_1\| = 2$
- (f) $\|P_2\| = 1$
- (g) $\|P_3\| = 5$

[Ans: (a),(b),(d),(g)]

3. Let $f(x)$ be a function defined on a domain D which contains the interval $[a, b]$. Let P be a partition of the interval $[a, b]$ defined as $a = x_0 < x_1 < \dots < x_n = b$, $x_i^* \in [x_{i-1}, x_i]$ and $\Delta x_i = x_i - x_{i-1}$. Which of the following option represents the Riemann sum?
 - (a) $\sum_{i=1}^n f(x_i^*) \Delta x_i$
 - (b) $\sum_{i=1}^n \Delta x_i$
 - (c) $\sum_{i=1}^n f(x_i^*)$
 - (d) None of the above.

[Ans: (a)]

4. Which of the following curves shown in the following figures shows negative area on the X axis in the interval $[1]$?

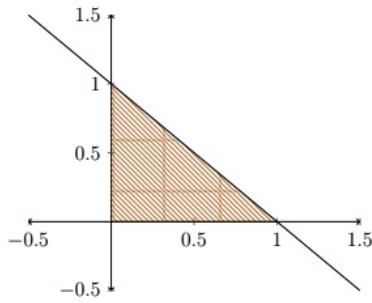


Figure: Curve 1

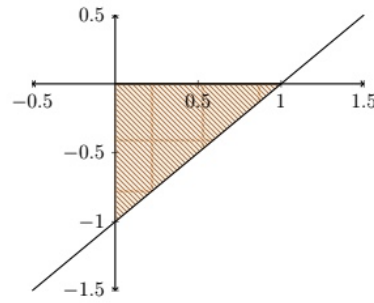


Figure: Curve 2

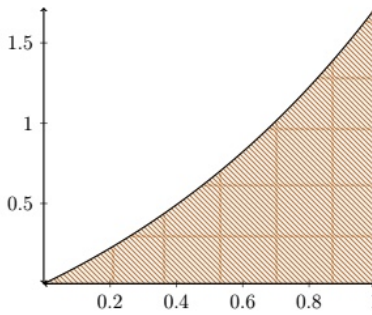


Figure: Curve 3

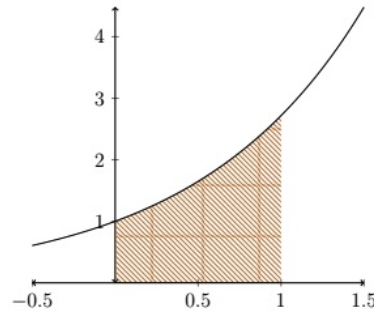


Figure: Curve 4

- (a) Curve 1
- (b) Curve 2
- (c) Curve 3
- (d) Curve 4

[Ans: (b)]

5. Consider an interval $[10, 35]$ and the following partition: $P : 10 = x_0 < x_1 < \dots < x_5 = 35$, where $x_1 = 15, x_2 = 19, x_3 = 26$, and $x_4 = 31$. If $\Delta x_i = x_i - x_{i-1}$ and for the partition P , define $\|P\| = \max_i \{\Delta x_i\}$, then $\|P\| =$ [Ans: 7]

AQ 9.4 Level 2:

1. Consider an interval $[3, 7]$ and the following partitions $P_1 : 3 = x_0 < x_1 < \dots < x_n = 7$, where $x_i = 4 + \frac{i \times 3}{n}, i = 1, 2, \dots, n$ $P_2 : 3 = x_0 < x_1 < \dots < x_n = 7$, where $x_i = 3 + \frac{i \times 4}{n}, i = 1, 2, \dots, n$ $P_3 : 3 = x_0 < x_1 < \dots < x_n = 7$, where $x_i = 5 + \frac{i \times 2}{n}, i = 1, 2, \dots, n$

Which of the following partitions has norm ($\|P_k\| = \max_i \{\Delta x_i\}$) tends to 0 as $n \rightarrow \infty$ (i.e., $\|P_k\| \rightarrow 0$ as $n \rightarrow \infty$)?

[Hint: First calculate Δx_1 , then compare with other $\Delta x_i, i = 2, 3, \dots, n$]

- (a) P_1
- (b) P_2
- (c) P_3
- (d) None of the above.

[Ans: (b)]

2. Let $f(x) = 3x + 1$ then find the value of the integral $\int_0^2 f(x)dx$ using limit of Riemann sums as $n \rightarrow \infty$, for the given partition $P = \{0 = x_0, x_1 = \frac{2}{n}, \dots, x_i = \frac{2 \times i}{n}, \dots, x_n = 2\}, i = 1, 2, \dots, n$ and $x_i^* \in [x_{i-1}, x_i]$, where $x_i^* = \frac{2 \times i}{n}$. [Ans: 8]

3. (Use the below information for question 3, 4 and 5)

Let $f(x) = x$ be a function defined on the domain D which contains the interval $[0, 5]$. Let P be a partition of the interval $[0, 5]$ defined as $0 = x_0 < x_1 < \dots < x_n = 5$ where $x_i = \frac{i \times 5}{n}$ i.e., $x_0 = 0 < \frac{5}{n} < \frac{2 \times 5}{n} < \frac{3 \times 5}{n} < \dots < \frac{5(n-1)}{n} < \frac{5n}{n} = 5 = x_n, i = 1, 2, \dots, n$ and $x_i^* \in [x_{i-1}, x_i], \Delta x_i = x_i - x_{i-1}$.

Which of the following option(s) is(are) true?

- (a) If $x_i^* = x_i$, then the Riemann sum $\sum_{i=1}^n f(x_i^*)\Delta x_i$ equals $\frac{25(n+1)}{2n}$.
- (b) If $x_i^* = x_i$, then the Riemann sum $\sum_{i=1}^n f(x_i^*)\Delta x_i$ equals $\frac{25(n+1)}{n}$.
- (c) If $x_i^* = x_{i-1}$, then the Riemann sum $\sum_{i=1}^n f(x_i^*)\Delta x_i$ equals $\frac{25(n-1)}{2n}$.
- (d) If $x_i^* = x_{i-1}$, then the Riemann sum $\sum_{i=1}^n f(x_i^*)\Delta x_i$ equals $\frac{25(n-1)}{n}$.
- (e) If $x_i^* = x_{i-1}$, then the Riemann sum $\sum_{i=1}^n f(x_i^*)\Delta x_i$ equals $\frac{25}{2}$.

[Ans: (a),(c)]

4. If $x_i^* = \frac{x_i + x_{i-1}}{2}$, then find the Riemann sum $\sum_{i=1}^n f(x_i^*)\Delta x_i$.

[Hint: Observe that n will get canceled from $\sum_{i=1}^n f(x_i^*)\Delta x_i$.]

[Ans: 12.5]

5. If $n = 10$, then find the absolute difference between the Riemann's left-hand sum and the Riemann's right-hand sum. [Ans: 2.5]

AQ 9.5 Level 1:

1. Suppose f is a continuous function on the domain D which includes interval $[4, 9]$ and F is the anti-derivative of f such that $F(4) = 3$ and $F(9) = 6$.

Then which of the following option(s) is (are) true?

- (a) $F'(x) = f(x)$
- (b) $F(x) - F(4) = \int_4^x f(x)dx$ where $x \in (4, 9)$.
- (c) $\int_4^9 f(x)dx = 9$
- (d) $\int_4^9 f(x)dx = 3$

[Ans: (a),(b),(d)]

2. Let $f(x) = x^3 - x^2 + x$. Let $F(x)$ be the anti-derivative of $f(x)$ such that $F(2) = 6$. Then $F(x)$ equals

- (a) $\frac{x^4}{4} - \frac{x^3}{3} + \frac{x^2}{2} + 8$
- (b) $\frac{x^5}{5} - \frac{x^4}{4} + \frac{x^3}{3} + \frac{8}{3}$
- (c) $\frac{x^3}{3} - \frac{x^2}{2} + x + \frac{8}{3}$
- (d) $\frac{x^4}{4} - \frac{x^3}{3} + \frac{x^2}{2} + \frac{8}{3}$

[Ans: (d)]

3. Let $F(x)$ be the anti-derivative of a function $f(x)$ and $F(x)$ is defined as $F(x) = \sin 2x + \tan 5x + 5^{9x} + \ln 3x + 5$, then $f(x) =$

- (a) $2 \cos 2x + 5 \sec^2 5x + 9(5^{9x}) \ln 5 + \frac{1}{x}$
- (b) $\cos 2x + \sec^2 5x + \ln 5 \cdot 5^{9x} + \frac{1}{x}$
- (c) $2 \cos 2x + 5 \sec^2 5x + 9 \ln 5 \cdot 5^{9x} + x$
- (d) $2 \cos 2x + 5 \sec^2 5x + 9 \ln 5 \cdot 5^{9x} + \frac{1}{x} + 5$

[Ans: (a)]

4. Which of the following option(s) is(are) true?

- (a) $\int_2^3 x^2 dx = \frac{3}{2}$
- (b) $\int_1^2 \frac{1}{x} dx = \ln 2$
- (c) $\int_0^{\frac{\pi}{3}} \tan x \sec x dx = 1$
- (d) $\int_0^2 \frac{1}{\sqrt{4-x^2}} dx = \frac{\pi}{2}$
- (e) $\int_{2020}^{2021} 1 dx = 2021$

(f) $\int_1^2 x^{-5} dx = \frac{15}{64}$

(g) $\int_0^1 \sqrt{x} dx = \frac{2}{3}$

(h) $\int_0^1 x^{\frac{3}{2}} dx = \frac{2}{5}$

[Ans: (b),(c),(d),(f),(g),(h)]

5. Suppose $F(x) = xe^{ax}$, where $a \in R$. If F is the anti-derivative of f such that $f(4) = F(4)$, then find the value a . [Ans: 0.75]

AQ 9.5 Level 2:

6. Match the functions $f(x)$ in column A with their anti-derivatives in column B in Table

	Function		Anti-derivative
	(Column A)		(Column B)
a)	$f(x) = -\sin 5x + \cos 2x$	i)	$F(x) = \frac{e^{2x}}{2} + \frac{2^{3x}}{3 \ln 2} + \frac{\ln 2x}{2} + 7$
b)	$f(x) = x^4 + \frac{2}{x^3}$	ii)	$F(x) = \frac{\cos 5x}{5} + \frac{\sin 2x}{2} + 2$
c)	$f(x) = e^{2x} + 2^{3x} + \frac{1}{2x}$	iii)	$F(x) = x \ln 3x - x + \sin^{-1} \frac{x}{4} + \tan^{-1} \frac{x}{5} - 1$
d)	$f(x) = \ln 3x + \frac{1}{\sqrt{16-x^2}} + \frac{5}{25+x^2}$	iv)	$F(x) = \frac{x^5}{5} - \frac{1}{x^2} + 10$

M2W3AQ2.

Table : M2W3AQ2

- (a) $a \rightarrow ii, b \rightarrow iv, c \rightarrow iii, d \rightarrow i$
 (b) $a \rightarrow ii, b \rightarrow iii, c \rightarrow i, d \rightarrow iv$.
 (c) $a \rightarrow ii, b \rightarrow iv, c \rightarrow i, d \rightarrow iii$.
 (d) $a \rightarrow iv, b \rightarrow iii, c \rightarrow i, d \rightarrow ii$

[Ans: (c)]

Let $f(x) = 2x^2$ be a function defined on the domain D which contains the interval $[0, 4]$. Let P be a partition of the interval $[0, 4]$ defined as $0 = x_0 < x_1 < \dots < x_n = 4$ where $x_i = \frac{i \times 4}{n}$ i.e., $x_0 = 0 < \frac{4}{n} < \frac{2 \times 4}{n} < \frac{3 \times 4}{n} < \dots < \frac{4(n-1)}{n} < \frac{4n}{n} = 4 = x_n, i = 1, 2, \dots, n$ and $x_i^* \in [x_{i-1}, x_i], \Delta x_i = x_i - x_{i-1}$.

Answer the questions 7 and 8 based on the above information.

7. Which of the following option(s) is(are) true?

- (a) $\Delta x_i = \frac{4}{n}$
 (b) $\|P\| = \frac{1}{n}$
 (c) If $x_i^* = x_{i-1}$, then the Riemann sum $\sum_{i=1}^n f(x_i^*) \Delta x_i$ equals the expression $\frac{2 \times 4^3}{n^3} [1^2 + 2^2 + \dots + (n-1)^2]$
 (d) If $x_i^* = x_{i-1}$, then the Riemann sum $\sum_{i=1}^n f(x_i^*) \Delta x_i$ equals the expression $\frac{2 \times 4^3}{n^3}$

- (e) If $x_i^* = x_{i-1}$, then the Riemann sum $\sum_{i=1}^n f(x_i^*)\Delta x_i$ equals the expression $1^2 + 2^2 + \dots + (n-1)^2$

[Ans: (a),(c)]

8. Which of the following option(s) is (are) true?

- (a) If $x_i^* = x_i$, then the Riemann sum $\sum_{i=1}^n f(x_i^*)\Delta x_i$ equals the expression $\frac{2 \times 4^3}{n^3}[1^2 + 2^2 + \dots + (n-1)^2]$
- (b) If $x_i^* = x_i$, then the Riemann sum $\sum_{i=1}^n f(x_i^*)\Delta x_i$ equals the expression $\frac{2 \times 4^3}{n^3}[1^2 + 2^2 + \dots + (n-1)^2 + n^2]$
- (c) If $x_i^* = \frac{x_i + x_{i-1}}{2}$, then the Riemann sum $\sum_{i=1}^n f(x_i^*)\Delta x_i$ equals the expression $1^2 + 2^2 + \dots + (n-1)^2 + n^2$
- (d) If $x_i^* = \frac{x_i + x_{i-1}}{2}$, then the Riemann sum $\sum_{i=1}^n f(x_i^*)\Delta x_i$ equals the expression $\sum_{i=1}^n \frac{32}{n^3}(4i^2 - 4i + 1)$
- (e) If $x_i^* = \frac{x_i + x_{i-1}}{2}$, then the Riemann sum $\sum_{i=1}^n f(x_i^*)\Delta x_i$ equals the expression $\frac{32}{n^3}$

[Ans: (b),(d)]

9. Which of the following option(s) is(are) true?

- (a) $\int_1^\infty e^{-x} dx = \frac{1}{e}$
- (b) $\int_0^\infty \frac{1}{4+x^2} dx = \pi$
- (c) $\int_1^\infty \frac{1}{x} dx$ does not exist
- (d) $\int_1^\infty x^{-3} dx = -\frac{1}{2}$

[Ans: (a),(d)]

10. If $\int_0^p (2x+3)dx = \int_p^{10} (5x-1)dx$, where $1 < p < 10$, then find the value of p . [Ans: 8]

AQ 9.6 Level 1:

1. Which of the following option(s) is (are) true for some constant c ?

(a) $\int x^3 \ln x dx = x^4 [\ln x - 1] + c$

(b) $\int x^3 \ln x dx = \frac{x^4}{4} [\ln x + \frac{1}{4}] + c$

(c) $\int x^2 2^{3x} dx = \frac{6x(2^{3x})}{\ln 2} - \frac{18(2^{3x})}{(\ln 2)^2} + c$

(d) $\int 3 \operatorname{cosec} 2x + 2 \cot 3x - x^4 dx = \frac{3 \ln |\operatorname{cosec} 2x - \cot 2x|}{2} + \frac{2 \ln |\sin 3x|}{3} - \frac{x^5}{5} + c$

(e) $\int 5 \tan 3x - \sec 4x dx = \frac{5 \ln |\sec 3x|}{3} - \frac{\ln |\sec 4x + \tan 4x|}{4} + c$

[Ans: (d),(e)]

2. Use basic properties of definite integrals to choose which of the following option(s) is (are) true?

(a) $\int_{-\pi}^{\pi} x^2 \cos 3x dx = \int_0^{\pi} x^2 \cos 3x dx$

(b) $\int_2^3 \cot 5x + \ln(\sin 3x) dx = - \int_3^2 \cot 5x + \ln(\sin 3x) dx$

(c) $\int_0^3 x^{100} dx = \int_0^1 x^{100} dx - \int_1^2 x^{100} dx - \int_2^3 x^{100} dx$

(d) $\int_0^3 x^{100} dx = \int_0^1 x^{100} dx + \int_1^2 x^{100} dx + \int_2^3 x^{100} dx$

[Ans: (b),(d)]

3. If $f(x) = x^2$ then find the area under the curve represented by $f(x)$, which lie above to X - axis in the interval $[-1, 2]$. [Ans: 3]

4. Find the value of given definite integral $\int_{-2021}^{2021} (x^{2021} \cdot \cos 2021x + \sin 2021x) dx$. [Ans: 0]

5. Suppose $f(x) = \frac{a}{x}$, where $a \in R$. If the area under the curve represented by $f(x)$, which lie above to X -axis in the interval $[1, e^4]$ is 20, then find the value of a . [Ans: 5]

AQ 9.6 Level 2:

1. Match the definite integral expressions in column A with their values in column B in Table M2W3AQ3.

	Definite integral expression (Column A)		Definite integral value (Column B)
a)	$\int_1^2 3^{3x} dx$	i)	$\pi - 2$
b)	$\int_1^2 \frac{\ln 3x}{3x} dx$	ii)	$\frac{234}{\ln 3}$
c)	$\int_0^{\frac{\pi}{2}} 2x \cos x$	iii)	2
d)	$\int_0^{\frac{\pi}{2}} 2x \sin x$	iv)	$\frac{1}{6}[(\ln 6)^2 - (\ln 3)^2]$

Table : M2W3AQ3

Choose the correct option.

- (a) $a \rightarrow ii, b \rightarrow iv, c \rightarrow iii, d \rightarrow i$.
 (b) $a \rightarrow ii, b \rightarrow iv, c \rightarrow i, d \rightarrow iii$.
 (c) $a \rightarrow iv, b \rightarrow ii, c \rightarrow iii, d \rightarrow i$.
 (d) $a \rightarrow iv, b \rightarrow ii, c \rightarrow i, d \rightarrow iii$.

[Ans: (b)]

2. If $f_1(x) = [x] + 1$ and $f_2(x) = \lceil x \rceil + 2$, then which of the following option(s) is(are) true?

- (a) $\int_0^4 f_1(x) dx = 10$
 (b) $\int_0^4 f_1(x) dx = 6$
 (c) $\int_0^4 f_2(x) dx = 10$

- (d) $\int_0^4 f_2(x)dx = 18$
 (e) $\int_0^4 f_1(x)dx = \int_0^4 f_2(x)dx$
 (f) $\int_0^{100} f_2(x)dx = 5250$
 (g) $\int_0^{100} f_2(x)dx = 5050$

[Ans: (a),(d),(f)]

3. Which of the following option(s) is(are) true?

[Hint: Use the function $\frac{1}{1-x^2}$ and $\frac{1}{1+x^2}$]

- (a) $\int_0^1 \frac{1}{1+x^3}dx \leq \frac{\pi}{4}$
 (b) $\int_0^1 \frac{1}{1+x^3}dx \geq \frac{\pi}{4}$
 (c) $\int_0^1 \frac{1}{1+x^3}dx \geq 2 \ln 2$
 (d) $\int_0^1 \frac{1}{1+x^3}dx \leq 2 \ln 2$

[Ans: (b),(d)]

4. Let $f_1(x) = 3x^2$ and $f_2(x) = 4 - x^2$ represent two curves (see Figure M2W3AQ7 for reference). If A is the area which is enclosed by the curves $f_1(x)$ and $f_2(x)$, then find the value of $3A$.

[Hint: Find the area under both curve $f(x) = 3x^2$ and $f(x) = 4 - x^2$, and between the intersection point of the curves and above to X - axis.]

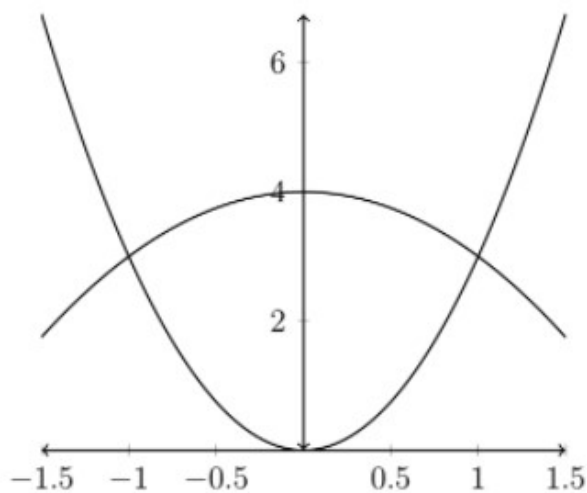


Figure: M2W3AQ7

[Ans: 16]

5. Suppose Rahul travels by bike and the following graph M2W3AQ8 shows the speed (in meter per second) vs time (in seconds) of the bike. Find the distance (in meters) travelled by Rahul in the first 40 seconds.

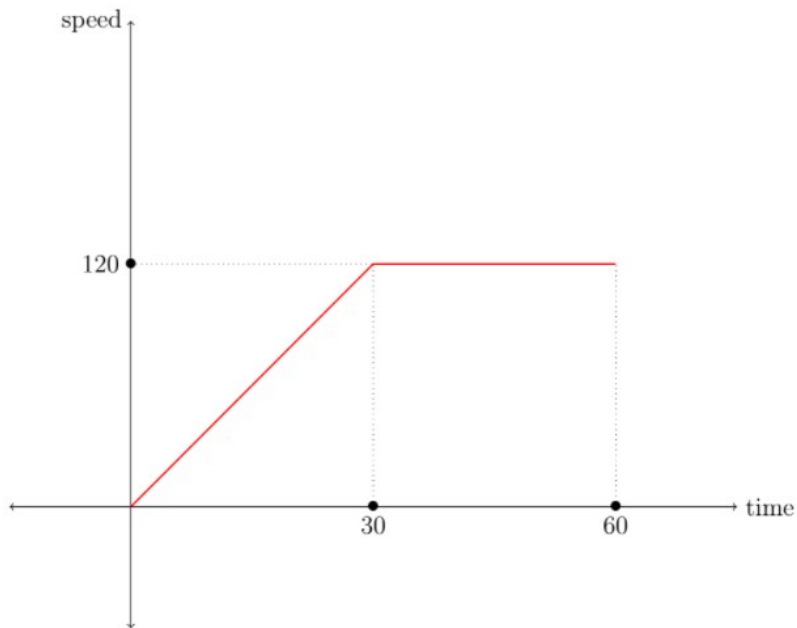


Figure: M2W3AQ8

[Ans: 3000]

PA W9:

1. Match the given functions in Column A with the (signed) area between its graph and the interval $[-1, 1]$ on the X -axis in column B and the pictures of their graphs and the highlighted region corresponding to the area computation in Column C, given in Table M2W3P1.

	Functions (Column A)		Area under the curve (Column B)		Graphs (Column C)
i)	$f(x) = 5x - 1$	a)	$\frac{\pi}{2}$	1)	
ii)	$f(x) = x^3$	b)	0	2)	
iii)	$f(x) = \frac{1}{x^2 + 1}$	c)	-2	3)	

Table: M2W3P1

- (a) i) $\rightarrow$ c) $\rightarrow$ 3).
 (b) ii) $\rightarrow$ a) $\rightarrow$ 1).
 (c) ii) $\rightarrow$ b) $\rightarrow$ 1).
 (d) iii) $\rightarrow$ a) $\rightarrow$ 2).
 (e) iii) $\rightarrow$ c) $\rightarrow$ 2).
 (f) i) $\rightarrow$ b) $\rightarrow$ 1).

[Ans: (a),(c),(d)]

2. Suppose $\int x \ln(1+x) dx = f(x) \ln(x+1) - \frac{x^2}{4} + Ax + B$, where B is the constant of integration. Which of the following are correct?

(a) $f(x) = \frac{x^2-1}{2}$

(b) $f(x) = \frac{x^2-1}{4}$

(c) $A = \frac{1}{4}$

(d) $A = \frac{1}{2}$

[Ans: (a),(d)]

3. Consider the function $f(x) = x^3 - 6x$. Which of the following options are correct?

(a) f has neither local maxima nor local minima.

(b) $\sqrt{2}$ is a local minimum.

(c) $\sqrt{2}$ is a local maximum.

(d) $-\sqrt{2}$ is a local maximum.

(e) $-\sqrt{2}$ is a local minimum.

(f) f has two critical points.

[Ans: (b),(d),(f)]

4. Choose the set of correct options.

(a) The left Riemann sum of the function $f(x) = x + 5$ on the interval $[1, 10]$ divided into three sub-intervals of equal length is 81.

(b) The middle Riemann sum of the function $f(x) = x^2$ on the interval $[0, 8]$ divided into four sub-intervals of equal length is 168.

(c) The left Riemann sum of the function $f(x) = x + 5$ on the interval $[3, 6]$ divided into n sub-intervals of equal length is $\frac{57}{2}$, as n tends to ∞ .

(d) The right Riemann sum of the function $f(x) = \frac{1}{x}$ on the interval $[1, 9]$ divided into four sub-intervals of equal length is $\frac{16}{15}$.

[Ans: (a),(b),(c)]

5. The value of $\int_0^{\frac{\pi^2}{4}} \sin \sqrt{x} dx$ is?

[Ans: 2]

6. Suppose $x + y = 16$. What is the value of xy when $x^3 + y^3$ is minimum?

[Ans: 64]

7. Suppose a wire of length 31 is cut into two pieces. One part is bent into a circle and other into a square. Let x be the minimum value of the combined area of the circle and the square. Then the value of $4x(\pi + 4)$ is:

[Ans: 961π]

8. **(Answer Questions 8,9, and 10 using the data given below.)**

A car manufacturer determines that in order to sell x number of cars, the price per car(in lakh) must be $f(x) = 1000 - x$, if $x \leq 800$, and the manufacturer also determines that the total cost(in lakh) of producing x number of cars is

$$g(x) = \begin{cases} 30000 + 300x & \text{if } x \leq 400, \\ 100x + 110000 & \text{if } 400 < x \leq 800 \end{cases} \quad (1)$$

Although in the above context, x can take only integer values, assume that x is a continuous variable in the interval $[800]$ and that the functions $f(x)$ and $g(x)$ are defined as above on this entire interval.

Suppose the company can produce a maximum of 400 cars due to a production issue. The number of cars the company should produce and sell in order to maximize profit is

[Ans: 350]

9. Suppose the company can produce a minimum of 401 cars and a maximum of 800 cars due to a production issue. The number of cars the company should produce and sell in order to maximize profit is

[Ans: 450]

10. Let $P(x)$ denotes the function representing the profit of the company. Choose the set of correct statements.

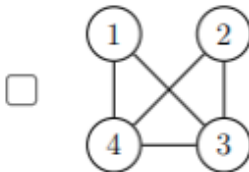
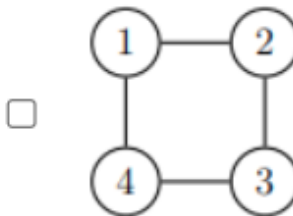
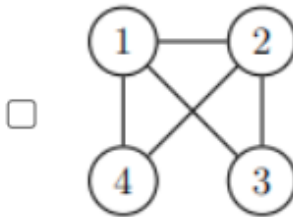
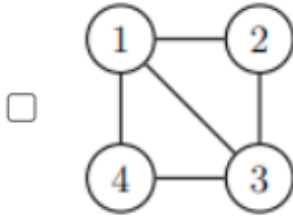
- (a) $P(x)$ is continuous in the interval $[0, 800]$
- (b) The function $P(x)$ has two local maxima in the interval $[0, 800]$.
- (c) The global maxima of $P(x)$ lies in the interval $[0, 400]$.
- (d) The global maxima of $P(x)$ lies in the interval $[300, 500]$

[Ans: (a),(b),(d)]

END OF WEEK 9

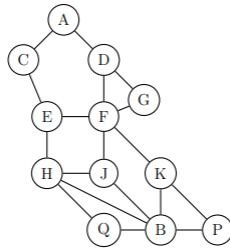
AQ 10.1

1. Which of the following options represents the graph $G = (V, E)$ where $V = \{1, 2, 3, 4\}$ and $E = \{(1, 2), (1, 3), (1, 4), (2, 3), (3, 4)\}$.



[Ans: (a)]

2. The person 'A' is a job applicant while another person 'B' is a hiring manager at a company. Although both of them are members of the website 'Linked-In', 'A' is not directly connected to 'B' via the website. The networks of 'A' and 'B' are shown in the graph below. Choose the option which shows one of the shortest paths of connecting 'A' and 'B'.



- (a) ACEHJB
- (b) ADFKB
- (c) ADGB
- (d) AKB

[Ans: (b)]

3. An electrical wire having a resistor can be represented as an edge of a graph i.e., a resistor is equivalent to two nodes connected with one edge (shown in figure AQ-10.1) and if there is no resistor, it will be counted as only one node.

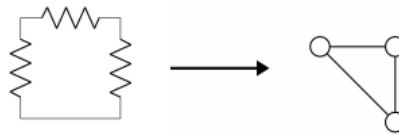


Figure AQ-10.1

Based on the figure AQ-10.1, find which graph represents the network shown in figure AQ-10.2.

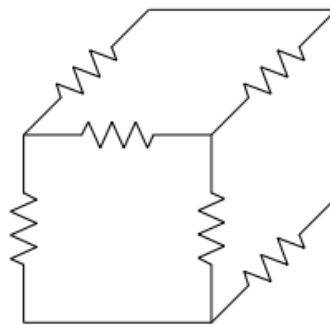
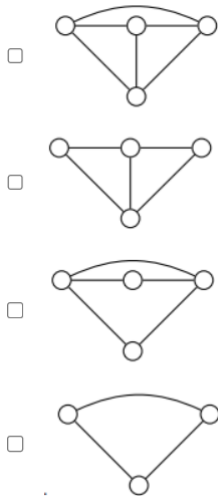


Figure AQ-10.2



[Ans: (a)]

4. A graph $G = (V, E)$ is shown in figure AQ-10.3, where V is the set of nodes in the graph and E is the set of edges.

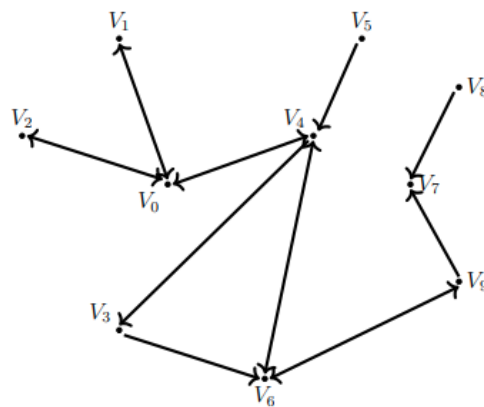


Figure AQ-10.3

Choose the set of correct options.

- (a) G is an undirected graph.
- (b) G is a directed graph.
- (c) V_4 is reachable from V_1 , V_2 , and V_5 .
- (d) V_4 is reachable from every node.
- (e) V_7 is reachable from every node.
- (f) Only V_7 is not reachable to V_4 .

[Ans: (b),(c),(e)]

AQ 10.2

1. In a factory, there are 10 buildings representing different departments. The factory wants some of the buildings as emergency assembly points such that any employee can reach an assembly point without having to go past another building. For that purpose, the factory decided to use the graph as shown in figure AQ-10.4. The nodes represents the buildings and the edges represents the path between two buildings. If V_a is a set of nodes such that V_a represents the minimum required nodes for the assembly points so that it can cover all the employees coming out from the buildings, then choose the correct option.

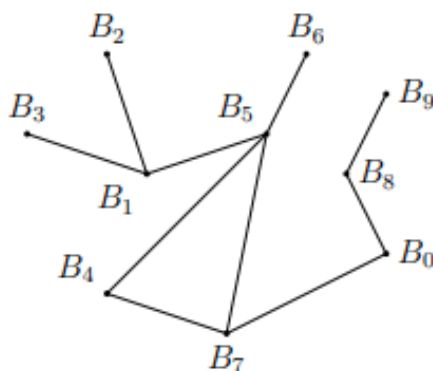


Figure AQ-10.4

- (a) $V_a = \{B1, B5, B7\}$
- (b) $V_a = \{B1, B5, B8\}$
- (c) $V_a = \{B1, B7, B9\}$
- (d) $V_a = \{B3, B2, B5\}$

[Ans: (b)]

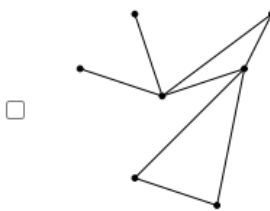
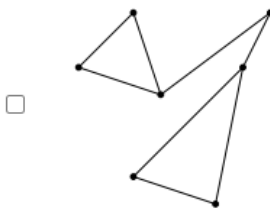
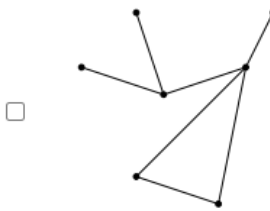
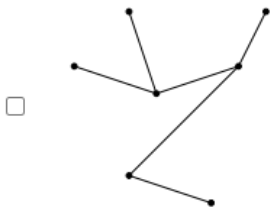
Use the following information for questions 2 and 3.

2. A college conducts an online gaming tournament for one day. There are seven games in that tournament and the time slots for those games are mentioned in the table AQ-10.4.

Game	Time slots
Fortnite	10:00 - 11:00
DoTA	10:30 - 11:30
Rocket League	10:40 - 11:40
Destiny	11:35 - 12:15
Battlefield	11:50 - 12:35
Titanfall	12:30 - 13:00
Hearthstone	12:20 - 13:00

Table AQ-10.4

Let games are represented as nodes and there is an edge between two nodes if there is a clash in timings of the corresponding games, then choose the correct option which represents best the above information.



[Ans: (c)]

3. Find the minimum possible number of computers for the tournament.

- (a) 7
- (b) 5
- (c) 4
- (d) 3

[Ans: (d)]

4. A college has sent a sports team for sport festival in other college. The team is multi-talented and each individual has the skills to take part in one or more than one sports. However, the same individual cannot be part of the team for two different sports because of a possible clash in timings. A graph (shown in figure AQ-10.5) is created where the nodes are the events and edges connect pairs of sports where the teams share a member. If the college wants the team members to take part in maximum number of sports, then choose the set of correct options.

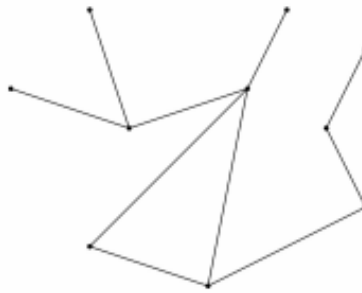


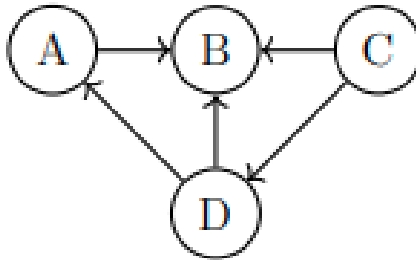
Figure AQ-10.5

- (a) The question can be solved by finding the maximum size independent set.
- (b) The question can be solved by finding the minimum size vertex cover.
- (c) The question can be solved by finding the minimal coloring.
- (d) Maximum number of sports in which the team can participate is 5.
- (e) Maximum number of sports in which the team can participate is 6.
- (f) Maximum number of sports in which the team can participate is 7.

[Ans: (a),(e)]

AQ 10.3

- Find the adjacency matrix for the following graph. (Take the sequence as A, B, C , and D).



(a)
$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

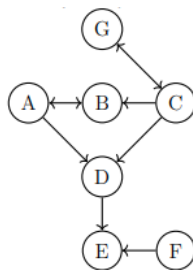
(b)
$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

(c)
$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

(d)
$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

[Ans: (b)]

- An adjacency list is created for the graph shown below. Find the value of c , d , and e .



A	{B, D}
B	{A}
C	c
D	d
E	e

- (a) $c = \{B, G, D\}, d = \{A, C, E\}$
- (b) $c = \{B, G, D\}, e = \{E\}$
- (c) $c = \{B, D\}, d = \{A, C\}$
- (d) $c = \{B, G, D\}, e = \phi$

[Ans: (d)]

3. Choose the set of correct options based on the adjacency matrix given below.

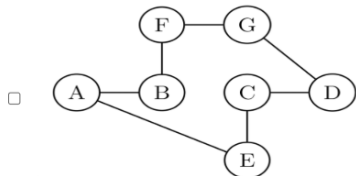
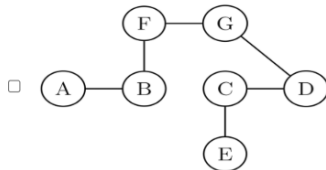
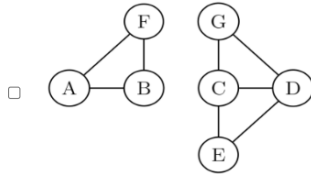
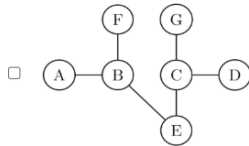
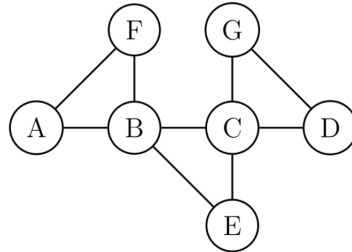
$$\begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix}$$

- (a) The graph has 5 nodes
- (b) There are 10 edges in the graph.
- (c) There are 5 edges in the graph.
- (d) The sum of the degrees of all the nodes is 20.
- (e) The sum of the degrees of all the nodes is 10.

[Ans: (a),(c),(e)]

AQ 10.4

- Suppose we perform BFS so that when we visit a vertex, we explore its unvisited neighbors in a random order. Which of the following graphs could represent the edges explored by BFS starting at vertex 'E'?



[Ans: (a)]

- The BFS algorithm uses a 'queue'. A 'queue' is a
 - LIFO (Last-In-First-Out) structure
 - FILO (First-In-Last-Out) structure
 - FIFO (First-In-First-Out) structure
 - The word 'queue' does not represent any particular structure.

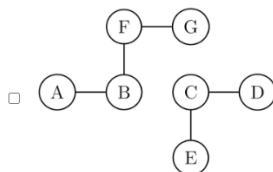
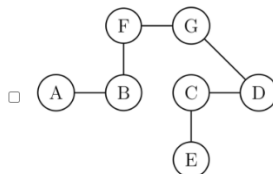
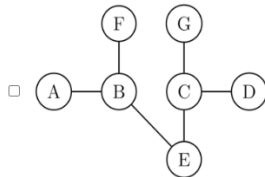
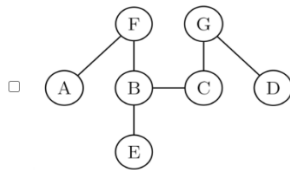
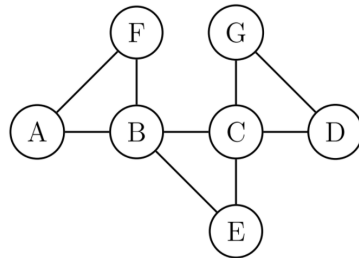
[Ans: (c)]

AQ 10.5

1. The DFS algorithm uses a 'stack'. A 'stack' is a
 - (a) FIFO (First-In-First-Out) structure
 - (b) LIFO (Last-In-First-Out) structure
 - (c) FILO (First-In-Last-Out) structure
 - (d) The word 'stack' does not represent any particular structure.

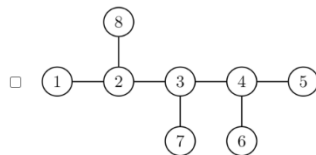
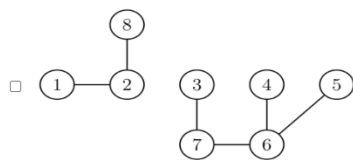
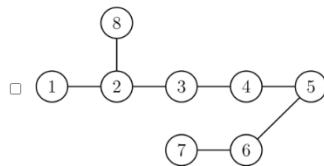
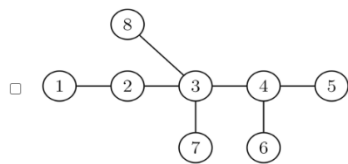
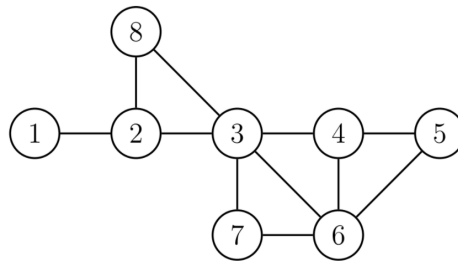
[Ans: (b)]

2. Suppose we perform DFS so that when we visit a vertex, we explore its unvisited neighbours in a random order. Which of the following graphs could represent the edges explored by DFS starting at vertex 'E'.



[Ans: (a)]

3. Suppose we perform DFS so that when we visit a vertex, we explore its unvisited neighbors in a random order. Which of the following graphs could represent the edges explored by DFS starting at vertex 3?



[Ans: (b)]

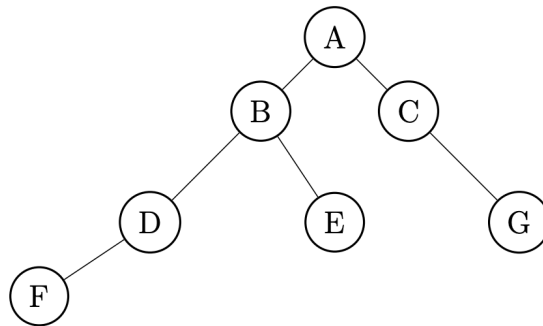
AQ 10.6

1. If we perform BFS in a directed graph G , how many times do we check whether a vertex v is visited?

- (a) 1
- (b) $\text{indegree}(v)$
- (c) $\text{outdegree}(v)$
- (d) 2

[Ans: (b)]

2. We obtain the following DFS tree rooted at node A for an undirected graph with vertices $\{A, B, C, D, E, F, G, H, I, J\}$

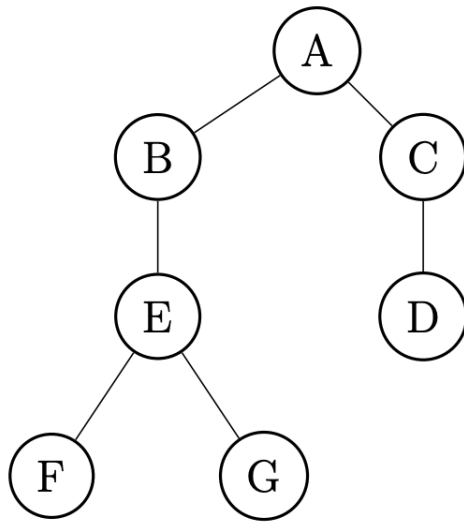


Which of the following cannot be an edge in the original graph?

- (a) (B, F)
- (b) (A, G)
- (c) (E, F)
- (d) (A, F)

[Ans: (c)]

3. Suppose we obtain the following BFS tree rooted at node A for an undirected graph with vertices $\{A, B, C, D, E, F, G\}$.



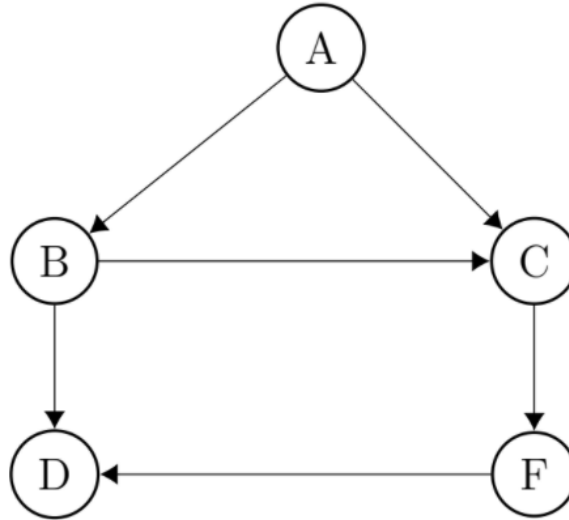
Which of the following cannot be an edge in the original graph?

- (a) (A, E)
- (b) (A, D)
- (c) (A, F)
- (d) (B, C)

[Ans: (a),(b),(c)]

AQ 10.7

1. Consider the following directed graph.



Suppose DFS of this graph is performed from node A , such that when we visit a vertex, we explore its unvisited neighbours in alphabetical order. Which of the following options are correct?

- (a) CF is a forward edge.
- (b) FE is a backward edge.
- (c) AC is a forward edge.
- (d) BD is a backward edge.

[Ans: (c)]

AQ 10.8

1. Which of the following statements is true for an undirected planar graph with 169 vertices?
 - (a) Number of edges will be less than or equal to 14196.
 - (b) Number of edges will be less than or equal to 14281.
 - (c) Number of edges will be less than or equal to 169.
 - (d) Number of edges will be less than or equal to 28382.

[Ans: (a)]

AQ 10.9

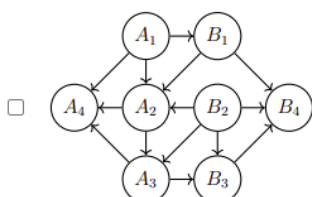
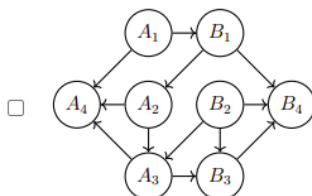
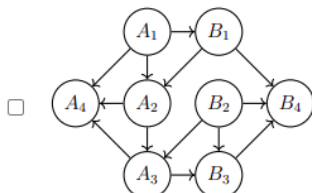
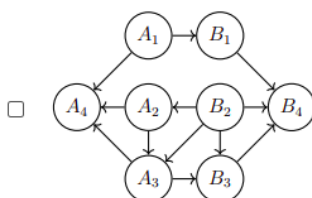
1. Use below information for Questions (1, 2 and 3)

The entry and its dependencies in a cell for any spread sheet can be represented using direct acyclic graphs. Table AQ-10.0 shows a small part of a spread sheet. The particular cell in a spread sheet can be represented as X_i , where X represents the column and i represents the row number. The entry A_1^2 in cell B_1 means B_1 has the value A_1^2 , where A_1 is the value in column A row 1.

	A	B
1	A_1	A_1^2
2	$B_1 - 10$	B_2
3	$A_2 + B_2$	B_2/A_3
4	$A_1 + A_2 + A_3$	$B_1 + B_2 + B_3$

Table AQ-10.0

Which of the following graphs represents the dependencies between cells in the table AQ-10.0



[Ans: (c)]

2. If the value of B_1 increases by 10 times, then the value of B_2 .

- (a) Will not change.
- (b) Will change but we do not know by how much.
- (c) Will also increase by 10 times.
- (d) Will decrease by 10 times.

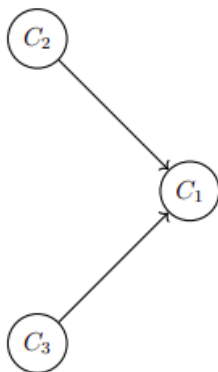
[Ans: (a)]

3. If there is a change in value of B_2 , then the reason is.

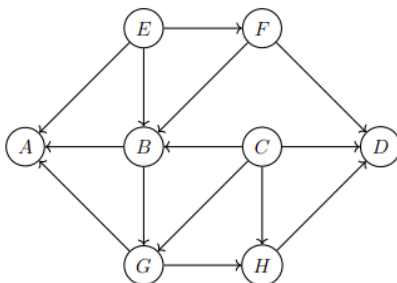
- (a) Change in value of B_4 .
- (b) Change in value of B_3 .
- (c) Change in value of A_2 .
- (d) None of the above.

[Ans: (d)]

4. If a chemical C_1 is prepared using chemicals C_2 and C_3 , then we can represent this using graphs as shown below.



A chart is shown using graphs for different types of chemicals being prepared in a factory. Choose the correct option(s) based on the graph.

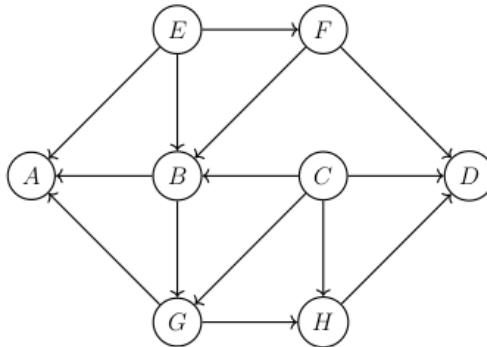


- (a) The given graph is an example of an directed acyclic graph.
- (b) The given graph is an example of a directed cyclic graph.
- (c) The given graph is an example of an undirected cyclic graph.
- (d) F and E are the initial raw material (which the factory does not produce).
- (e) C and E are the initial raw materials (which the factory does not produce).
- (f) The quality of chemical C depends on the qualities of D , G , and H .
- (g) The qualities of chemicals D , G , and H depend on the quality of C .
- (h) If there is an impurity in G , it means there was an impurity in E definitely
- (i) If there is an impurity in G , it means there might be an impurity in E .

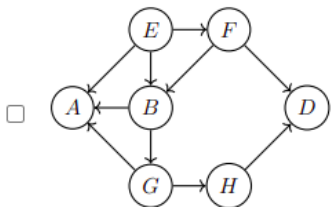
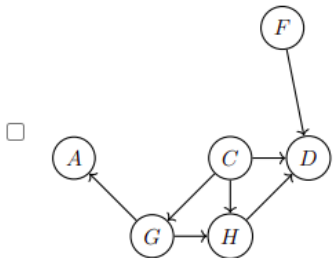
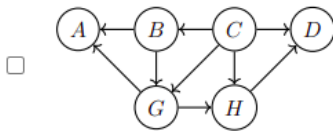
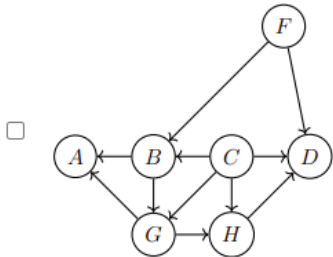
[Ans: (a),(e),(g),(i)]

AQ 10.10

- Answer the following questions based on the graph shown below.

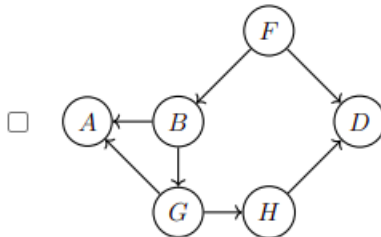
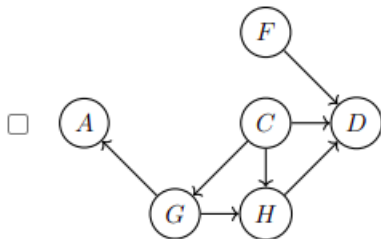
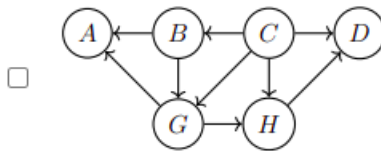
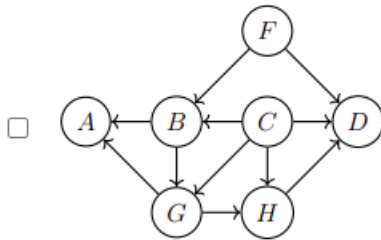


What are the resulting graphs after removing the first node during topological sorting.



[Ans: (a),(d)]

2. What is the resulting graph after removing the first two nodes during topological sorting.



[Ans: (b),(d)]

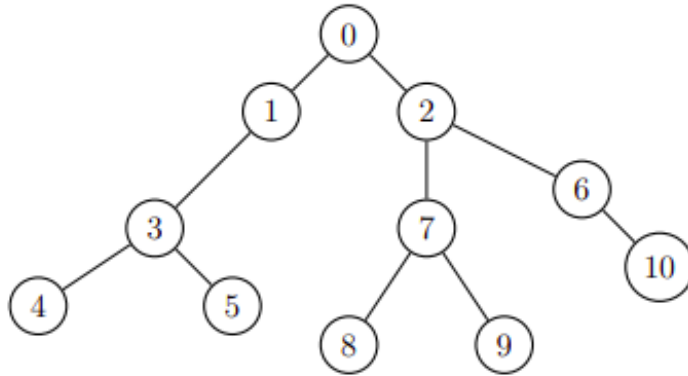
3. Which of the following are valid topological orders for this graph?

- (a) E, F, C, B, G, A, H, D .
- (b) E, C, B, F, A, G, H, D .
- (c) E, C, F, B, G, A, H, D .
- (d) C, F, E, B, A, G, H, D .
- (e) C, E, F, B, G, A, H, D .
- (f) E, C, F, B, D, H, H, F .

[Ans: (a),(c),(e)]

PA 10:

1. Suppose we obtain the following DFS tree rooted at node 0 for an undirected graph with vertices $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.



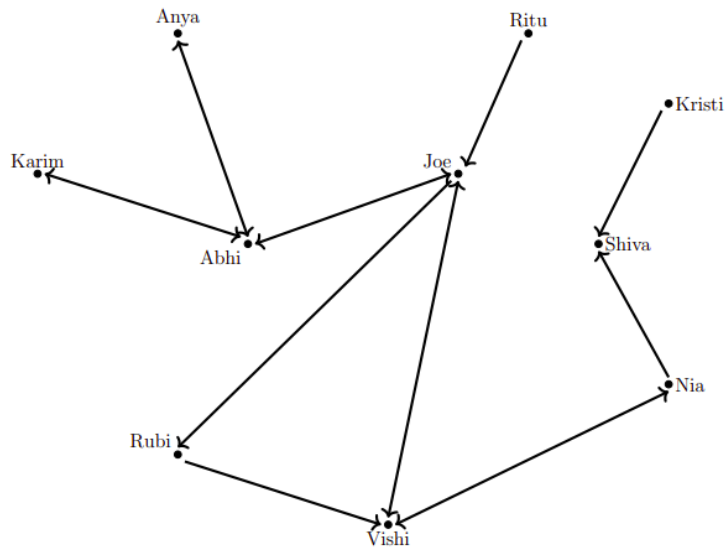
Which of the following cannot be an edge in the original graph?

- (a) $(1, 4)$
- (b) $(0, 4)$
- (c) $(7, 10)$
- (d) $(2, 9)$

[Ans: (c)]

2. Use the following information for the questions 2-4:

Ten friends in a college decided to have a night party at the home of one of them. Unfortunately at D-day the government closes many of the routes of the city. The below graph shows the location of their homes and the open routes as the graph $G = (V, E)$, where V is the set of nodes and E is the edges representing the open routes.



The possible place for the party is

- (a) Shiva's house.
- (b) Abhi's house.
- (c) Joe's house.
- (d) Vishi's house.

[Ans: (a)]

3. Let $V_1 = \{\text{Kristi, Shiva, Nia}\}$ and $E_1 = E \cap (V_1 \times V_1)$, that is, E_1 is the subset of edges of G with both end points in V_1 . Choose the correct option.

- (a) $G_1 = (V_1, E_1)$ is an undirected graph.
- (b) $G_1 = (V_1, E_1)$ is a cyclic graph.
- (c) $G_1 = (V_1, E_1)$ will not be a graph.
- (d) $G_1 = (V_1, E_1)$ is a directed graph.

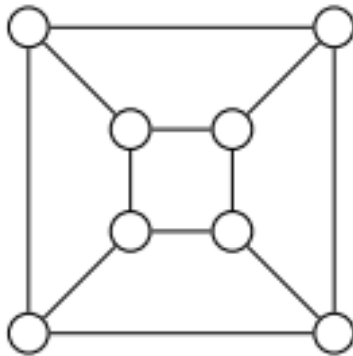
[Ans: (d)]

4. If Joe wants to have the party on his home, then at most how many members can join the party.

- (a) 5
- (b) 6
- (c) 7
- (d) 8

[Ans: (d)]

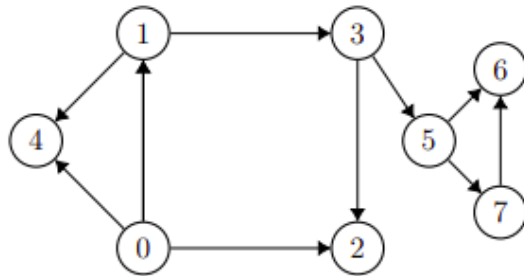
5. Suppose G be a graph (shown in the below figure). Let V be the set of vertices of G , V_i be the maximum independent set and V_c be the minimum vertex cover. Which of the followings is(are) true?



- (a) Cardinality of V_i is 4.
- (b) Cardinality of V_c is 3.
- (c) Cardinality of V_i is 5.
- (d) Cardinality of V_c is 4.

[Ans: (a), (d)]

6. Consider the graph given below.



Suppose we perform BFS/DFS so that when we visit a vertex, we explore its unvisited neighbours in a random order. Which of the following options are correct?

- (a) If we perform Breadth First Search at node 0, then one of the possible order in which the nodes will be visited is 01423567.
- (b) If we perform Depth First Search at node 0, then one of the possible order in which the nodes will be visited is 04123576.
- (c) If we perform Breadth First Search at node 0, then one of the possible order in which the nodes will be visited is 01423576.

- (d) If we perform Depth First Search at node 0, then one of the possible order in which the nodes will be visited is 04132567.

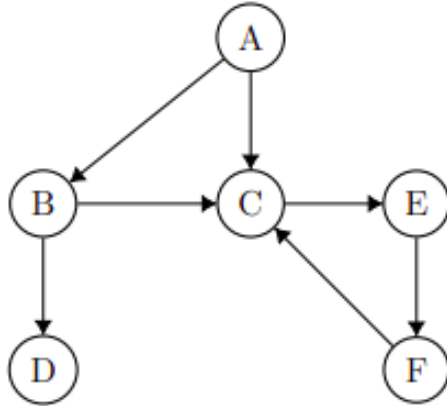
[Ans: (a), (c), (d)]

7. Which of the followings options are correct?

- (a) In Depth First search of a directed graph only back edges generate cycles.
- (b) If the maximum independent set of a graph G contains only 1 element, then the graph must be connected.
- (c) If we add an edge to a tree T , then the resulting graph becomes cyclic.
- (d) In a connected graph G having n vertices, at least two vertices have same degree.

[Ans: (a), (b), (c), (d)]

8. Consider the following directed graph.



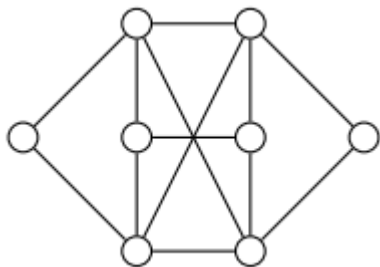
Suppose DFS of this graph is performed from node A such that when we visit a vertex we explore its unvisited neighbours in alphabetical order.

Which of the following options are correct?

- (a) AC is a forward edge.
- (b) CE is a backward edge.
- (c) BD is a forward edge.
- (d) FC is a backward edge.

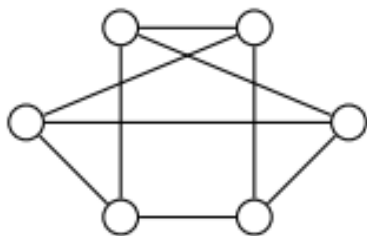
[Ans: (a), (d)]

9. The cardinality of the maximum independent set of the graph given below is



[Ans: 4]

10. The minimum colouring of the below graph is

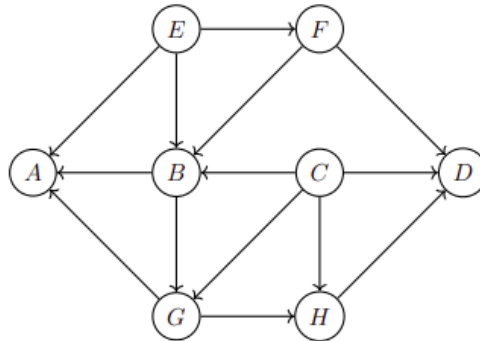


[Ans: 2]

END OF WEEK 10

AQ 11.1

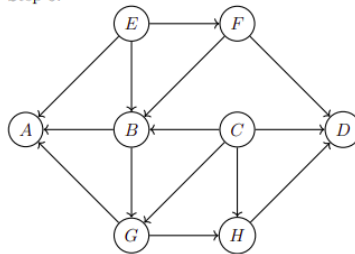
1. Answer the following questions based on the graph shown below.



One possible topological order could be C, E, F, B, G, A, H, D , and the steps are mentioned below along with the longest path computation.

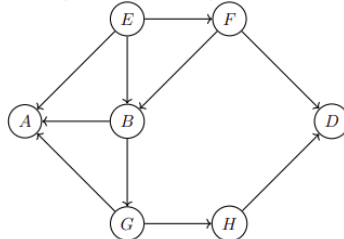
Note: L.P. refers to longest path.

Step 0:



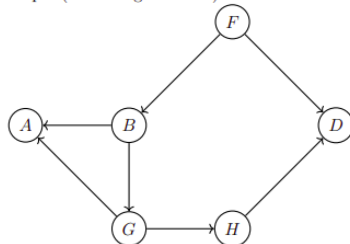
Nodes	A	B	C	D	E	F	G	H
Indegree	3	3	0	3	0	1	2	2
L. P.	0	0	0	0	0	0	0	0

Step 1 (Removing node 'C'):



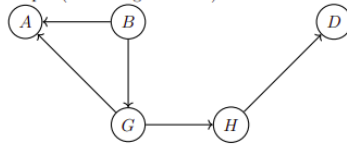
Nodes	A	B	C	D	E	F	G	H
Indegree	3	2	-	2	0	1	1	1
L. P.	0	1	0	1	0	0	1	1

Step 2 (Removing node 'E'):



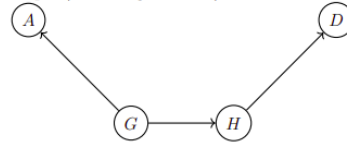
Nodes	A	B	C	D	E	F	G	H
Indegree	2	1	-	2	-	0	1	1
L. P.	1	1	0	1	0	1	1	1

Step 3 (Removing node 'F'):



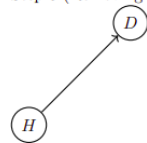
Nodes	A	B	C	D	E	F	G	H
Indegree	2	0	-	1	-	-	1	1
L. P	1	2	0	2	0	1	1	1

Step 4 (Removing node 'B'):



Nodes	A	B	C	D	E	F	G	H
Indegree	1	-	-	1	-	-	1	1
L. P	2	2	0	2	0	1	3	1

Step 5 (Removing node 'G'):



Nodes	A	B	C	D	E	F	G	H
Indegree	-	-	-	1	-	-	-	0
L. P	4	2	0	2	0	1	3	4

Step 6 (Removing node 'H'):



Nodes	A	B	C	D	E	F	G	H
Indegree	-	-	-	-	-	-	-	-
L. P	4	2	0	5	0	1	3	4

Which of the following is an error in the computation above?

- (a) Step 4- indegree of A
- (b) Step 4- Longest path of A
- (c) Step 5- indegree of D
- (d) Step 5- Longest path of D

[Ans: (b)]

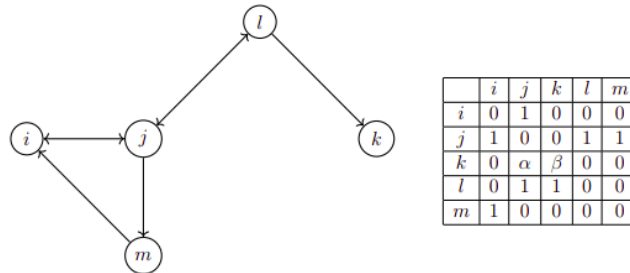
2. There are two possible topological order E, C, F, B, G, A, H, D and C, E, F, B, G, A, H, D . The longest path for G will be.

- (a) Different in both the cases.
- (b) Same in both the cases
- (c) impossible to find.
- (d) None of the above.

[Ans: (b)]

AQ 11.2

1. Answer the following questions based on the given information below. A directed graph and corresponding adjacency matrix (A or A^1) is shown in the figure.



What are the values of α and β ?

- (a) $\alpha = 1$ and $\beta = 0$
- (b) $\alpha = 1$ and $\beta = 1$
- (c) $\alpha = 0$ and $\beta = 1$
- (d) $\alpha = 0$ and $\beta = 0$

[Ans: (d)]

2. A^2 (as discussed in lecture) is created based on the adjacency matrix. What are the values of α and β ?

	i	j	k	l	m
i	1	0	0	α	1
j	1	1	1	0	0
k	0	0	β	0	0
l	α	0	0	1	1
m	0	α	0	0	0

- (a) $\alpha = 1$ and $\beta = 0$
- (b) $\alpha = 0$ and $\beta = 0$
- (c) $\alpha = 1$ and $\beta = 1$
- (d) $\alpha = 0$ and $\beta = 1$

[Ans: (a)]

AQ 11.3

1. Given the following matrices A and B .

$$A = \begin{pmatrix} 1 & 0 & 3 \\ 2 & 5 & 0 \\ 0 & 8 & 0 \end{pmatrix}$$

$$B = \begin{pmatrix} 0 & 2 & 3 \\ 4 & 0 & 6 \\ 7 & 0 & 4 \end{pmatrix}$$

Which of the following represents the product $C = A \times B$?

☐ $C = \begin{pmatrix} 21 & 2 & 15 \\ 20 & 4 & 36 \\ 3 & 0 & 9 \end{pmatrix}$

☐ $C = \begin{pmatrix} 21 & 2 & 15 \\ 20 & 4 & 36 \\ 32 & 0 & 9 \end{pmatrix}$

☐ $C = \begin{pmatrix} 21 & 2 & 15 \\ 20 & 4 & 36 \\ 32 & 0 & 48 \end{pmatrix}$

☐ $C = \begin{pmatrix} 21 & 2 & 15 \\ 20 & 4 & 36 \\ 32 & 0 & 52 \end{pmatrix}$

[Ans: (c)]

2. Answer the questions 2 and 3 based on the adjacency matrix (A or A^1) given below.

	i	j	k	l	m
i	0	α	0	0	1
j	0	0	1	0	1
k	0	0	β	1	0
l	0	0	0	0	1
m	0	0	0	0	0

We use matrix multiplication to find A^2 (shown below). What are the values of α and β ?

	i	j	k	l	m
i	0	0	1	0	1
j	0	0	0	1	0
k	0	0	0	0	1
l	0	0	0	0	0
m	0	0	0	0	0

- (a) $\alpha = 1, \beta = 0$
- (b) $\alpha = 0, \beta = 0$
- (c) $\alpha = 1, \beta = 1$
- (d) $\alpha = 0, \beta = 1$

[Ans: (a)]

3. Mark all correct statements below.

Hint: $A^3 = A^2 \times A$ and $A^4 = A^2 \times A^2$.

- (a) There is path of length 5 from i to reach m .
- (b) There is path of length 4 from i to reach m .
- (c) There is path of length 3 from i to reach m
- (d) There is path of length 2 from i to reach m
- (e) There is path of length 1 from i to reach m

[Ans: (b),(d),(e)]

AQ 11.4

1. Which of the following statements is (are) correct?
 - (a) If a graph has a negative cycle, shortest paths are not defined.
 - (b) A negative cycle is one which has only negative edge weights.
 - (c) In a weighted graph, the adjacency matrix records the weight where ever there is an edge and 0 if there is no edge.
 - (d) Shortest path in a weighted graph need not be minimum in terms of number of edges.

[Ans: (a),(c),(d)]

AQ 11.5

1. (Use the following information for questions 1, 2 and 3).

The Ministry of Earth Sciences (MoES) gave flood warning to the cities A, B, C, D, E, F and G which are very near to the foothills of Himalaya. At time 0 minutes, city B was completely flooded. In the graph (Figure AQ-12.1), vertices represent cities and edges represent how cities are connected.

Note: The weight of each edge indicates the time in minutes by which water reaches different cities.

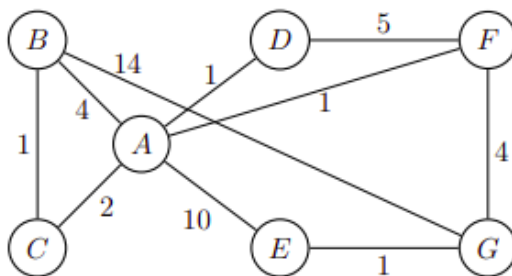


Figure AQ-12.1

At what time, will city G start flooding?

- (a) 8 min
- (b) 15 min
- (c) 14 min
- (d) 6 min

[Ans: (a)]

2. What is the shortest path to reach city D from the source of flood (B)?

- (a) $B \rightarrow A \rightarrow D$
- (b) $B \rightarrow C \rightarrow A \rightarrow F \rightarrow D$
- (c) $B \rightarrow C \rightarrow A \rightarrow D$
- (d) $B \rightarrow A \rightarrow F \rightarrow D$

[Ans: (c)]

3. At what time will all the cities be flooded?

- (a) 15 min

(b) 14 min

(c) 8 min

(d) 9 min

[Ans: (d)]

AQ 11.6

1. (Use the following information for questions 1 and 2).

While using Bellman-Ford Algorithm for the graph shown below, let $D(v)$ be the shortest distance of vertex v from the source vertex after 7 iterations.

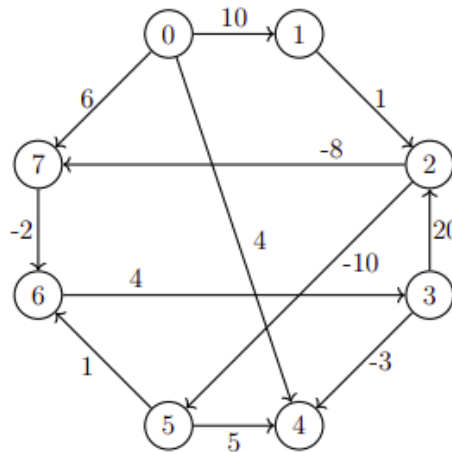


Figure AQ-12.2

Suppose the source vertex is 0. Which of the following options is (are) correct?

- (a) $D(5) = D(6) = 1$
- (b) $D(3) = 5$
- (c) $D(4) = 2$
- (d) $D(4) = 5$

[Ans: (a),(b),(c)]

2. If the source vertex is changed from vertex 0 to vertex 4. then which of the following options is (are) correct?

- (a) Bellman-Ford algorithm stabilizes after the first iteration.
- (b) Bellman-Ford algorithm will not be applicable.
- (c) $D(v)$ is finite for some vertex v other than source vertex.
- (d) None of the above.

[Ans: (a)]

3. For what values of x can we use the Bellman-Ford algorithm to find the shortest path from a source vertex 0 to every other vertex in the graph given below?

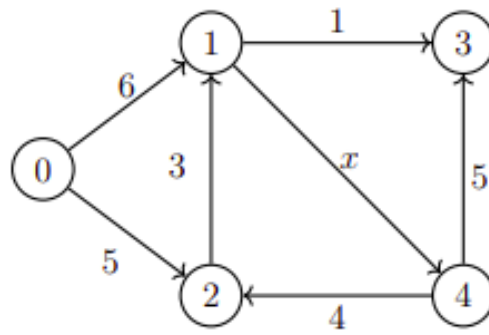


Figure AQ-12.3

- (a) $(-6, \infty)$
- (b) $(-14, \infty)$
- (c) $(-7, \infty)$
- (d) $(-21, \infty)$

[Ans: (c)]

AQ 11.7

1. Which of the following is(are) matched correctly?
 - (a) For transitive closure - Warshall algorithm.
 - (b) For all pair shortest path - Floyd-Warshall algorithm.
 - (c) For single source shortest distance for non-negative edge weights - Dijkstra's algorithm.
 - (d) For single source shortest distance for negative or non-negative edge weights - Bellman-Ford algorithm.

[Ans: (a),(b),(c),(d)]

(Use the graph (Figure AQ-12.4) for questions 2, 3, 4 and 5). (Hint: Use Floyd-Warshall Algorithm.)

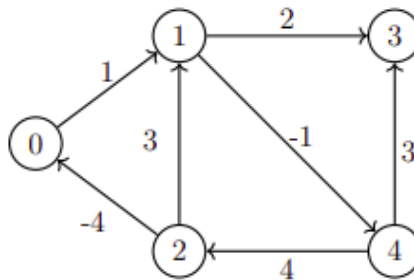


Figure AQ-12.4

2. Which of the following matrices represents SP_0 ?

☐

SP^0	0	1	2	3	4
0	∞	1	∞	∞	∞
1	∞	∞	∞	2	-1
2	-4	3	∞	∞	∞
3	∞	∞	-4	∞	∞
4	∞	∞	4	3	∞

☐

SP^0	0	1	2	3	4
0	∞	1	∞	∞	∞
1	∞	∞	∞	2	-1
2	-4	3	∞	∞	∞
3	∞	∞	∞	∞	∞
4	∞	∞	4	3	∞

☐

SP^0	0	1	2	3	4
0	∞	∞	-4	∞	∞
1	1	∞	3	∞	∞
2	∞	∞	∞	∞	4
3	∞	2	∞	∞	3
4	∞	-1	∞	∞	∞

☐

SP^0	0	1	2	3	4
0	∞	1	∞	∞	∞
1	∞	∞	∞	2	-1
2	-4	3	∞	∞	∞
3	∞	∞	∞	∞	∞
4	∞	∞	3	4	∞

[Ans: (b)]

3. Which of the following matrices represents SP_3 ?

☐

SP^3	0	1	2	3	4
0	∞	1	∞	3	0
1	∞	∞	∞	2	-1
2	-4	-3	∞	-1	-4
3	∞	∞	∞	∞	∞
4	0	1	4	3	0

☐

SP^3	0	1	2	3	4
0	∞	1	∞	∞	∞
1	∞	∞	∞	2	-1
2	-4	-3	∞	-1	-4
3	∞	∞	∞	∞	∞
4	∞	1	4	3	∞

☐

SP^3	0	1	2	3	4
0	∞	1	∞	∞	∞
1	∞	∞	∞	2	-1
2	-4	-3	∞	-1	-4
3	∞	∞	∞	∞	∞
4	0	1	4	3	∞

☐

SP^3	0	1	2	3	4
0	∞	1	∞	∞	∞
1	∞	∞	∞	2	-1
2	-4	-3	∞	∞	-4
3	∞	∞	∞	∞	∞
4	∞	1	4	3	0

[Ans: (a)]

4. Which of the following matrices represents SP_5 ?

☐

SP^5	0	1	2	4	3
0	0	1	4	3	0
1	-1	0	3	2	-1
2	-4	-3	0	-1	-4
3	∞	∞	-4	∞	∞
4	0	1	4	3	0

☐

SP^5	0	1	2	3	4
0	0	1	4	3	0
1	-1	0	3	2	-1
2	-4	-3	0	-1	-4
3	∞	∞	-4	∞	∞
4	0	1	4	3	0

☐

SP^5	0	1	2	3	4
0	0	1	4	3	0
1	-1	0	3	2	-1
2	-4	-3	0	-1	-4
3	∞	∞	∞	∞	∞
4	0	1	4	3	0

☐

SP^5	0	1	2	3	4
0	0	1	4	3	0
1	∞	∞	∞	2	-1
2	-4	3	∞	∞	∞
3	∞	∞	∞	∞	∞
4	∞	∞	3	4	∞

[Ans: (c)]

5. If $SP_i = SP_j$, for some $i, j \in \{0, 1, 2, \dots, 5\}$, then which of the following may be the values of i and j ?

- (a) $i = 4$ and $j = 3$
- (b) $i = 3$ and $j = 4$
- (c) $i = 2$ and $j = 3$
- (d) $i = 3$ and $j = 2$

[Ans: (a),(b)]

AQ 11.8

1. Which of the following is (are) correct with respect to the spanning tree G' ?

- (a) G is connected.
- (b) G is acyclic.
- (c) G has n edges.
- (d) Among the different spanning trees, one with minimum cost is minimum cost spanning tree.

[Ans: (a),(b),(d)]

2. Use the graph (Figure AQ-12.5) for questions 2 and 3.

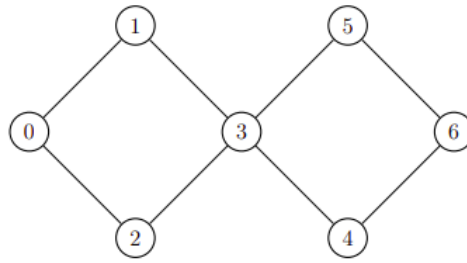
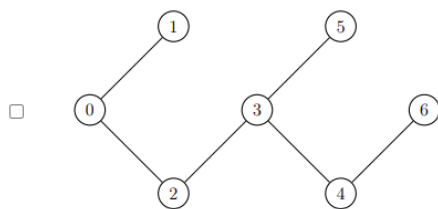
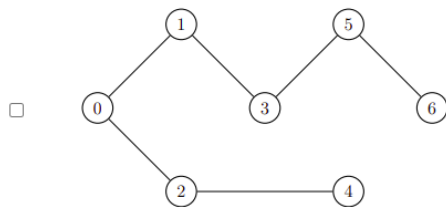
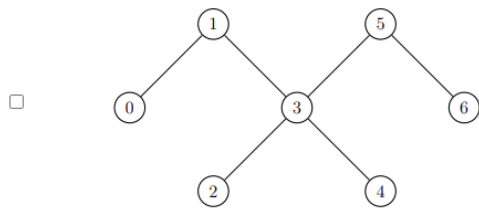
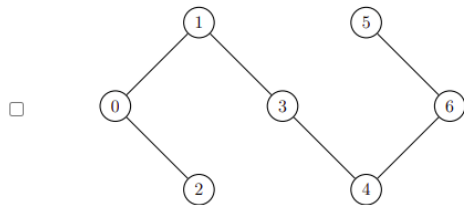


Figure AQ-12.5

Which of the following graphs is not a spanning tree of the graph given in Figure AQ-12.5?



[Ans: (c)]

3. How many spanning trees are possible for the graph shown in Figure AQ-12.5? -----
 [Ans: 16]
4. Suppose the graph shown in Figure AQ-12.6 is a spanning tree of a graph G .

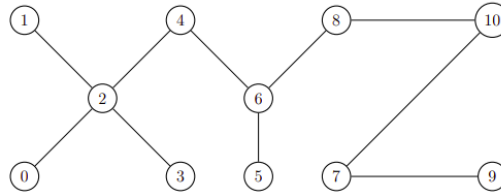
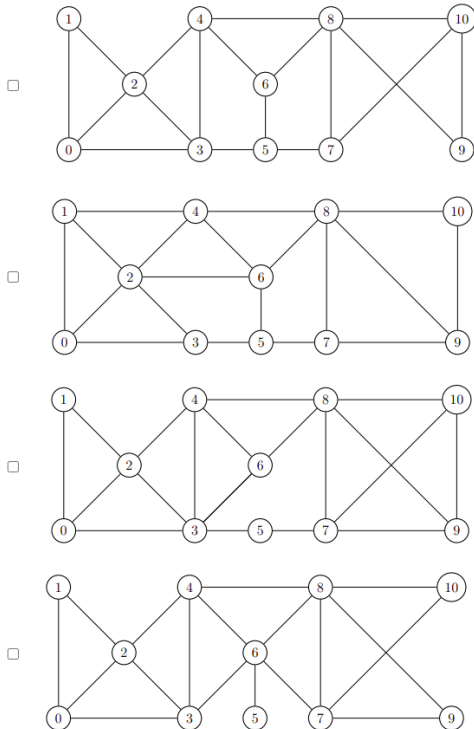


Figure AQ-12.6

Which of the following graphs may represent G ?



[Ans: (d)]

AQ 11.9

1. (Use the following information for questions 1, 2 and 3).

An undirected weighted graph G is given Figure AQ-12.7.

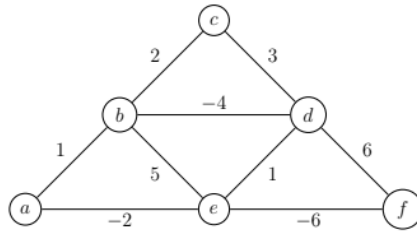
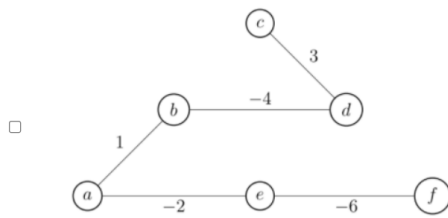
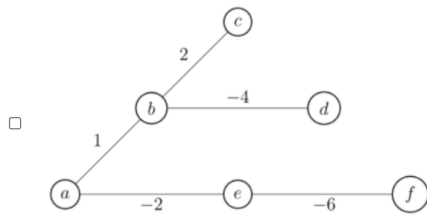
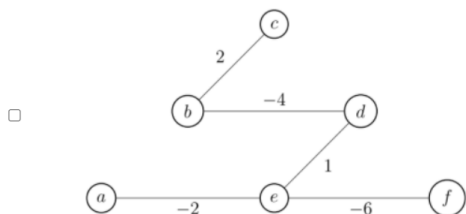
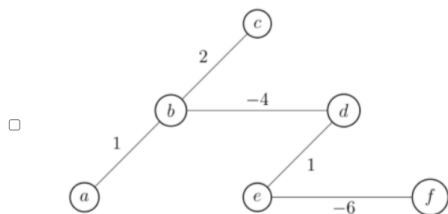


Figure AQ-12.7

If we perform Prim's algorithm on G , then which of the following options may represent a minimum cost spanning tree?





[Ans: (a),(d)]

2. Which of the following is the order in which edges are added to the minimum cost spanning tree?

- (a) $(e, f), (a, e), (e, d), (b, d), (b, c)$
- (b) $(e, f), (b, d), (a, e), (a, b), (b, c)$
- (c) $(e, f), (a, e), (b, d), (e, d), (b, c)$
- (d) $(e, f), (e, d), (b, d), (a, b), (b, c)$

[Ans: (a)]

3. What is the weight of the minimum cost spanning tree? _____

[Ans: -9]

AQ 11.10

- (Use the following information for questions 1, 2 and 3).

An undirected weighted graph G is given in Figure AQ-12.8.

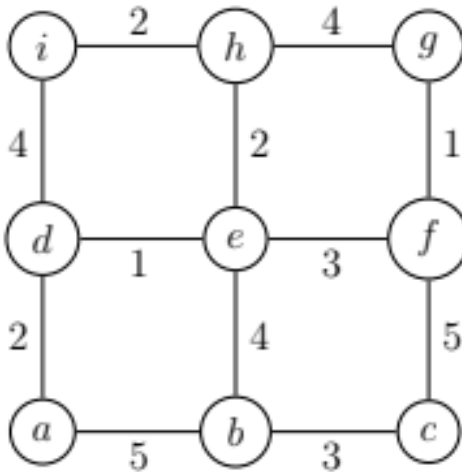
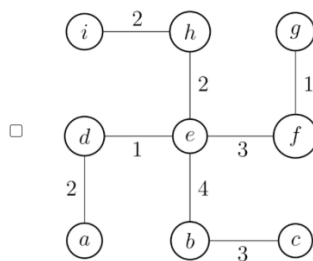
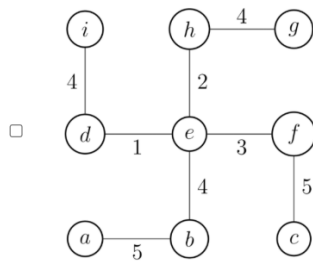
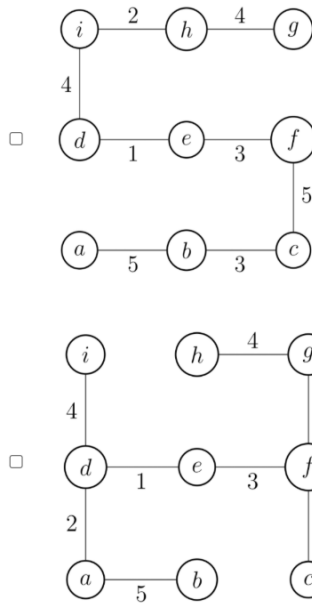


Figure AQ-12.8

If we perform Kruskal's algorithm on G , (Figure AQ-12.8), then which of the following options may represent the minimum cost spanning tree?





[Ans: (b)]

2. Which of the following options is (are) correct w.r.t the graph G (Figure AQ-12.8)?
- (a) The minimum cost spanning tree of the graph G is unique.
 - (b) There can be more than one minimum cost spanning tree of the graph G because there are some edges with equal weights.
 - (c) The weight of the minimum cost spanning tree of the graph G is 18.
 - (d) The order in which the edges are added to the minimum cost spanning tree is not unique.

[Ans: (a),(c),(d)]

3. Find the number of edges that are removed from the graph G (Figure AQ-12.8) to obtain the minimum cost spanning tree. -----
4. Suppose Prim's algorithm and Kruskal's algorithm are performed on a graph G separately to find the minimum cost spanning tree. Which of the following options will always be same for both the algorithms?
- (a) The weight of the minimum cost spanning tree of the graph G .
 - (b) The order in which the edges are added to the minimum cost spanning tree.
 - (c) Number of edges in the minimum cost spanning tree.
 - (d) The minimum weight edge in the minimum cost spanning tree.

[Ans: (a),(c),(d)]

PA W11

1. Consider the following weighted graph in Figure *PA – 12.1*.

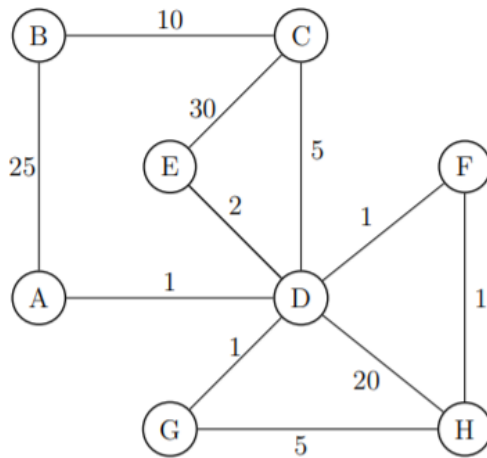


Figure PA-12.1

The shortest weighted path from H to B is

- (a) $H \rightarrow G \rightarrow D \rightarrow A \rightarrow B$
- (b) $H \rightarrow D \rightarrow C \rightarrow B$
- (c) $H \rightarrow F \rightarrow D \rightarrow E \rightarrow C \rightarrow B$
- (d) $H \rightarrow F \rightarrow D \rightarrow C \rightarrow B$

[Ans: (d)]

2. Suppose Dijkstra's algorithm is run on the graph below (Figure *PA – 12.2*), starting at node A . In what order do the shortest distances to the other vertices get finalized?

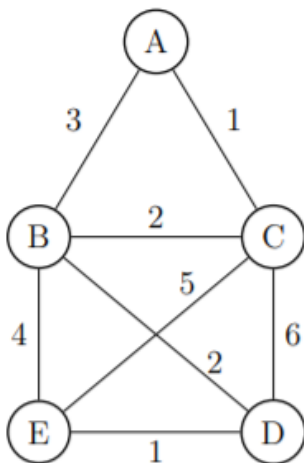


Figure PA-12.2

- (a) A, C, B, D, E
- (b) A, C, B, E, D
- (c) A, C, D, B, E
- (d) A, C, D, E, B

[Ans: (a)]

3. If we perform Floyd-Warshall algorithm for the graph shown below (Figure PA-12.3),

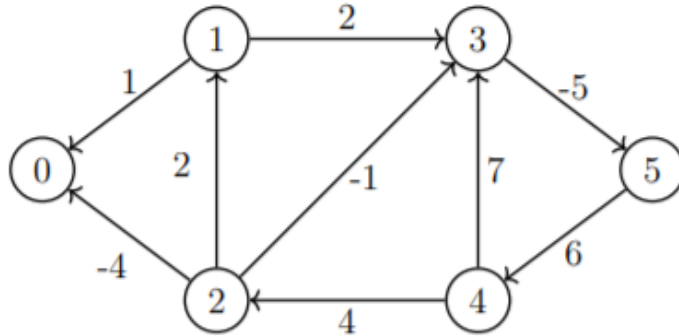


Figure PA-12.3

then which of the following matrices represents SP_4 ?

☐

SP^4	0	1	2	3	4	5
0	∞	∞	∞	∞	∞	∞
1	1	∞	∞	2	∞	∞
2	-4	2	∞	-1	∞	∞
3	∞	∞	$-\infty$	∞	∞	-5
4	∞	∞	4	7	∞	∞
5	∞	∞	∞	∞	6	∞

☐

SP^4	0	1	2	3	4	5
0	∞	∞	∞	∞	∞	∞
1	1	∞	∞	2	∞	∞
2	-4	2	∞	-1	∞	∞
3	∞	∞	∞	∞	∞	-5
4	0	6	4	3	∞	∞
5	∞	∞	∞	∞	6	∞

□

SP^4	0	1	2	3	4	5
0	∞	∞	∞	∞	∞	∞
1	1	∞	∞	2	∞	-3
2	-4	2	∞	-1	∞	-6
3	∞	∞	∞	∞	∞	-5
4	0	6	4	3	∞	-2
5	∞	∞	∞	∞	6	∞

□

SP^4	0	1	2	3	4	5
0	∞	∞	∞	∞	∞	∞
1	1	∞	∞	2	∞	-3
2	∞	2	∞	-1	∞	-6
3	∞	∞	∞	∞	∞	-5
4	0	6	4	3	∞	∞
5	∞	∞	∞	∞	6	∞

[Ans: (c)]

4. Using the graph below answer the following questions [Question 4 and 5]

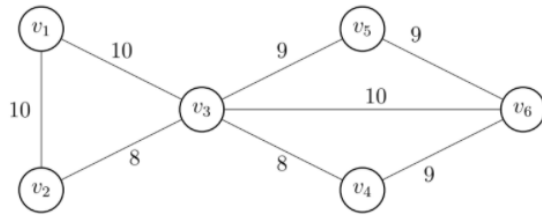
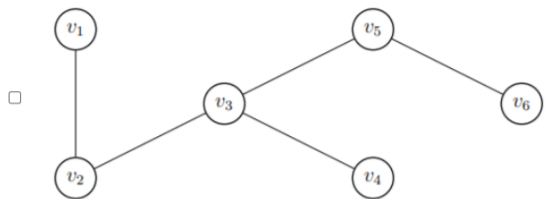
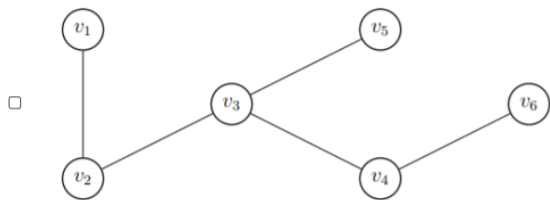
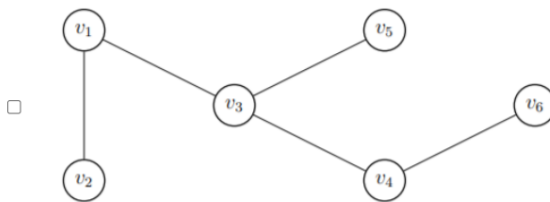
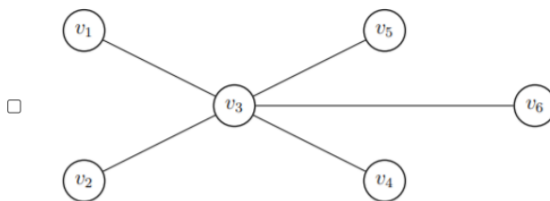


Figure PA-12.4

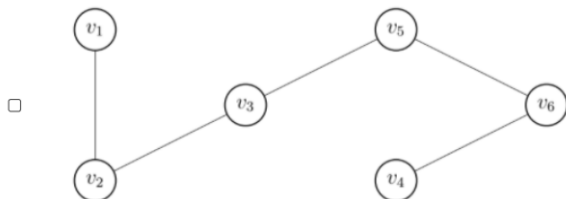
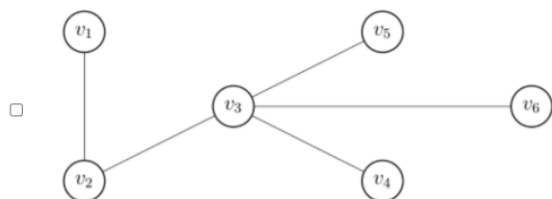
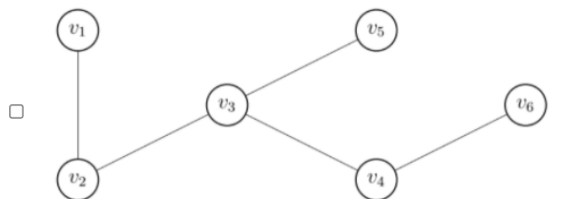
]

Which of the following could be the minimum cost spanning tree computed by running Prim's algorithm on the graph in Figure *PA* – 12.4



[Ans: (c), (d)]

5. Which of the following could be the minimum cost spanning tree computed by running Kruskal's algorithm on the graph in Figure *PA – 12.4*?



[Ans: (a), (d)]

6. While using Bellman-Ford Algorithm for the graph shown below (Figure PA – 12.5), let $D(v)$ be the shortest distance of vertex v from the source vertex 4 after 7 iterations.

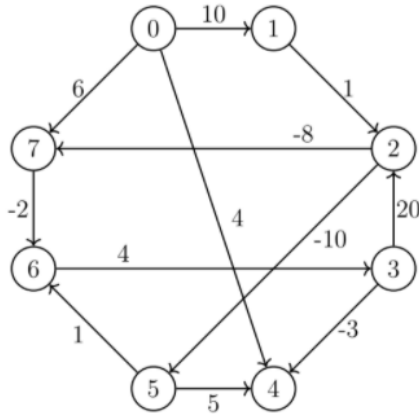


Figure PA-12.5

If the direction of edges in the graph are reversed, then which of the following is (are) CORRECT?

- (a) $D(0) = 2$
- (b) $D(2) = 9$
- (c) Bellman-Ford is not applicable for the new graph.
- (d) $D(v)$ is negative for some vertex v .

[Ans: (a)]

7. Which of the following options are correct?

- (a) Let G be a weighted graph and in which the weights of all the edges are different. If we run a shortest path algorithm on G , then we will get a unique shortest path from the starting vertex to every other vertex.
- (b) Suppose $G = (V, E)$ is a weighted graph, where $V = v_1, v_2, \dots, v_n$. Let P be a shortest path from v_i to v_j ($i \neq j$). If we increase the weight of each edge in the graph by one, then P will still be the shortest path from v_i to v_j .
- (c) A graph G can have more than one spanning tree.
- (d) Suppose $G = (V, E)$ is a weighted graph and the weights of all the edges are positive. Let P be a shortest path from $a \in V$ to $b \in V$. If we double the weight of every edge in the graph G , then the shortest path remains same but the total weight of path changes.

[Ans: (c), (d)]

8. Which of the following options are correct?

- (a) Dijkstra's algorithm works for graphs having no negative weight edge.
- (b) Floyd–Warshall algorithm works for graphs with negative weight cycles.
- (c) Dijkstra's algorithm works on any graph without negative weight cycles.
- (d) Shortest path problem is not applicable for a graph with a negative weight cycle.

[Ans: (a), (d)]

9. What is the weight of a minimum cost spanning tree of the graph given below (Figure PA – 12.6)?

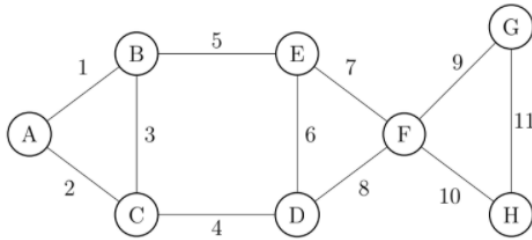


Figure PA-12.6

[Ans: 38]

END OF WEEK 11—————